

Asymmetric Information in Wage Contracts: Evidence from an Online Experiment

Daniel Herbst
University of Arizona

February 2025

Insuring Labor Product

- Most workers' labor product combines effort and risk
- Would likely value **insurance** against that risk
- What would be the premium for a contract that...
 - Pays restaurant servers when they're stiffed on the tip?
 - Pays lawyers when they lose a case?
 - Pays academics when they're rejected from journals?
- Employers provide *implicit insurance* through **fixed wages**

Piece-rate Pay

- Paid by unit of output
- Worker sells labor product at market price
- Examples:
 - Rideshare driver paid per mile
 - Self-employed freelance photographer

Fixed Wages

- Paid by unit of time
- Worker sells *claim* on labor product
- Examples:
 - Limo driver paid per hour
 - Staff photographer with annual salary

← Earnings Risk

Insurance →

Hourly wage + tips

Annual salary + bonus

Base pay + commission

Are these contracts socially optimal?



Piece-rate Pay

- Paid by unit of output
- Worker sells labor product at market price
- Examples:
 - Rideshare driver paid per mile
 - Self-employed freelance photographer

Fixed Wages

- Paid by unit of time
- Worker sells *claim* on labor product
- Examples:
 - Limo driver paid per hour
 - Staff photographer with annual salary

Earnings Risk

Insurance

Asymmetric information $\Rightarrow$ market distortions:

- **Moral hazard:** fixed wage induces less effort
- **Adverse selection:** less productive workers sort into fixed wages

$\Rightarrow$ Suboptimal wage contracts

Piece-rate Pay

- Paid by unit of output
- Worker sells labor product at market price
- Examples:
 - Rideshare driver paid per mile
 - Self-employed freelance photographer

Fixed Wages

- Paid by unit of time
- Worker sells *claim* on labor product
- Examples:
 - Limo driver paid per hour
 - Staff photographer with annual salary

Earnings Risk

Insurance

Asymmetric information $\Rightarrow$ market distortions:

- **Moral hazard:** fixed wage induces less effort
- **Adverse selection:** less productive workers sort into fixed wages

$\Rightarrow$ Suboptimal wage contracts

Public policies can mitigate these distortions:

- Taxes tips, commissions, or bonuses
- Hourly wage subsidies
- Employment classification rules
- Portable benefits programs
- Minimum wage

$\Rightarrow$ Promote implicit insurance in wage contracts

Piece-rate Pay

- Paid by unit of output
- Worker sells labor product at market price
- Examples:
 - Rideshare driver paid per mile
 - Self-employed freelance photographer

Fixed Wages

- Paid by unit of time
- Worker sells *claim* on labor product
- Examples:
 - Limo driver paid per hour
 - Staff photographer with annual salary

Earnings Risk

Insurance

Asymmetric information $\Rightarrow$ market distortions:

- **Moral hazard:** fixed wage induces less effort
- **Adverse selection:** less productive workers sort into fixed wages

$\Rightarrow$ Suboptimal wage contracts

Public policies can mitigate these distortions:

- Taxes tips, commissions, or bonuses
- Hourly wage subsidies
- Employment classification rules
- Portable benefits programs
- Minimum wage

$\Rightarrow$ Promote implicit insurance in wage contracts

How do moral hazard and adverse selection influence wage contracts?

This Paper:

1. Experimentally identify treatment effects and selection into hourly pay
 - Two-stage RCT separates moral hazard, adverse selection, and wage effects
 1. Offer workers a *choice* between randomized hourly wage or piece rate
 2. Randomly increase hourly workers' wages to match highest wage offer
 - Find evidence of both moral hazard and adverse selection into hourly wages
 - MH $\equiv$ Treatment effect: take-up of hourly contract $\Rightarrow$ output $\downarrow$ by 6.3% relative to the mean
 - AS $\equiv$ Selection into treatment: wage offer $\uparrow$ by 10% $\Rightarrow$ marginal worker's productivity $\uparrow$ by 1.4%

This Paper:

1. Experimentally identify treatment effects and selection into hourly pay
2. Develop model of fixed-wage labor markets under asymmetric info
 - Clarifies roles of AS and MH in shaping equilibrium, allowing for monitoring costs
 - Equilibrium and efficient shares of hourly work are determined by three curves:
 1. $\bar{w}(\theta)$: **reservation wage** for an hourly position at quantile θ
 2. $MV(\theta)$: **marginal value** of output among workers at a given reservation wage
 3. $AV(\theta)$: **average value** of output among workers with lower reservation wage

This Paper:

1. Experimentally identify treatment effects and selection into hourly pay
2. Develop model of fixed-wage labor markets under asymmetric info
3. Use MTE framework to estimate equilibrium and efficient hourly wages
 - Let experimental wage offers serve as **instrument** for hourly contract take-up $\Rightarrow$
 - Hourly labor supply at given wage, $S(w)$ = propensity score
 - Marginal value under hourly pay, $MV_1(\theta)$ = potential outcome in treated state
 - Marginal value under piece rate, $MV_0(\theta)$ = potential outcome in untreated state $\Rightarrow$ MTE of hourly contracts map directly into model

This Paper:

1. Experimentally identify treatment effects and selection into hourly pay
2. Develop model of fixed-wage labor markets under asymmetric info
3. Use MTE framework to estimate equilibrium and efficient hourly wages
4. Calculate DWL of inefficient wages and welfare impact of tax/subsidy
 - Welfare loss of \$0.04 per hour of labor
 - Hourly subsidy $\leq \$1.00 \Rightarrow$ generates \$1.15 of welfare per \$1 of govt expenditure
 - Piece-rate tax $\leq 15\% \Rightarrow$ raises \$1 of govt revenue at social cost of \$0.85 or less

Existing Literature

- Asymmetric Information in Labor Markets
 - Spence (1973), Weiss (1980), Gibbons and Katz (1991), Farber and Gibbons (1996), Shimer (2005), Landers et al. (1996), Ales and Maziero (2016), Pallais (2016), Sztutman (2024)
- Selection and Incentive Effects of Wage Contracts
 - Lazear (2000), Nagin et al. (2002), Gaynor et al. (2004), Shearer (2006), DellaVigna and Pope (2018), Kantarevic and Kralj (2016), Angrist et al. (2017), Brown and Andrabi (2021), Emanuel and Harrington (2024)
- Measuring Adverse Selection & Moral Hazard
 - Chiappori and Salanie (2000), Karlan and Zinman (2009), Einav, Finkelstein, Cullen (2010), Kowalski (2023), Herbst and Hendren (2024)
- Selection and Marginal Treatment Effects
 - Black et al. (2022), Mogstad et al. (2018), Huber (2013), Carneiro et al. (2011), Heckman and Vytlacil (2005, 2007), Björklund and Moffitt (1987)

Outline

- 1 Experimental Design**
- 2 Main Results**
- 3 Model of Asymmetric Information**
- 4 Estimates of Marginal Value and Welfare Loss**
- 5 Optimal Fixed-Wage Subsidy and Piece-Rate Tax**

Detecting Moral Hazard and Adverse Selection

Existing methods of testing for moral hazard and adverse selection:

1. Test for correlation between take-up and realized risk (Chiappori and Salanie 2000)
 - Cannot separate adverse selection from moral hazard
2. Compare take-up and risk across exogenous price changes in existing markets (Einav et al. 2010, Einav et al. 2013; Hackmann et al. 2012)
 - “Under-the-lamppost” problem: Cannot observe *unraveled* contracts, which are unprofitable
3. Construct *hypothetical* contract choices from surveys on subjective beliefs (Herbst and Hendren 2024; Hendren 2013, 2017)
 - Strong parametric assumptions to predict choices across “missing” contracts

My strategy: if I can't observe missing contracts, offer them myself!

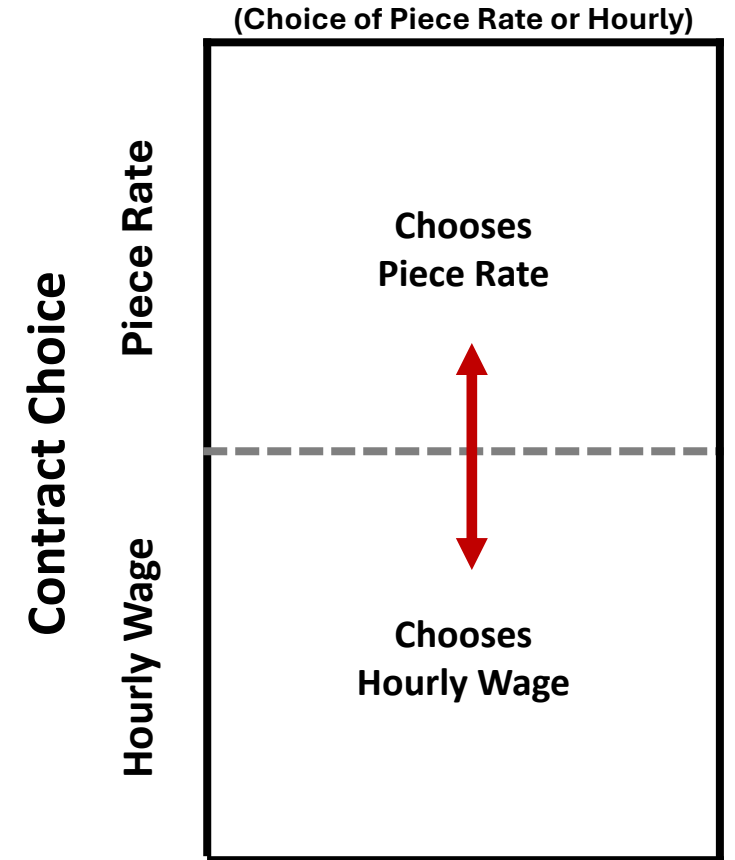
- Advantage #1: I can experimentally randomize contract prices (i.e., wages)
- Advantage #2: I can offer unraveled (i.e., unprofitable) contracts

Detecting Moral Hazard and Adverse Selection

- Suppose worker i is offered a choice:
 - Remain on piece rate ($H_i = 0$)
 - Switch to an hourly wage ($H_i = 1$)
- Potential outcomes:
 Y_{1i} : Worker i 's potential output under hourly wage
 Y_{0i} : Worker i 's potential output under piece rate
$$Y_i = H_i Y_{1i} + (1 - H_i) Y_{0i}$$
- Comparing across self-selected groups combines treatment and selection:

$$E[Y_i | H_i = 1] - E[Y_i | H_i = 0] =$$

$$\underbrace{E[Y_{1i} - Y_{0i} | H_i = 1]}_{\text{Treatment on the Treated}} + \underbrace{E[Y_{0i} | H_i = 1] - E[Y_{0i} | H_i = 0]}_{\text{Selection on } Y_0}$$

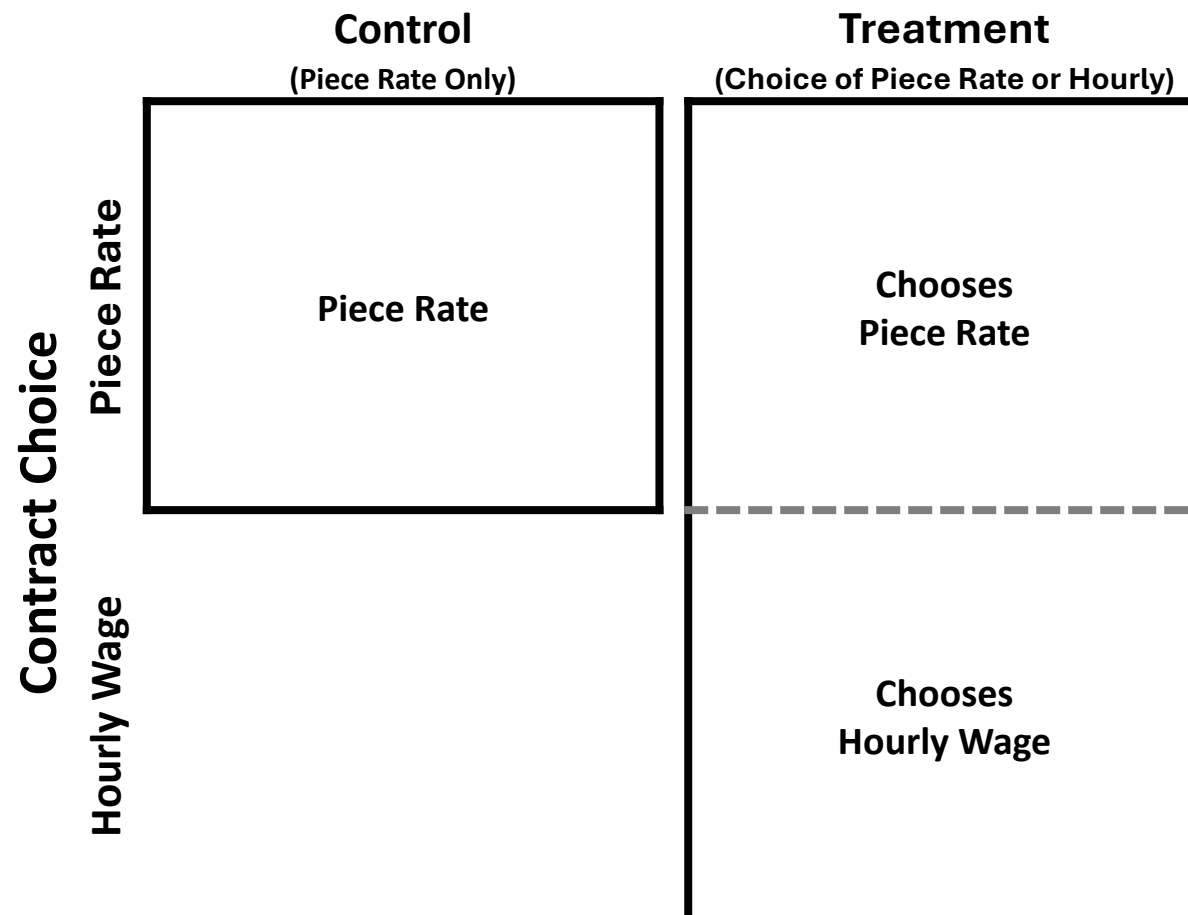


Experimental Design: Single Wage Offer

Simplified Experimental Design

Randomize workers into two offer sets:

- **Treated workers ($W_i = 1$):** offered choice
 - Remain on piece rate ($H_i = 0$)
 - Switch to an hourly wage ($H_i = 1$)
- **Control workers ($W_i = 0$):** no hourly offer
 - Remain on piece rate ($H_i = 0$)

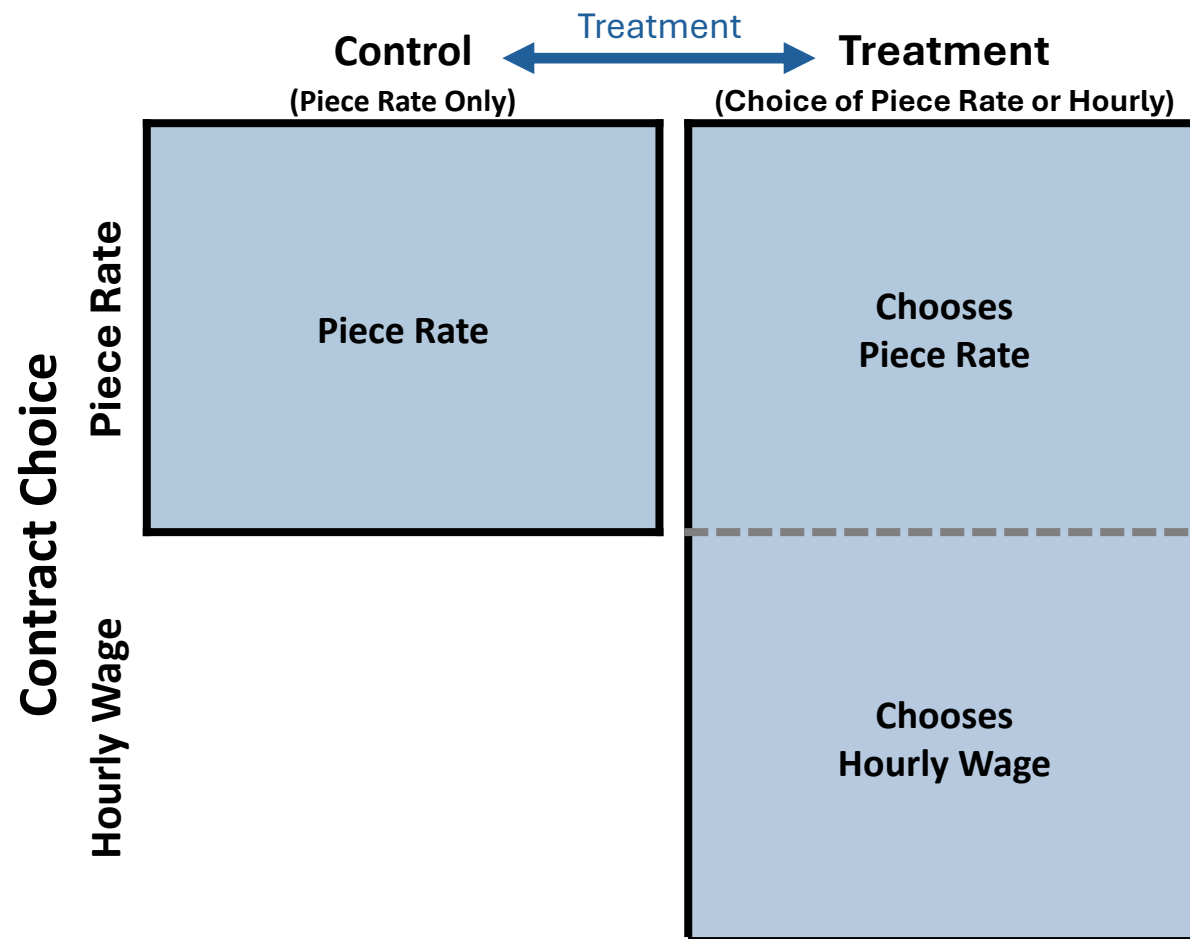


Experimental Design: Single Wage Offer

Comparing between workers facing different hourly offers identifies **treatment effect** of hourly wages:

$$\frac{E[Y_i|W_i = 1] - E[Y_i|W_i = 0]}{\Pr(H_i = 1|W_i = 1)} = \underbrace{E[Y_{1i} - Y_{0i}|H_i = 1]}_{\text{Local Average Treatment Effect (LATE)}}$$

Moral Hazard

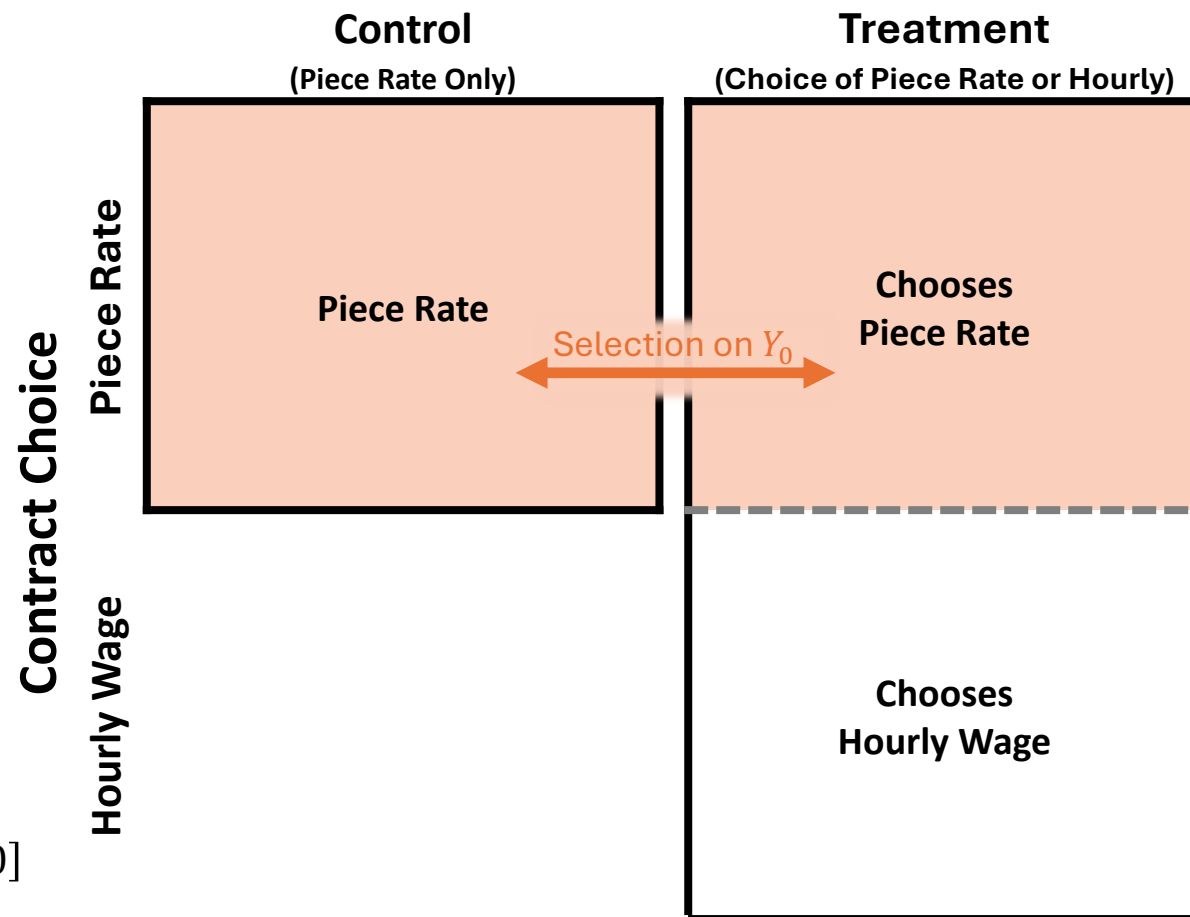


Experimental Design: Single Wage Offer

Comparing piece-rate workers who faced different offer sets identifies ***selection into treatment***:

$$\frac{E[Y_i|W_i = 0] - E[Y_i|H_i = 0, W_i = 1]}{\Pr(H_i = 1|W_i = 1)} = \underbrace{E[Y_{0i}|H_i = 1] - E[Y_{0i}|H_i = 0]}_{\text{Selection on } Y_0}$$

Adverse Selection



Experimental Design: Single Wage Offer

Comparing between workers facing different hourly offers identifies **treatment effect** of hourly wages:

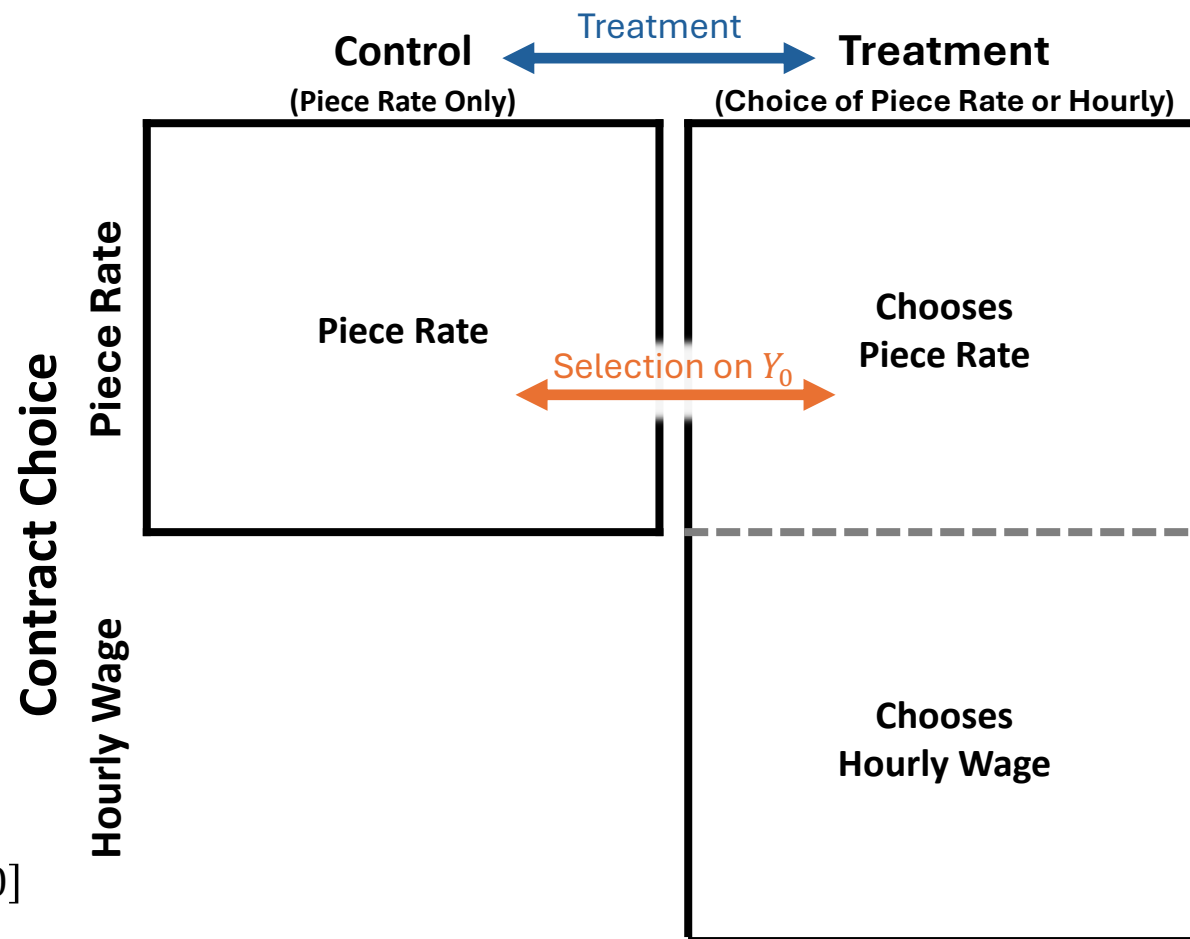
$$\frac{E[Y_i|W_i = 1] - E[Y_i|W_i = 0]}{\Pr(H_i = 1|W_i = 1)} = \underbrace{E[Y_{1i} - Y_{0i}|H_i = 1]}_{\text{Local Average Treatment Effect (LATE)}}$$

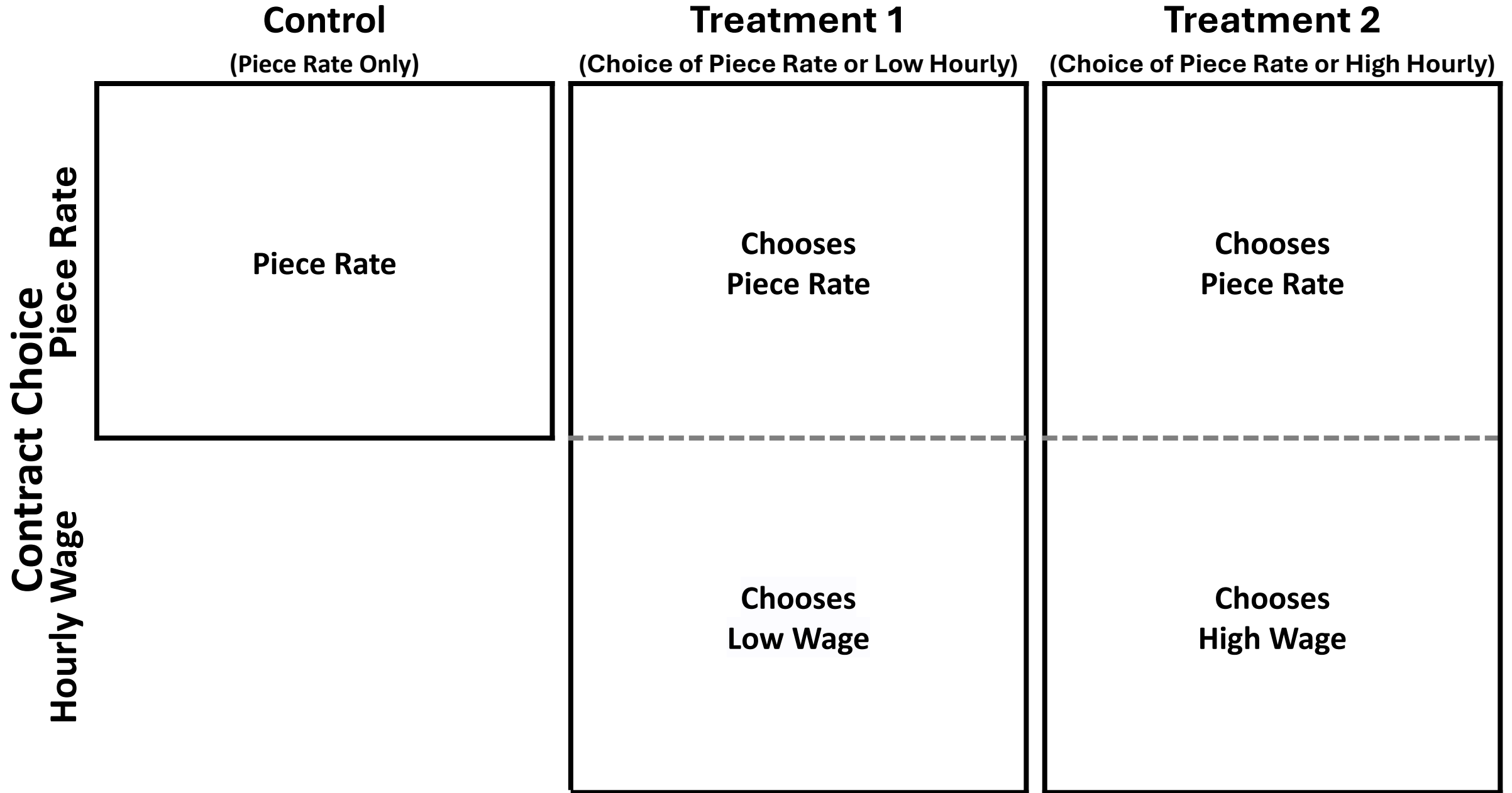
Moral Hazard

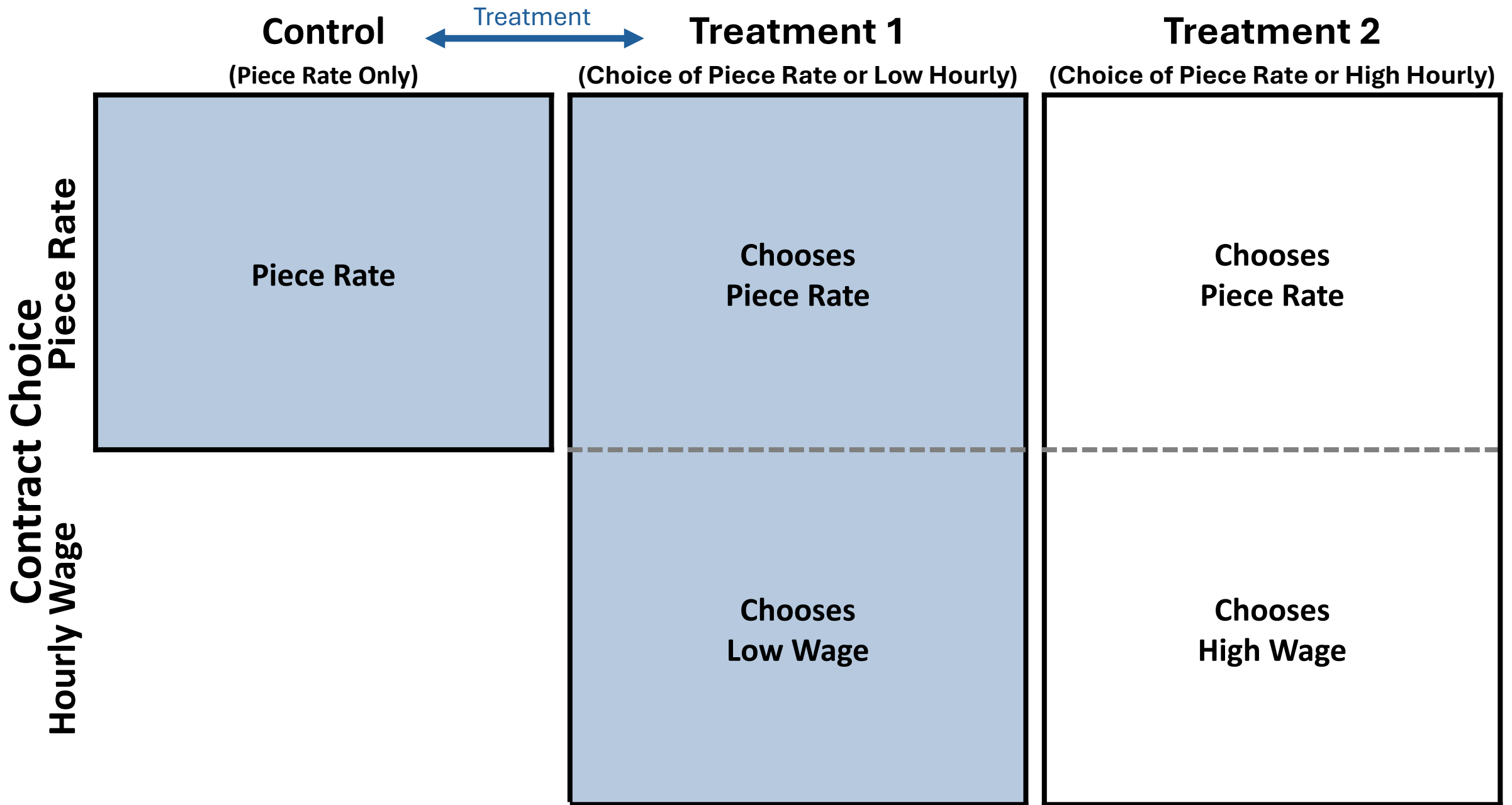
Comparing piece-rate workers who faced different offer sets identifies **selection into treatment**:

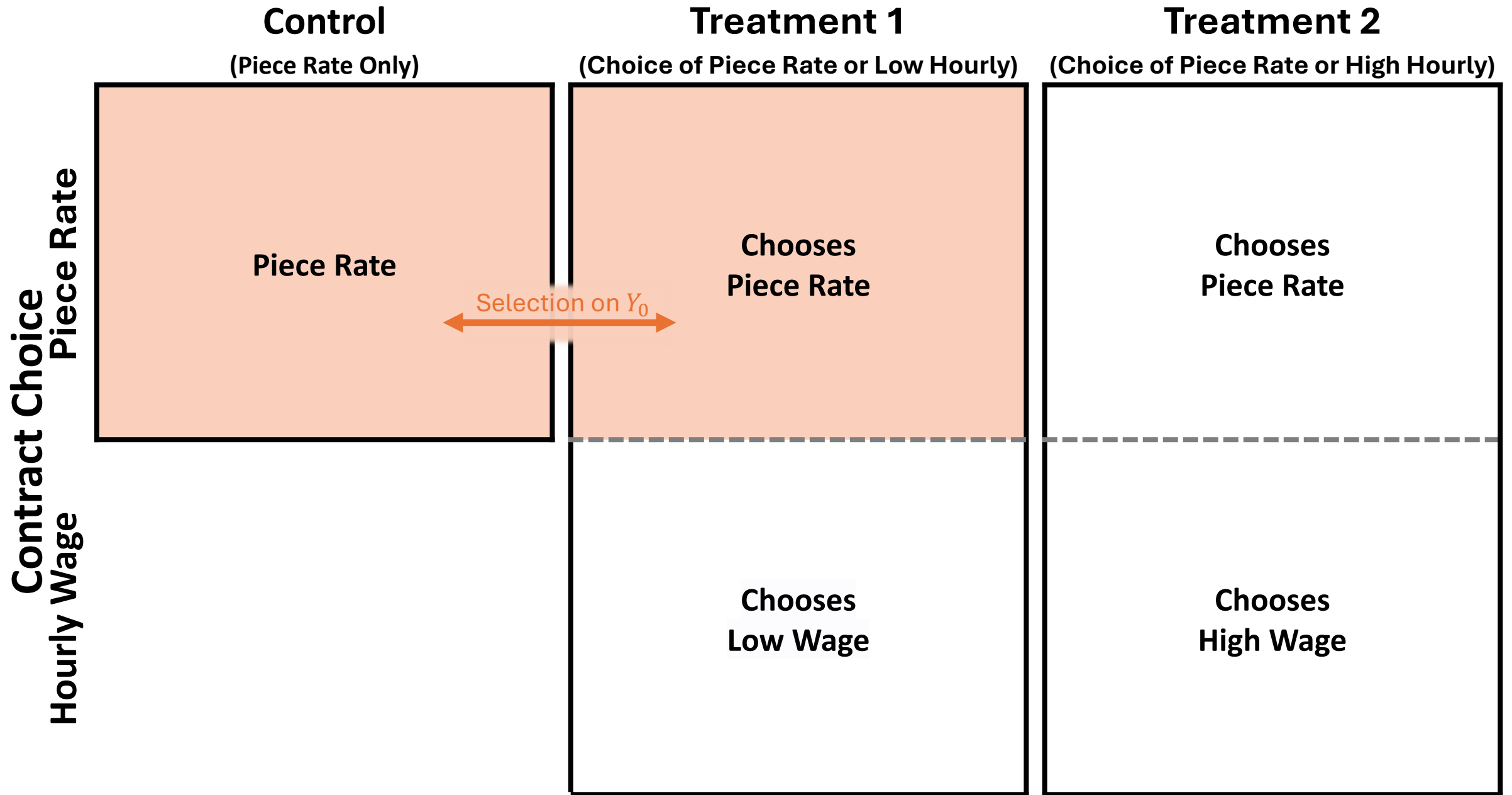
$$\frac{E[Y_i|W_i = 0] - E[Y_i|H_i = 0, W_i = 1]}{\Pr(H_i = 1|W_i = 1)} = \underbrace{E[Y_{0i}|H_i = 1] - E[Y_{0i}|H_i = 0]}_{\text{Selection on } Y_0}$$

Adverse Selection



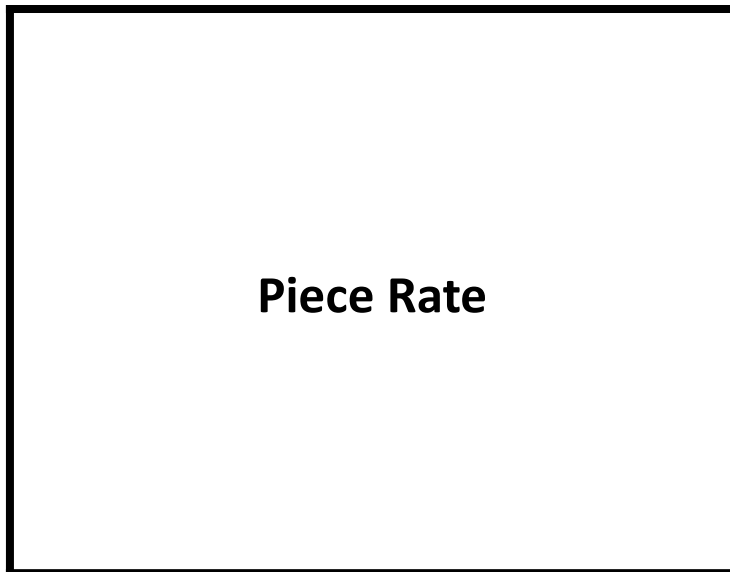






Contract Choice
Piece Rate
Hourly Wage

Control
(Piece Rate Only)



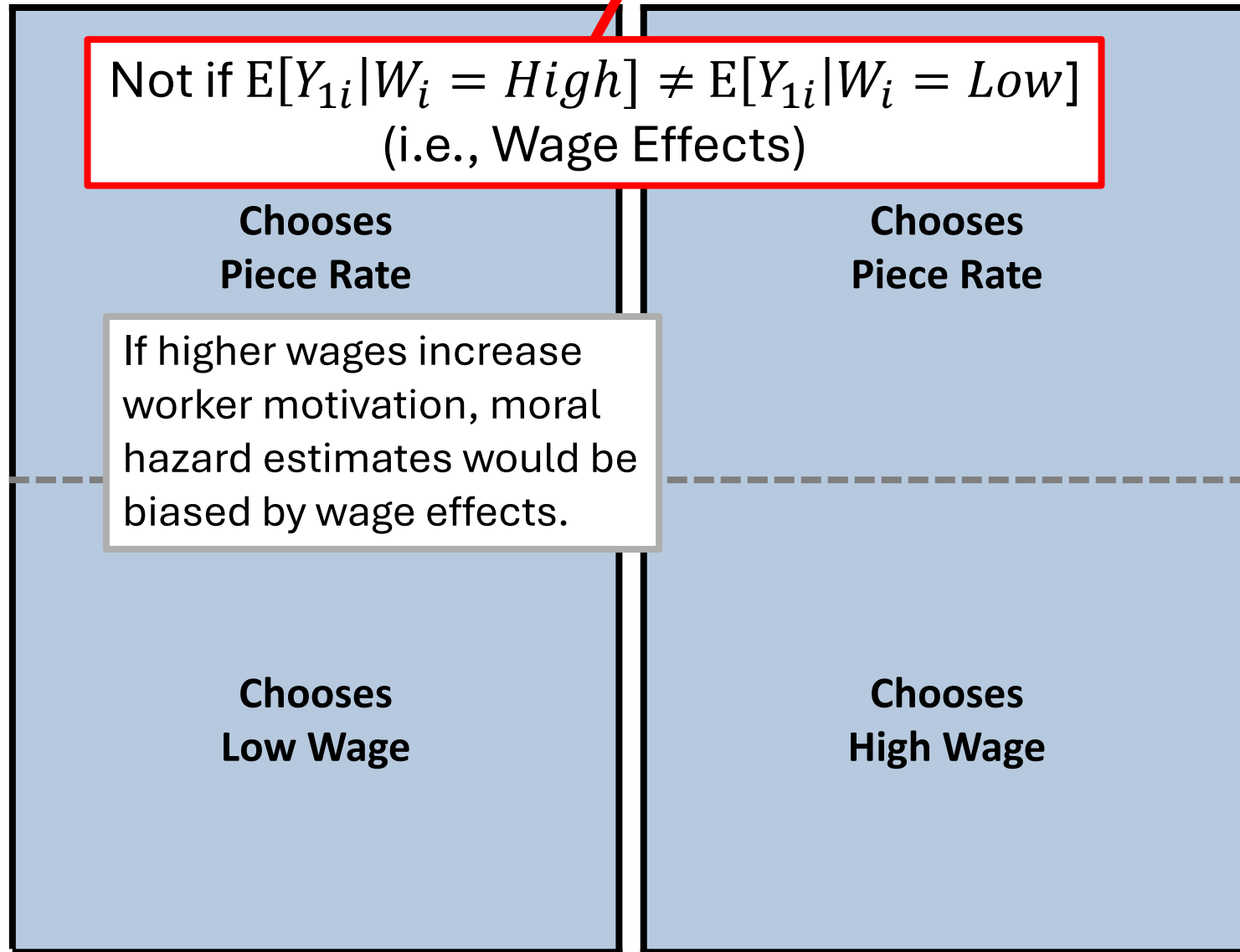
Treatment 1

(Choice of Piece Rate or Low Hourly)

~~Treatment ?~~

Treatment 2

(Choice of Piece Rate or High Hourly)



Not if $E[Y_{1i}|W_i = High] \neq E[Y_{1i}|W_i = Low]$
(i.e., Wage Effects)

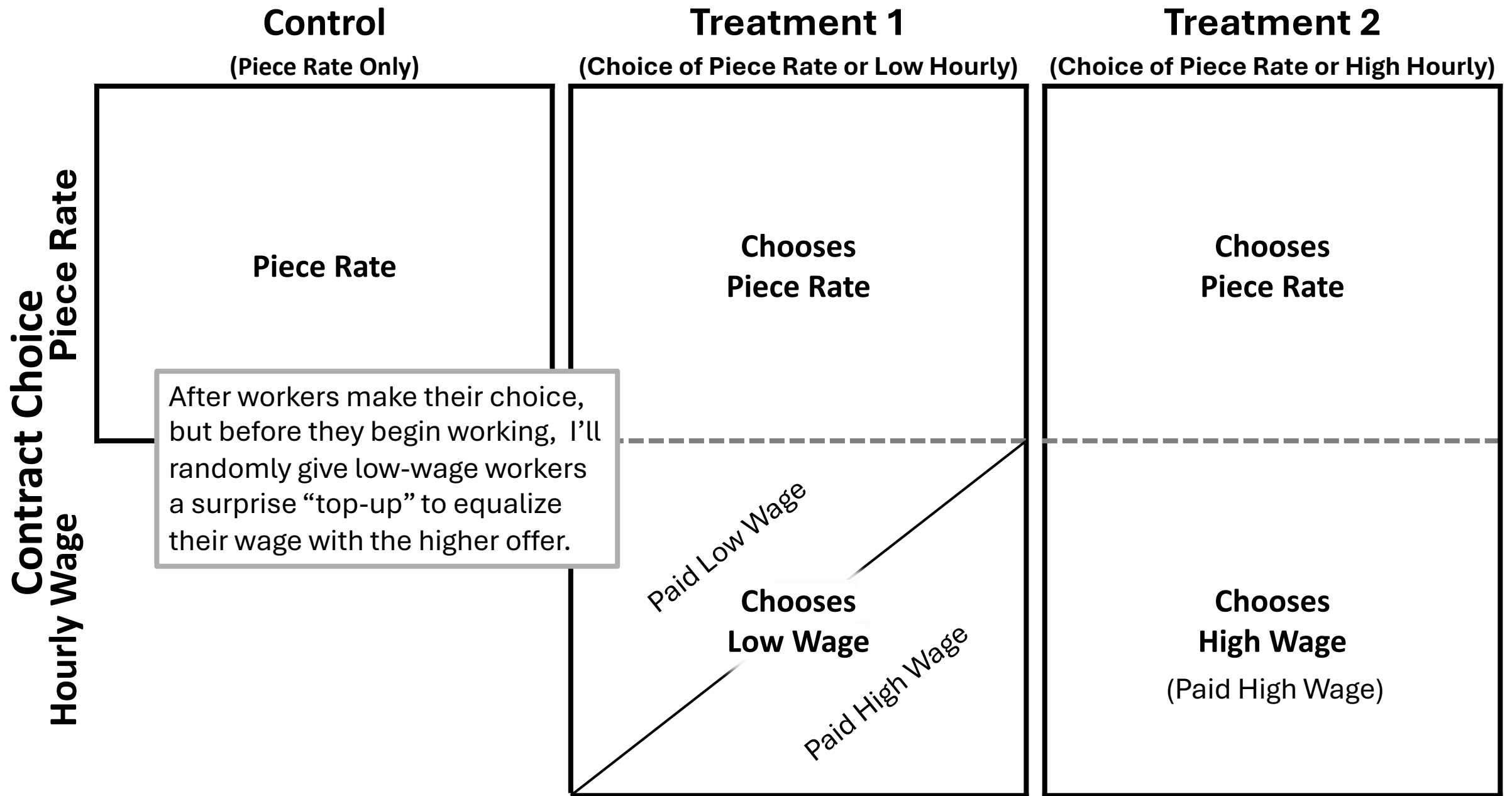
Chooses
Piece Rate

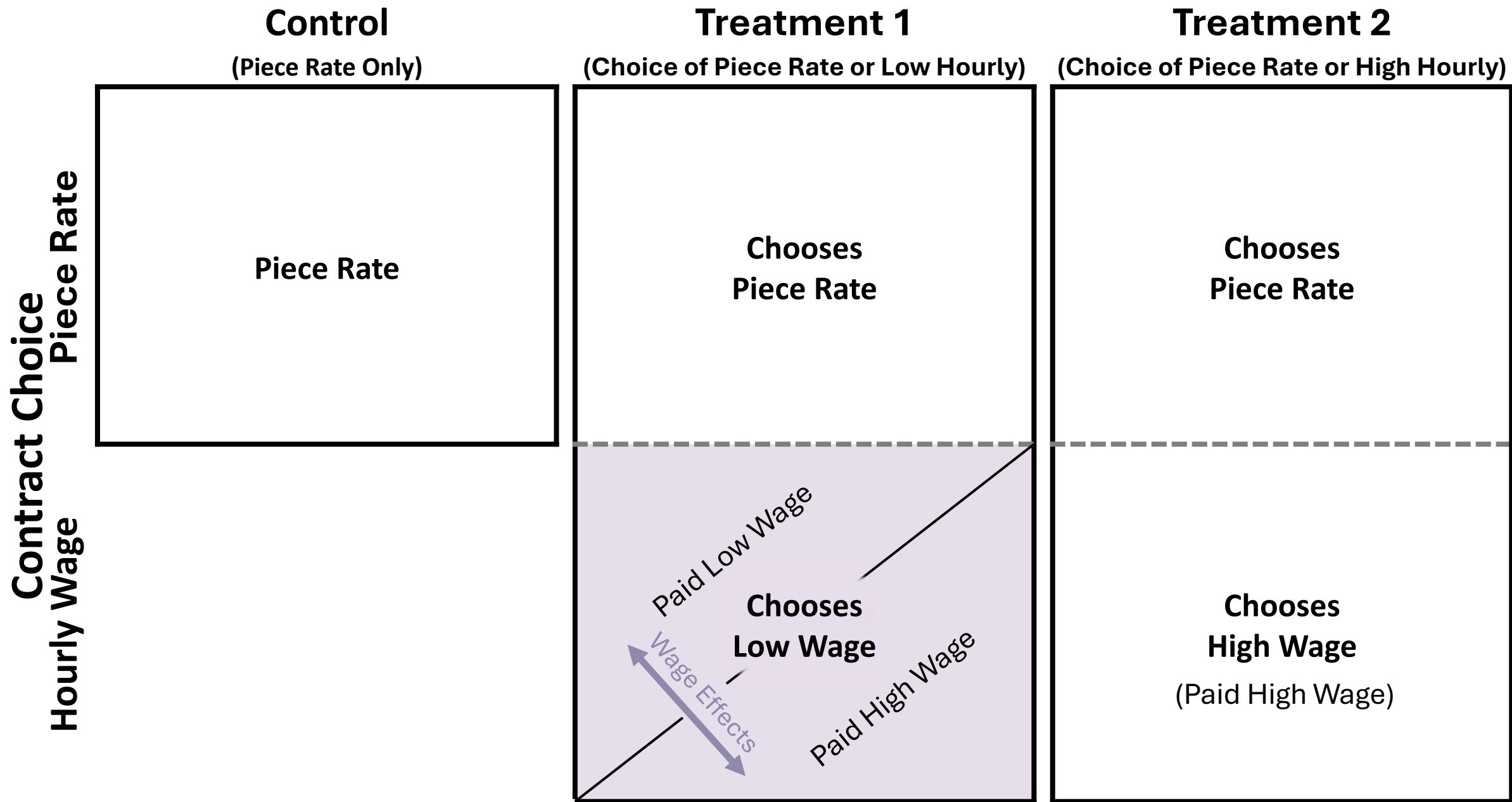
Chooses
Piece Rate

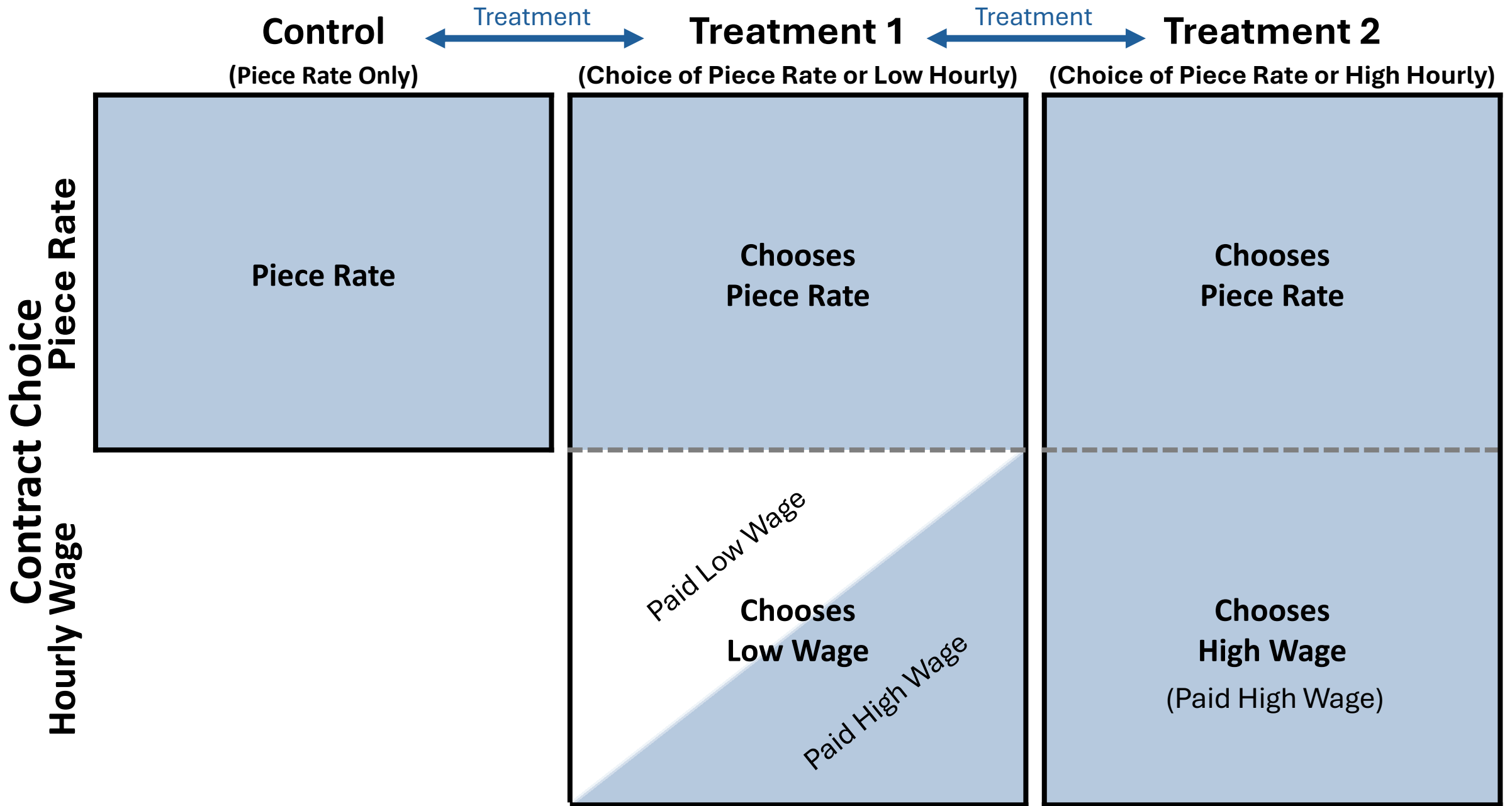
If higher wages increase
worker motivation, moral
hazard estimates would be
biased by wage effects.

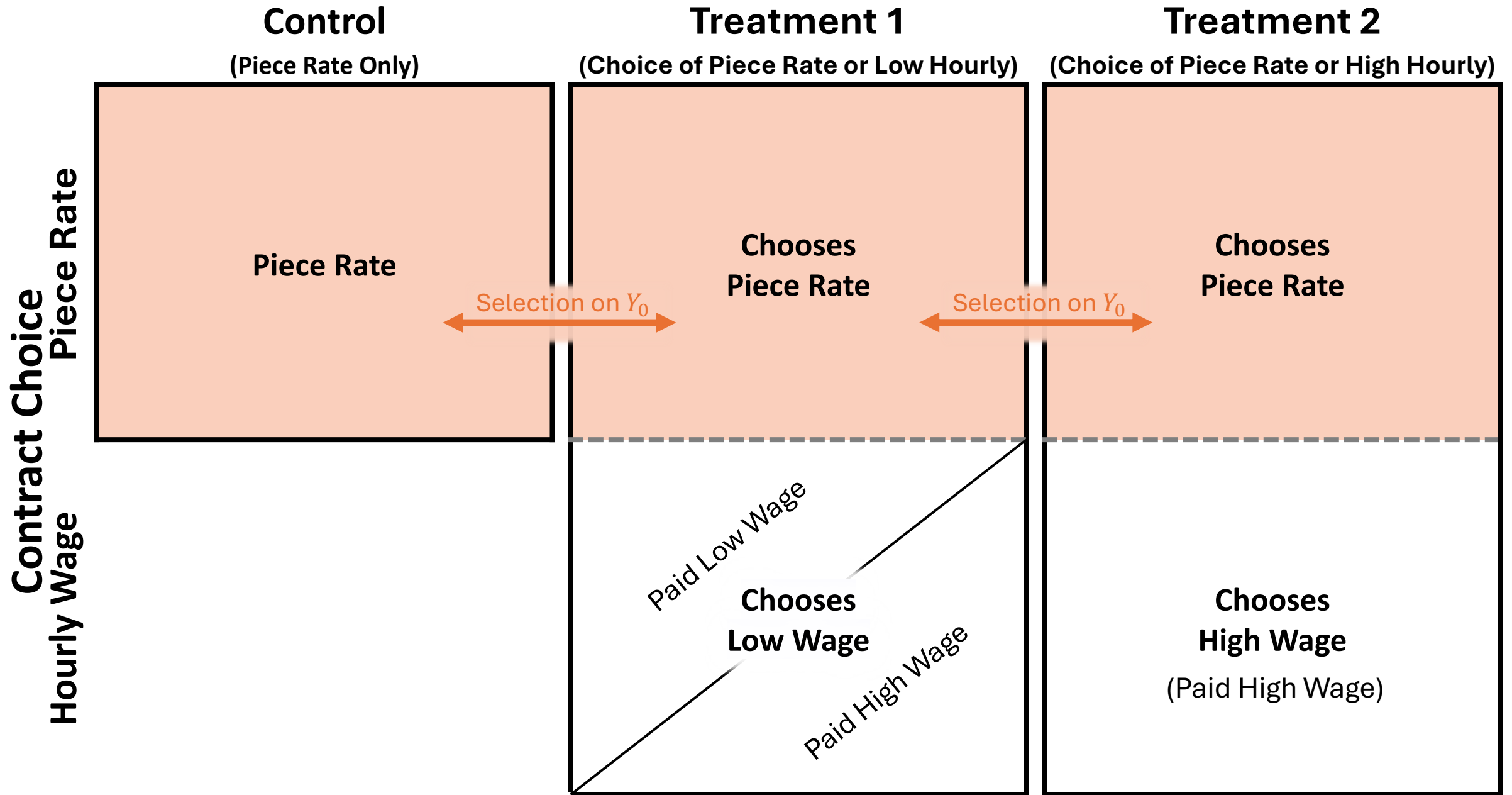
Chooses
Low Wage

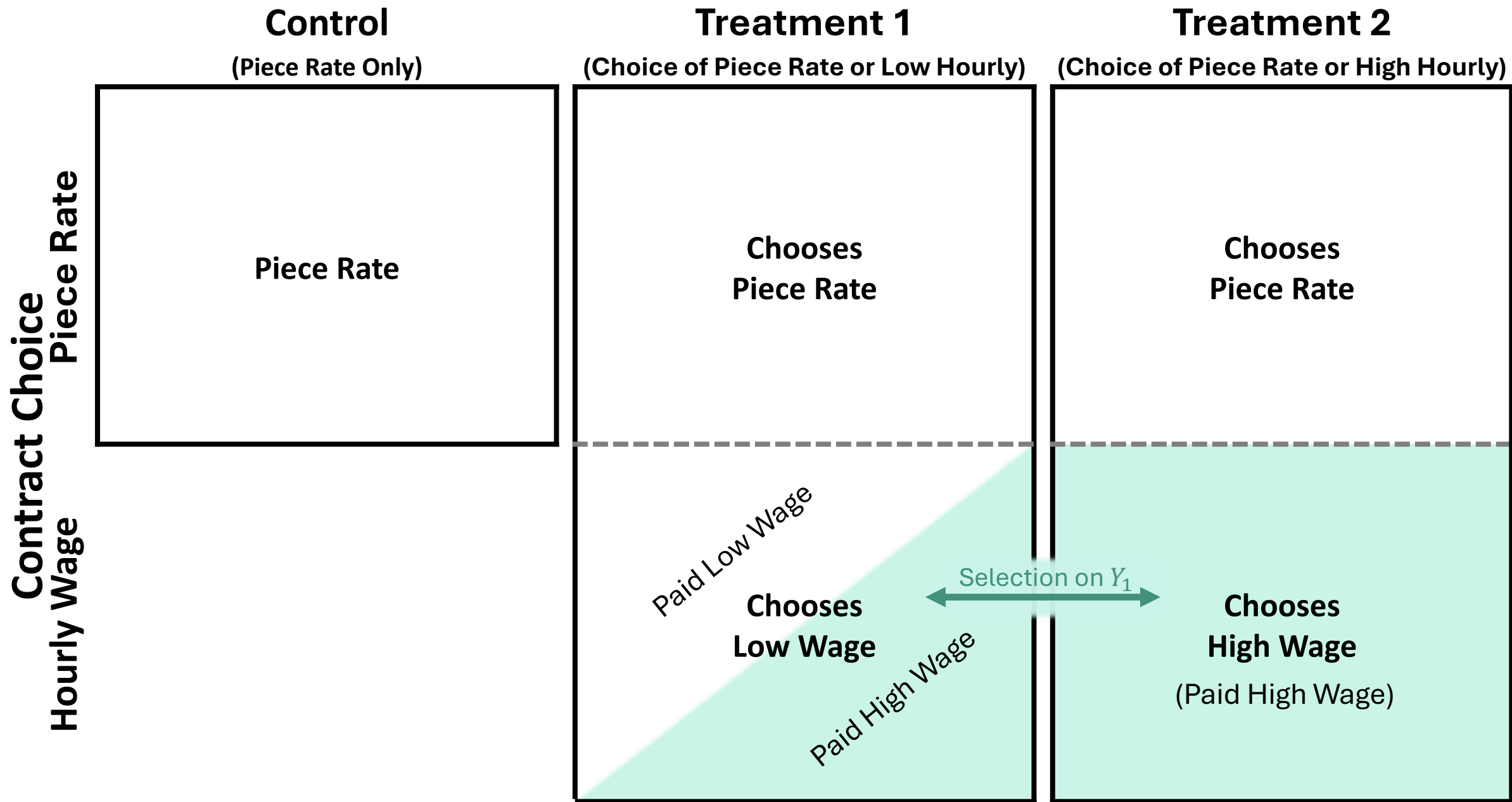
Chooses
High Wage

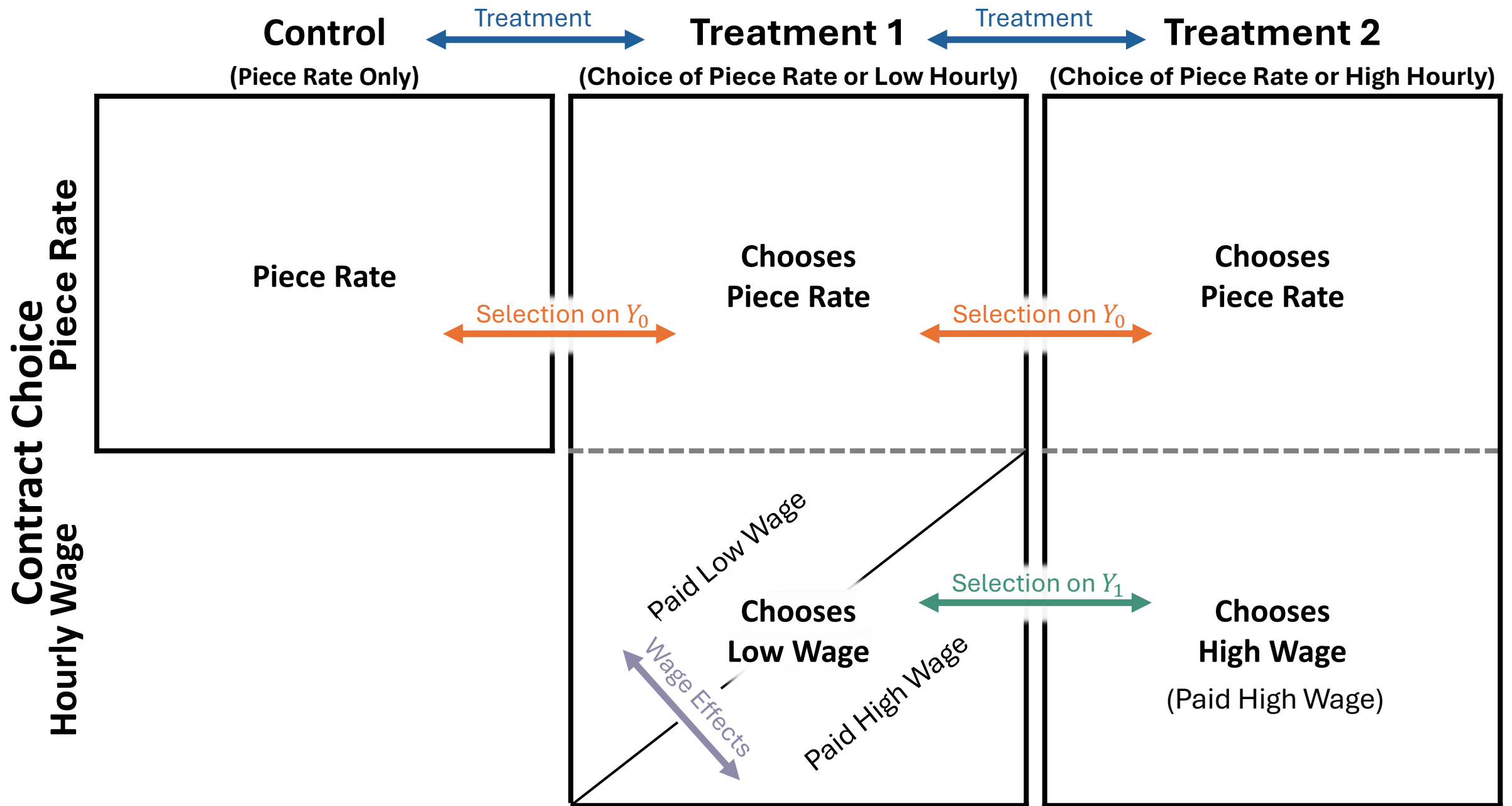












Experimental Procedure

Workers Recruited

Prolific job posting advertises a \$1.00 reward for a five-minute data-entry task, plus \$0.03 per correct entry.

- Posting screens for
 - Approval rating $\geq 98\%$
 - Number of approved tasks ≥ 10
 - Located within US
- In practice, most users who see the posting accept the job.
- Experiment takes place over ten waves of ≈ 300 workers

 **\$1.00 • \$12.00/hr**  5 mins  300 places  Writing

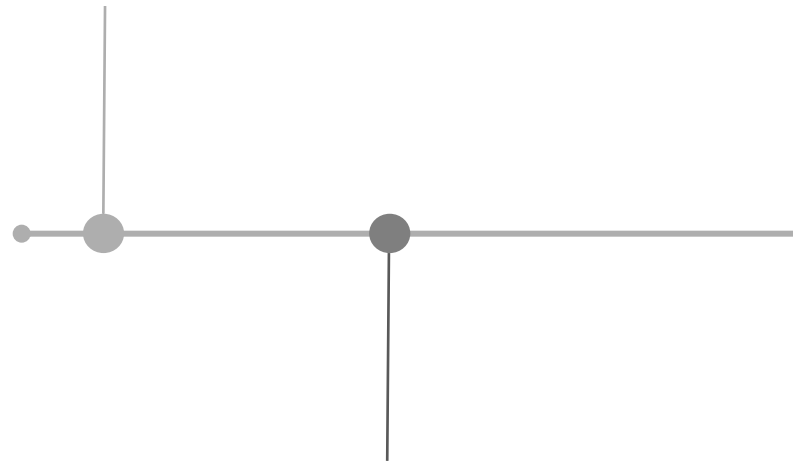
You will be shown a series of handwritten sentences for 5 minutes. Your task is to type each sentence into the corresponding text box.

*You can earn an additional \$0.03 in bonus compensation for each **correctly** typed sentence. Any bonus payments will be deposited within 24 hours of completion. You must reach the end of the 5-minute task to receive credit for your submission.*

Experimental Procedure

Workers Recruited

Prolific job posting advertises a \$1.00 reward for a five-minute data-entry task, plus \$0.03 per correct entry.



Randomization #1: Wage Offers

Workers are randomly assigned to one of eighteen experimental groups, each with a different menu of bonus wage offers.

Hourly Wage Offer	Piece-Rate Offer	Number of Participants
No Hourly Offer	\$0.03 per sentence	302
\$1.20/hr	\$0.03 per sentence	300
\$1.80/hr	\$0.03 per sentence	101
\$2.40/hr	\$0.03 per sentence	103
\$3.00/hr	\$0.03 per sentence	304
\$3.60/hr	\$0.03 per sentence	100
\$4.20/hr	\$0.03 per sentence	99
\$4.80/hr	\$0.03 per sentence	101
\$5.40/hr	\$0.03 per sentence	101
\$6.00/hr	\$0.03 per sentence	305
\$7.20/hr	\$0.03 per sentence	100
\$8.40/hr	\$0.03 per sentence	102
\$9.60/hr	\$0.03 per sentence	101
\$10.80/hr	\$0.03 per sentence	100
\$12.00/hr	\$0.03 per sentence	305
\$15.00/hr	\$0.03 per sentence	100
\$18.00/hr	\$0.03 per sentence	102
\$21.00/hr	\$0.03 per sentence	304

Total: 3030

Experimental Procedure

Workers Recruited

Prolific job posting advertises a \$1.00 reward for a five-minute data-entry task, plus \$0.03 per correct entry.

Choice of Compensation

Treated workers choose between a \$0.03 piece rate and hourly wage offers ranging from \$1.20 to \$21.00 per hour.

Randomization #1: Wage Offers

Workers are randomly assigned to one of eighteen experimental groups, each with a different menu of bonus wage offers.

Before you begin the task, we'd like to offer you a choice of how to receive your bonus payment. Please select your preferred method of compensation from the options below:

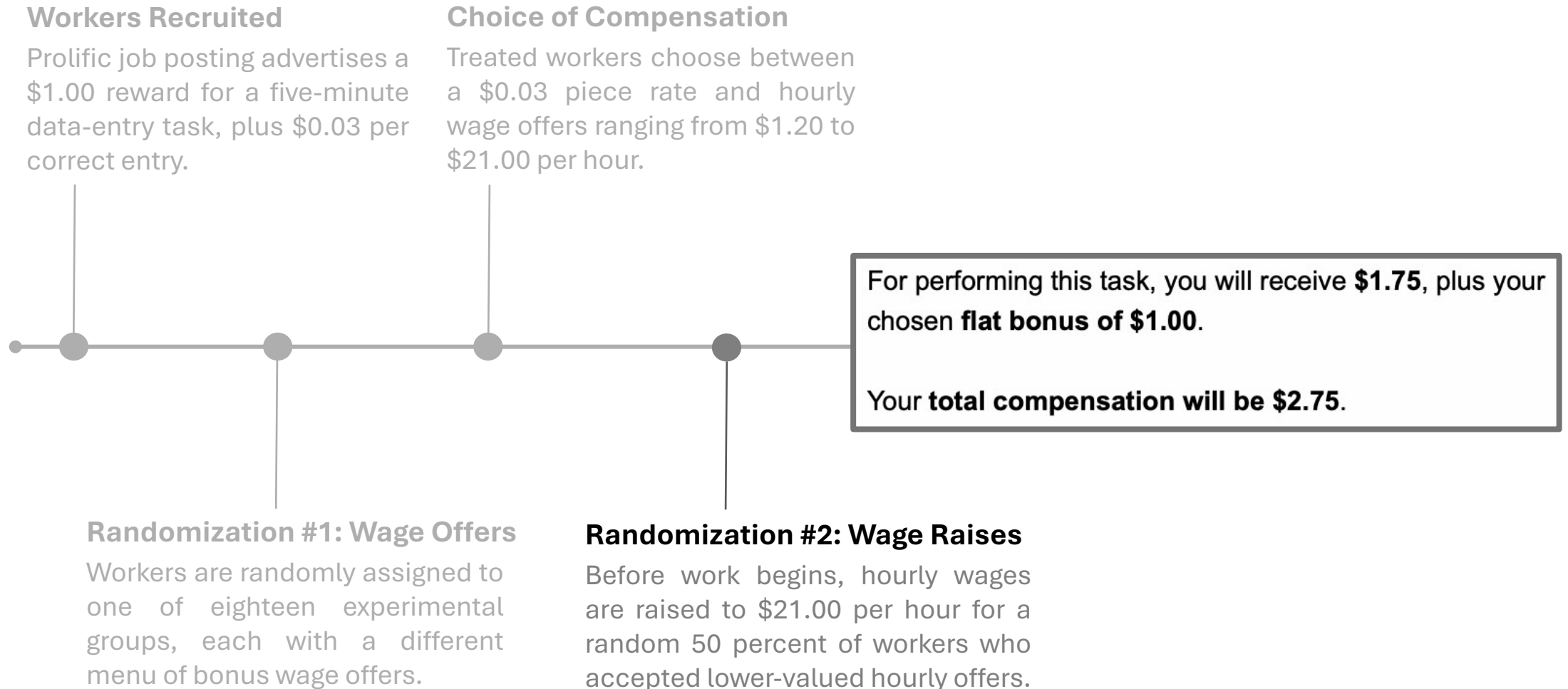
Get paid a flat bonus of \$1.00.



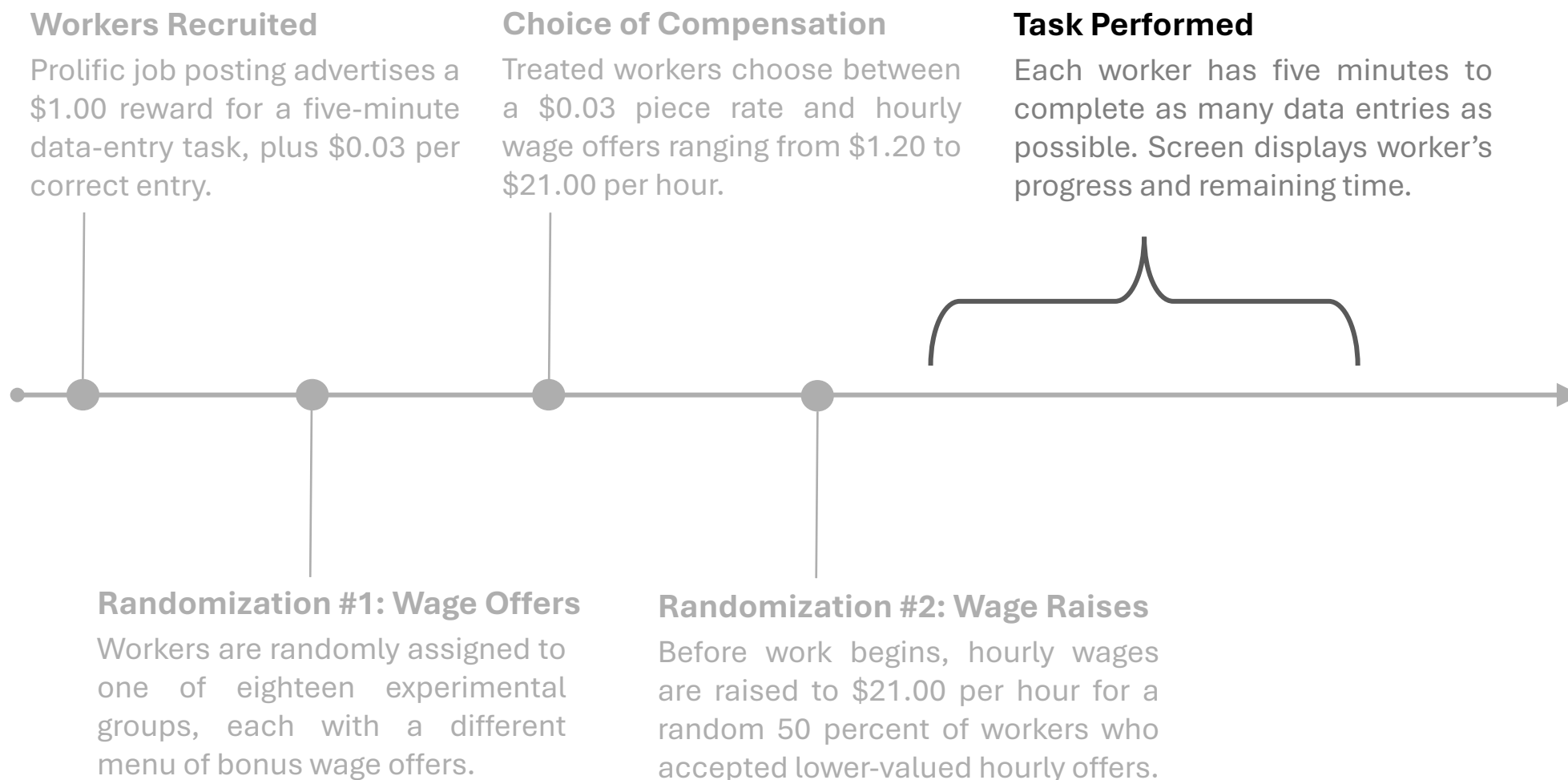
Get paid \$0.03 for each sentence you correctly complete.



Experimental Procedure



Experimental Procedure



You will be shown a series of handwritten sentences. On each page, your task is to type the sentence into the text box below.

Here is one example of a completed sentence:

The quick brown fox jumps over the lazy dog.

The quick brown fox jumps over the lazy dog.

Your answers should be as accurate as possible. Please be mindful of capitalization, spacing, and punctuation. When you've completed a sentence, click the "→" button to move on.

You will have 5 minutes to complete as many sentences as you can. You cannot start over, and you can only perform this task once.



Before you begin the task, we'd like to offer you a choice of how to earn your bonus payment. Please select your preferred bonus compensation from the options below:

Get paid a \$0.03 bonus for each sentence you correctly complete.



Get paid a flat bonus of \$1.00.



Time Remaining: 05:00

For performing this task, you will receive **\$1.00**, plus your chosen bonus of **\$0.03 per correct sentence**.

Click "Begin Task" to begin your 5-minute typing task.

Once you begin, the timer will start. Do not press the back button or refresh your browser—doing so may invalidate your results!

Begin Task

Score: 0
Earnings: \$1.00

Time Remaining: 02:23

The stars twinkled over the silent forest.

The stars twinkled over the silent fore|



Score: 15
Earnings: \$1.45

Time is up.

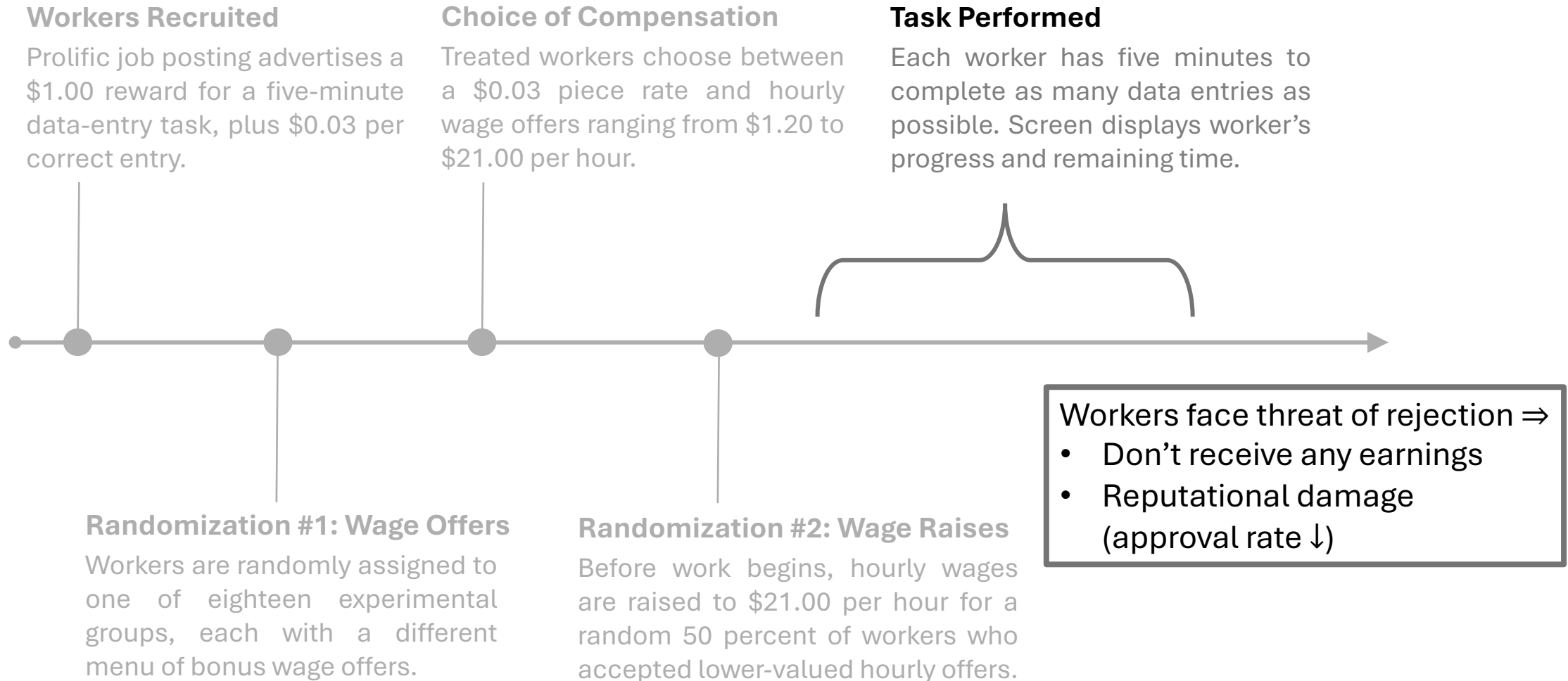
Thank you for participating!

You completed 31 sentences correctly.

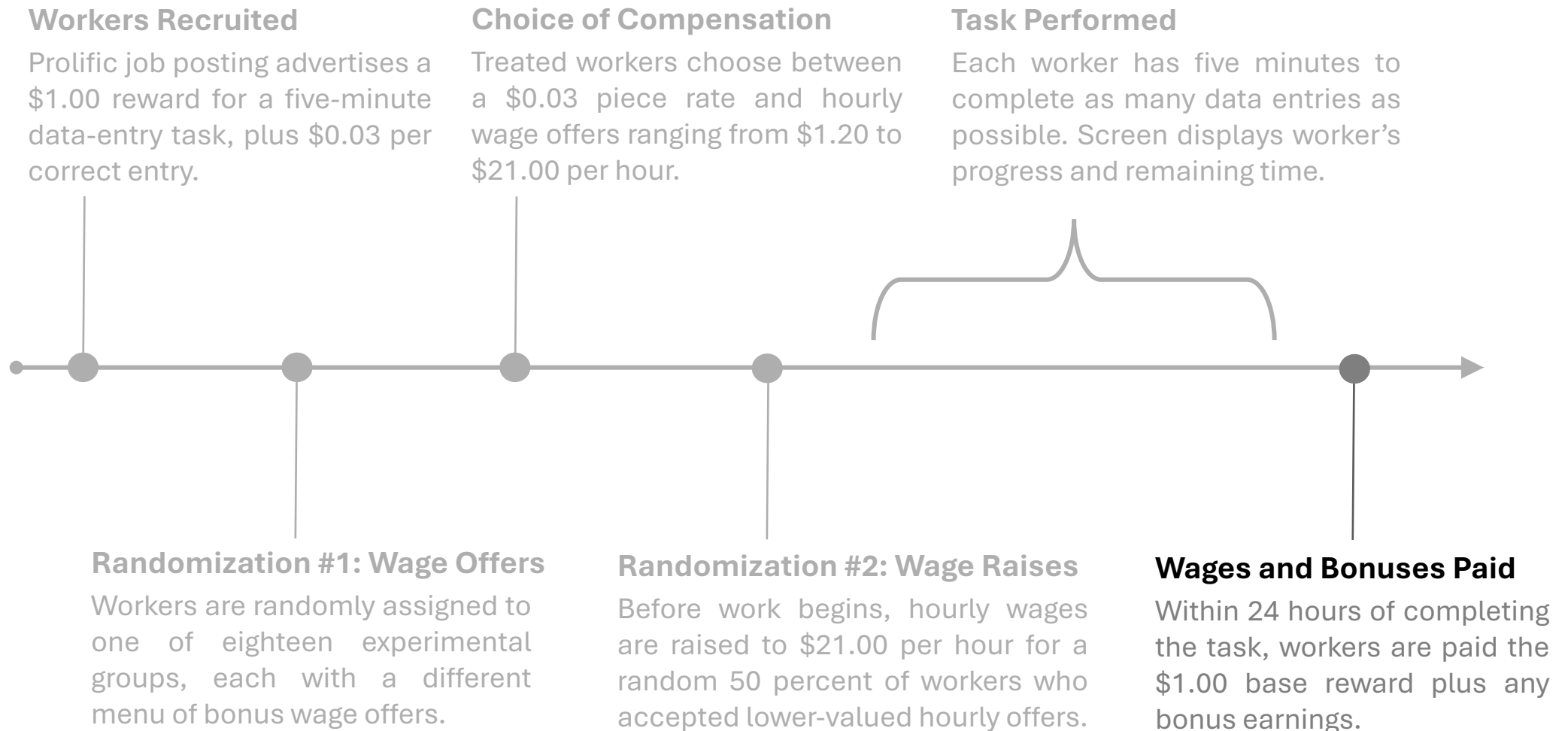
Please click [here](#) to receive your \$1.93 payment.

Score: 31
Earnings: \$1.93

Experimental Procedure



Experimental Procedure



External Validity

- Results are specific to online workers performing a data-entry task.
- Cannot claim to generalize to the broader labor market
- Do estimates from these settings generalize any better?
 - Rideshare drivers (Angrist et al., 2021; Cook et al., 2021)
 - Agricultural workers (Brune et al., 2022; Bandiera et al., 2010)
 - Cashiers (Mas and Moretti 2009)
 - Windshield repair (Lazear 2000)
 - Call centers (Mas and Pallais 2017; Nagin et al. 2002)
- Division of labor $\Rightarrow$ applied research on worker incentives usually comes from highly specialized settings
- Even so, I try to make the experimental task as generalizable as possible
 - 66% of US workers type on the job
 - Workers don't know it's an experiment (debriefed afterwards)

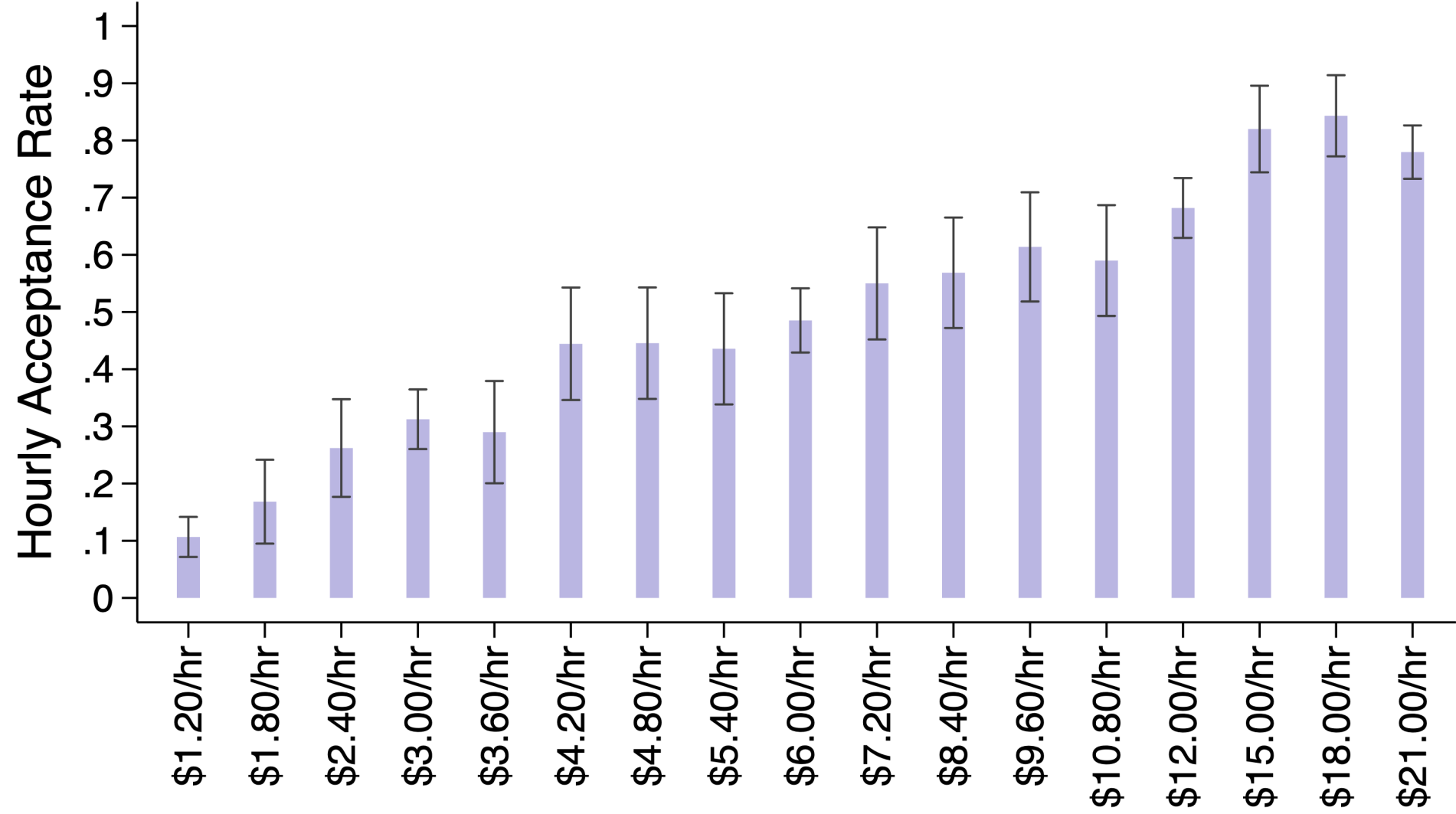
Outline

- ① **Experimental Design**
- ② **Main Results**
- ③ **Model of Asymmetric Information**
- ④ **Estimates of Marginal Value and Welfare Loss**
- ⑤ **Optimal Fixed-Wage Subsidy and Piece-Rate Tax**

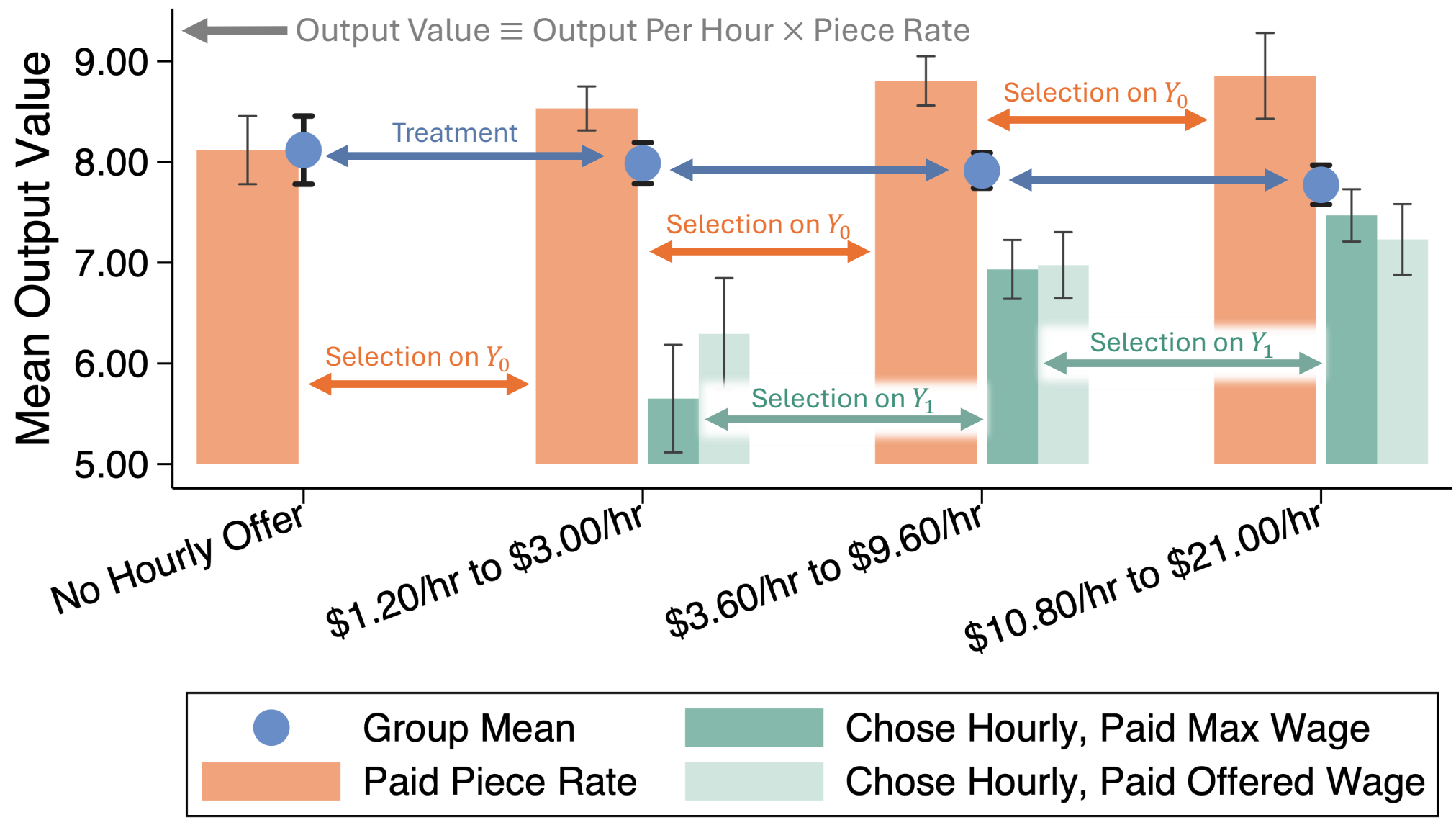
Summary Statistics

Category	Variable	Mean	SD
<i>Panel A: Task Performance</i>	Accepted Hourly Offer	0.438	0.496
	Completed Sentences	21.98	8.148
	Correct Sentences	17.79	9.360
	Output Value	7.912	2.933
	Finished	0.986	0.118
<i>Panel B: Demographics & Employment</i>	Age	37.23	12.18
	Female	0.643	0.479
	Minority	0.357	0.479
	Employed	0.685	0.465
	Student	0.187	0.390
	Number of Previous Tasks	1281.6	1746.4

Hourly Wage Take-up



Worker Output by Offer & Contract



2SLS Estimates of Treatment Effects

	(1)	(2)	(3)	(4)
	Output Value	Output Value	Output Value	Output Value
Accepted Hourly Offer	−0.506** (0.206)	−0.500** (0.200)	−0.488** (0.200)	−0.365** (0.185)
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes
R-squared	0.037	0.080	0.096	0.232
N	3030	3030	3030	3030

$$\tilde{Y}_i = \delta H_i + \boldsymbol{\eta} \mathbf{X}_i + \epsilon_i$$

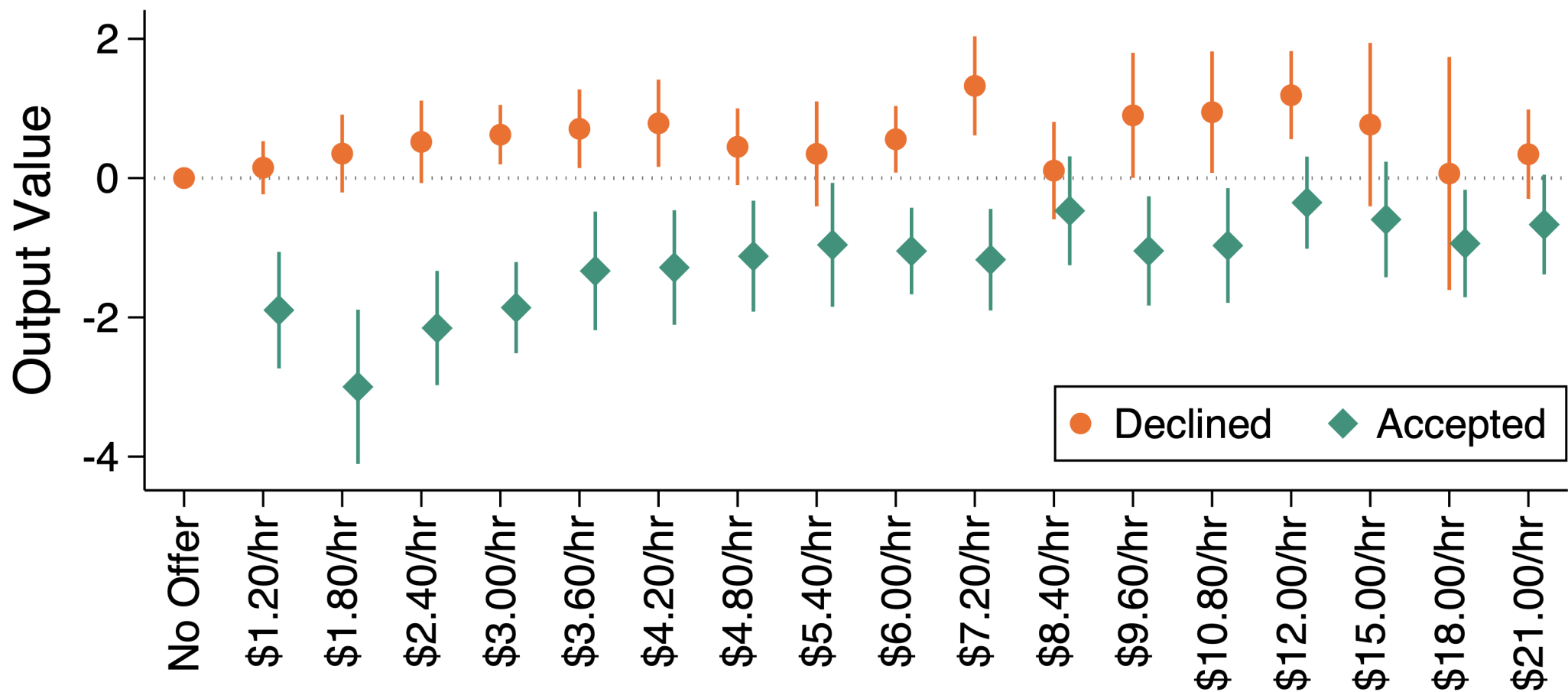
$$\tilde{Y}_i \equiv Y_i - \gamma H_i \times W_i^P$$

Selection by Log Wage Offer

	(1) Output Value	(2) Output Value	(3) Output Value	(4) Output Value
Accepted Hourly Offer	-2.598*** (0.329)	-2.481*** (0.319)	-2.439*** (0.320)	-2.256*** (0.300)
Declined × Log Hourly Wage Offer	0.167* (0.0960)	0.193** (0.0932)	0.210** (0.0925)	0.230*** (0.0855)
Accepted × Log Hourly Wage Offer	0.621*** (0.116)	0.570*** (0.112)	0.568*** (0.113)	0.501*** (0.104)
Accepted × Log Effective Hourly Wage	-0.0608 (0.122)	-0.0443 (0.118)	-0.0444 (0.118)	-0.0000829 (0.110)
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes
R-squared	0.082	0.123	0.139	0.273
N	3030	3030	3030	3030

$$Y_i = \alpha H_i + \beta_0(1 - H_i) \times W_i + \beta_1 H_i \times W_i + \gamma H_i \times W_i^P + \xi X_i + \epsilon_i$$

Selection by Log Wage Offer



$$Y_i = \sum_{w \in W} [(\beta_0^w(1 - H_i) + \beta_1^w H_i) \times \mathbf{1}\{W_i = w\}] + \gamma H_i \times W_i^P + \xi X_i + \epsilon_i$$

Summary

What have I shown so far?

Evidence of Moral Hazard

Hourly wages reduce average output among workers who accept.

$$E[Y_{1i} - Y_{0i} | H_i = 1, W_i = w] < 0$$

Local Average Treatment Effect

Evidence of Adverse Selection

Low-productivity workers are more likely to accept low hourly wage offers.

$$\frac{\partial}{\partial w} E[Y_{0i} | H_i = 0, W_i = w] > 0$$

Local Average Selection on Y_0

$$\frac{\partial}{\partial w} E[Y_{1i} | H_i = 1, W_i = w] > 0$$

Local Average Selection on Y_1

What are the implications for...

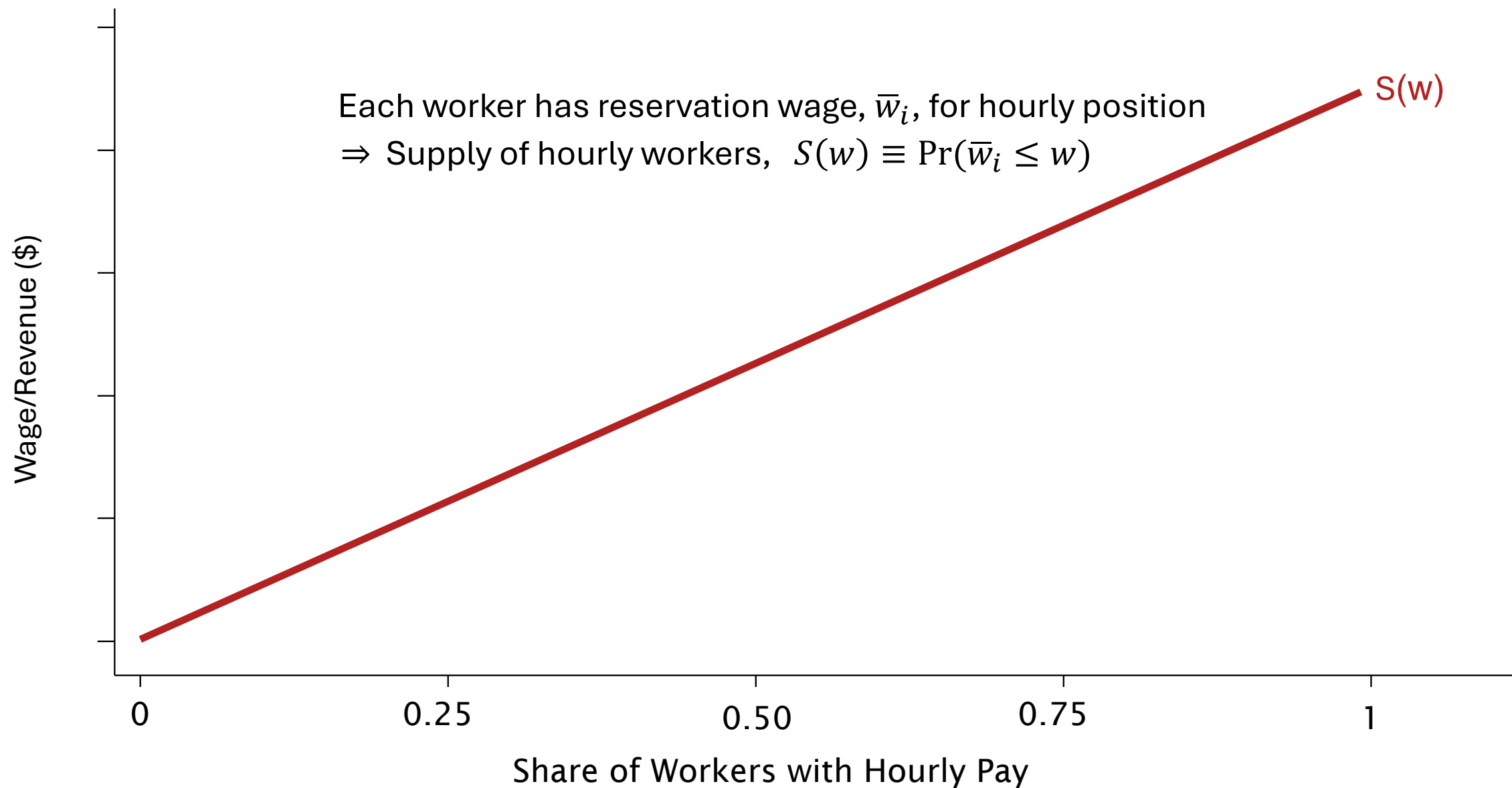
- Market equilibrium?
- Social welfare?
- Policy?

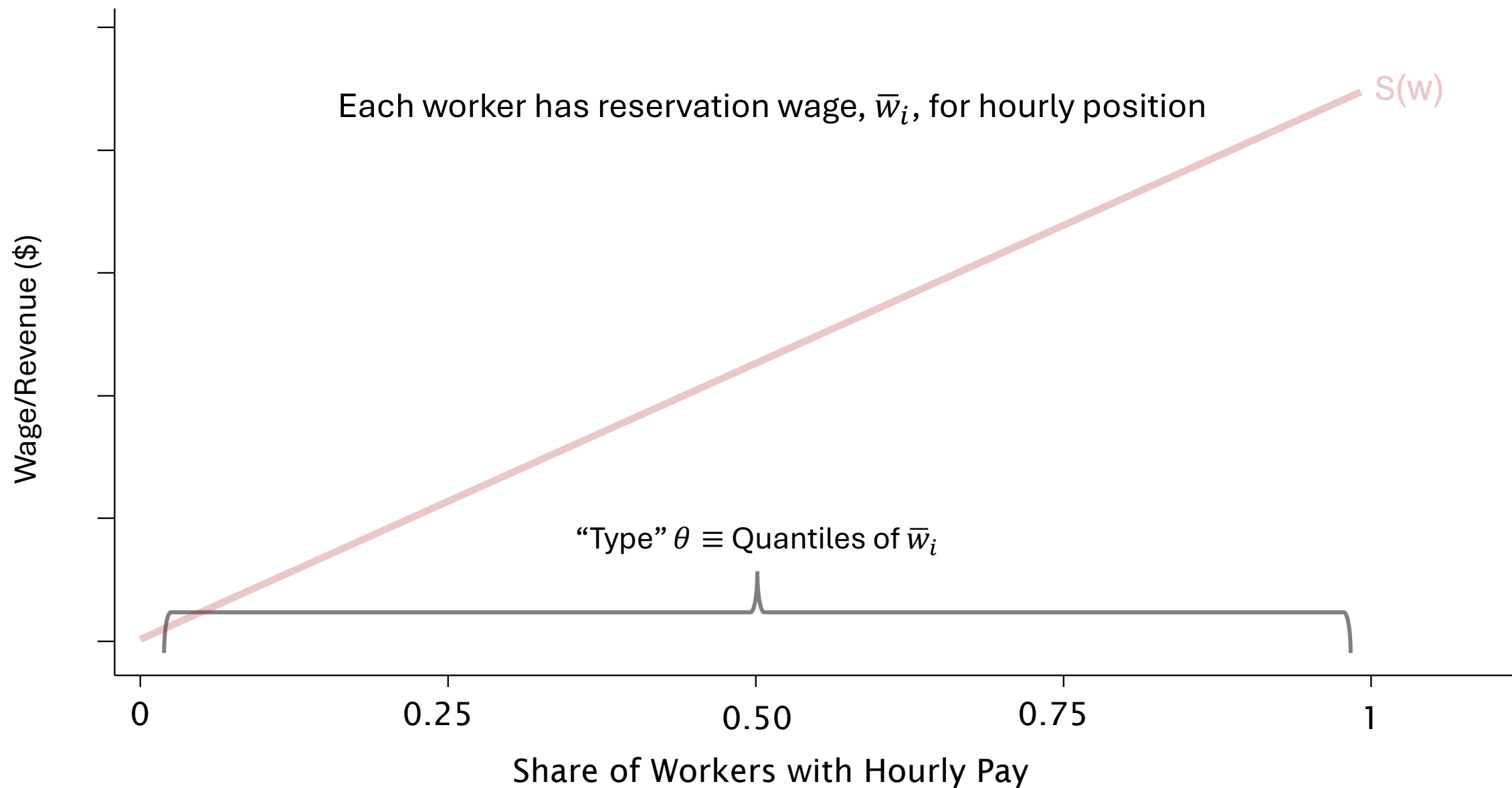
Outline

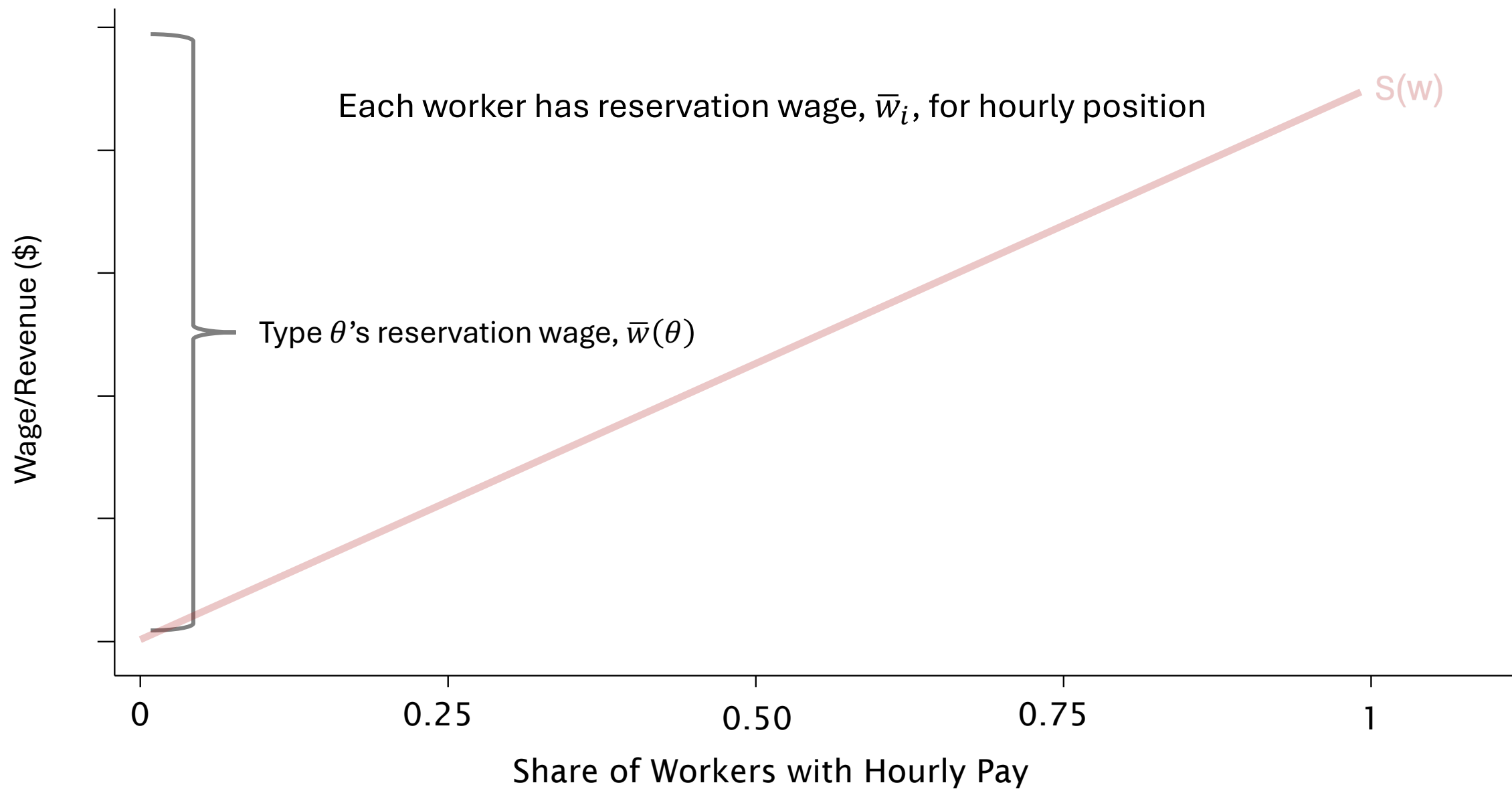
- ① **Experimental Design**
- ② **Main Results**
- ③ **Model of Asymmetric Information**
- ④ **Estimates of Marginal Value and Welfare Loss**
- ⑤ **Optimal Fixed-Wage Subsidy and Piece-Rate Tax**

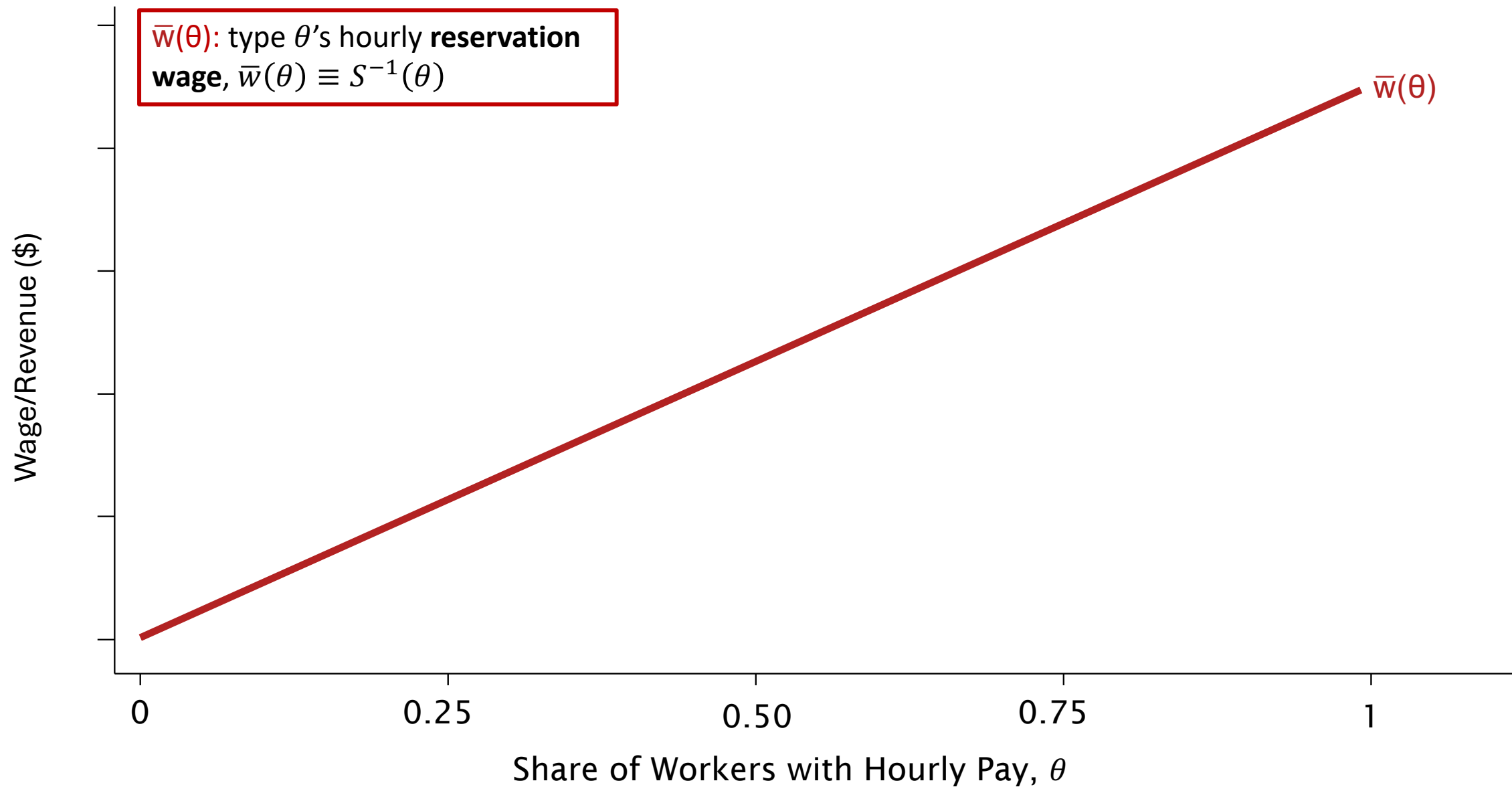
Model

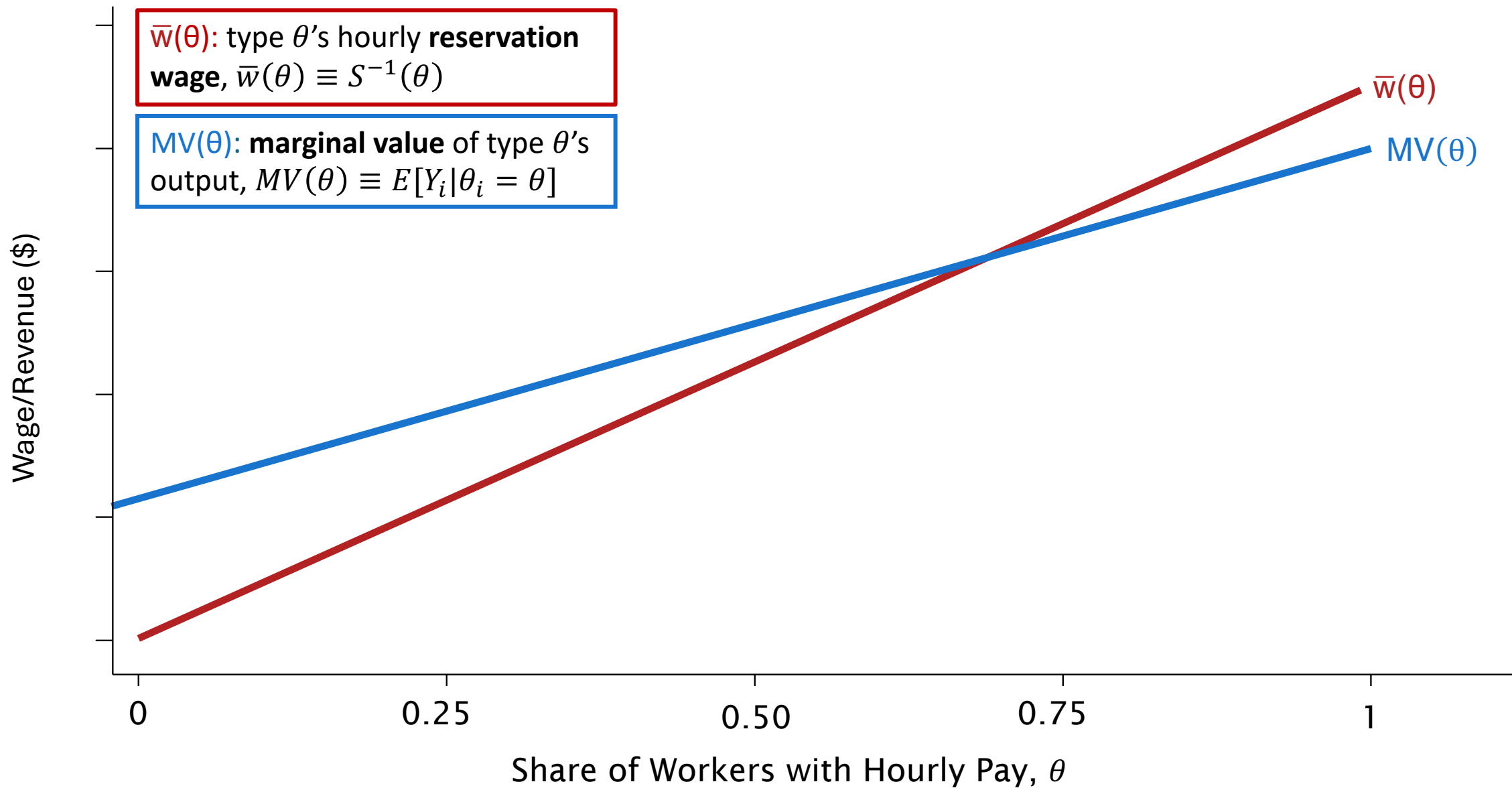
- Population of observationally equivalent (pre-screened) workers
- Each worker produces hourly output $q_i = f(\zeta_i, e_i, v_i)$
 - ζ_i : (unobserved) worker characteristics
 - e_i : individual effort, potentially influenced by the contract (moral hazard)
 - v_i : random noise
- Value of worker i 's output is $Y_i \equiv pq_i$, where $p \equiv$ price per unit q
- Firms have two options:
 1. Buy output at per-unit market price p (e.g., piece rate, freelance hire)
 2. Offer worker *ex-ante* hourly wage, w , in exchange for claim on realized q_i

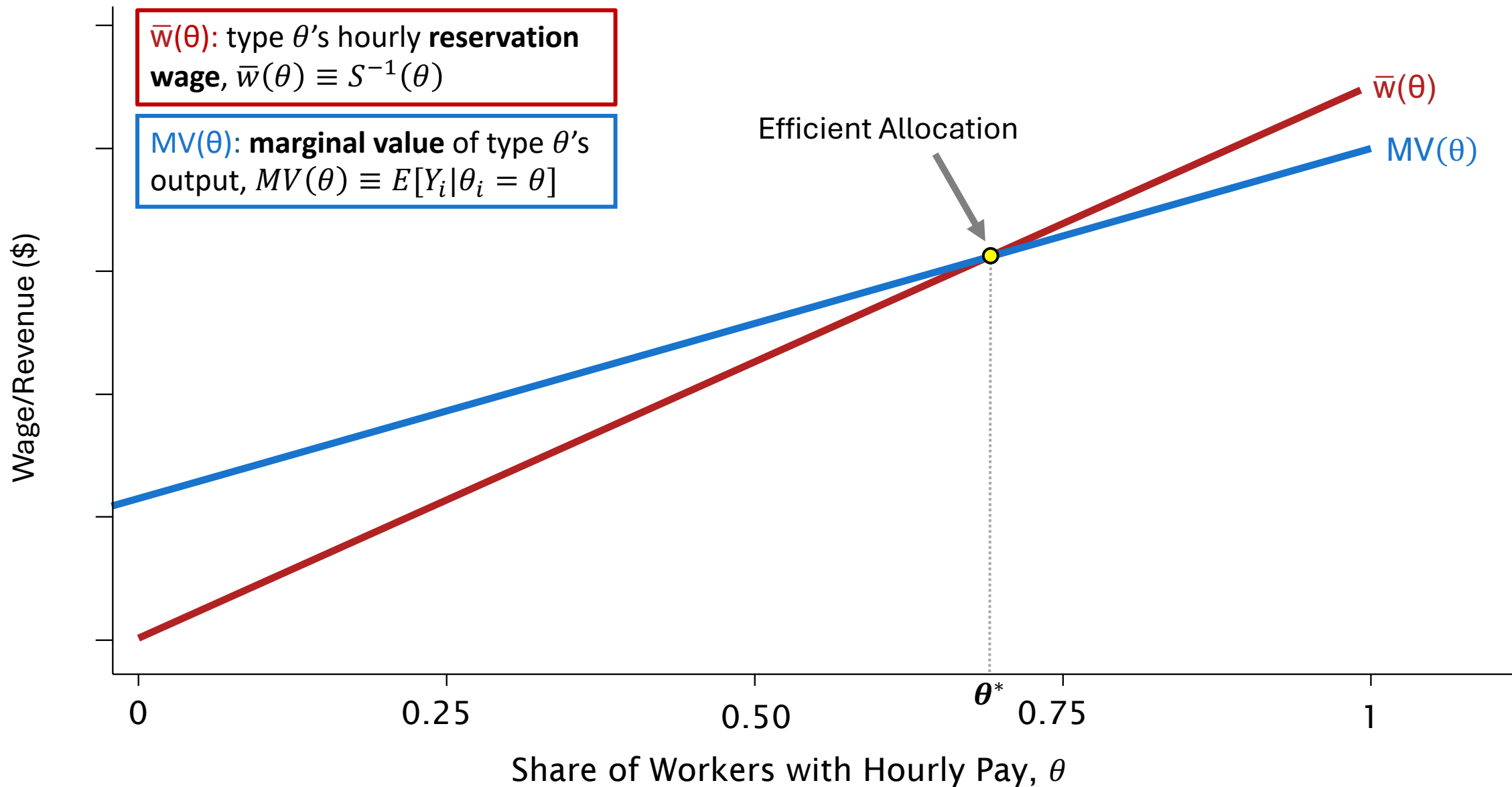


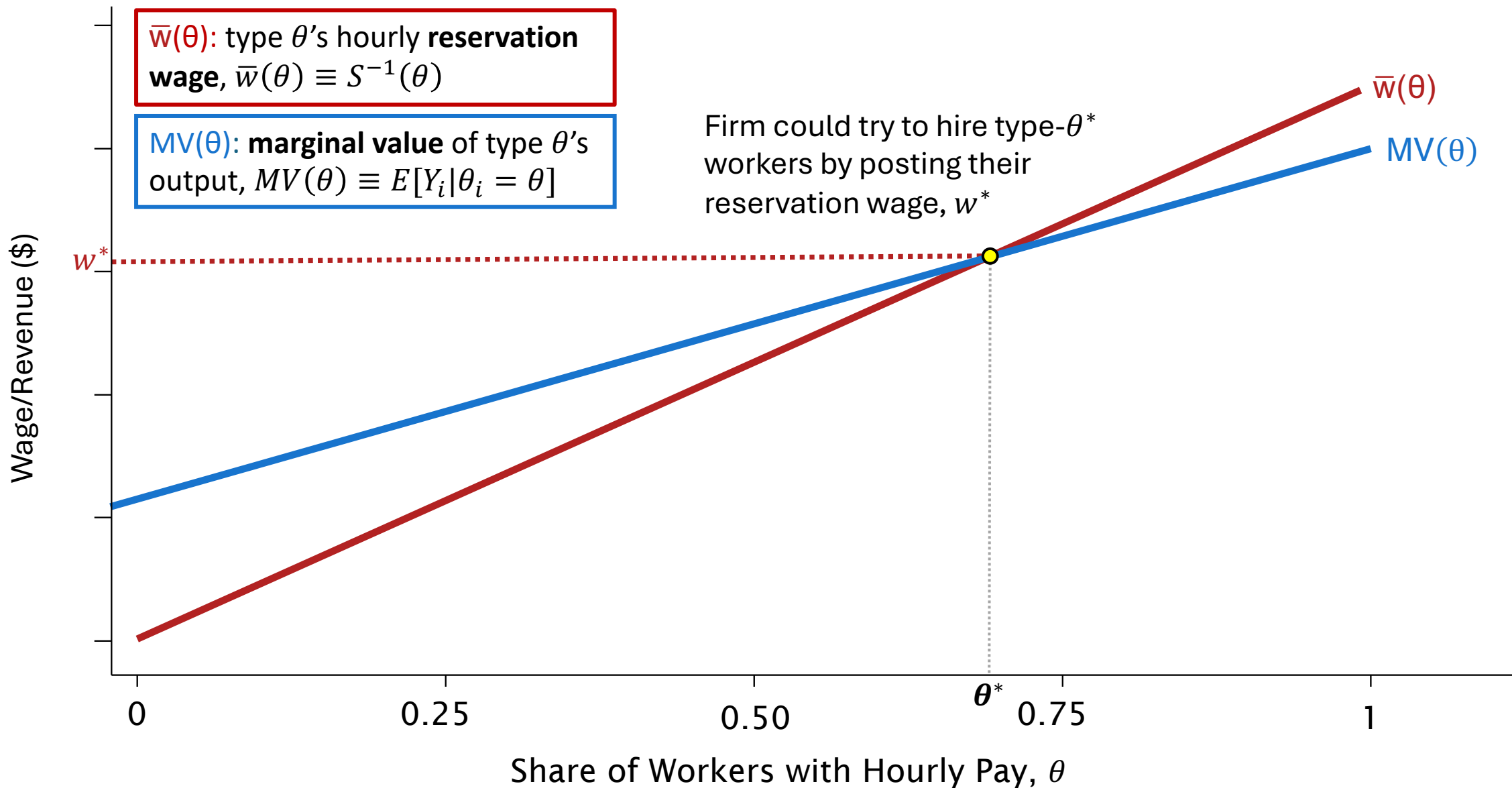


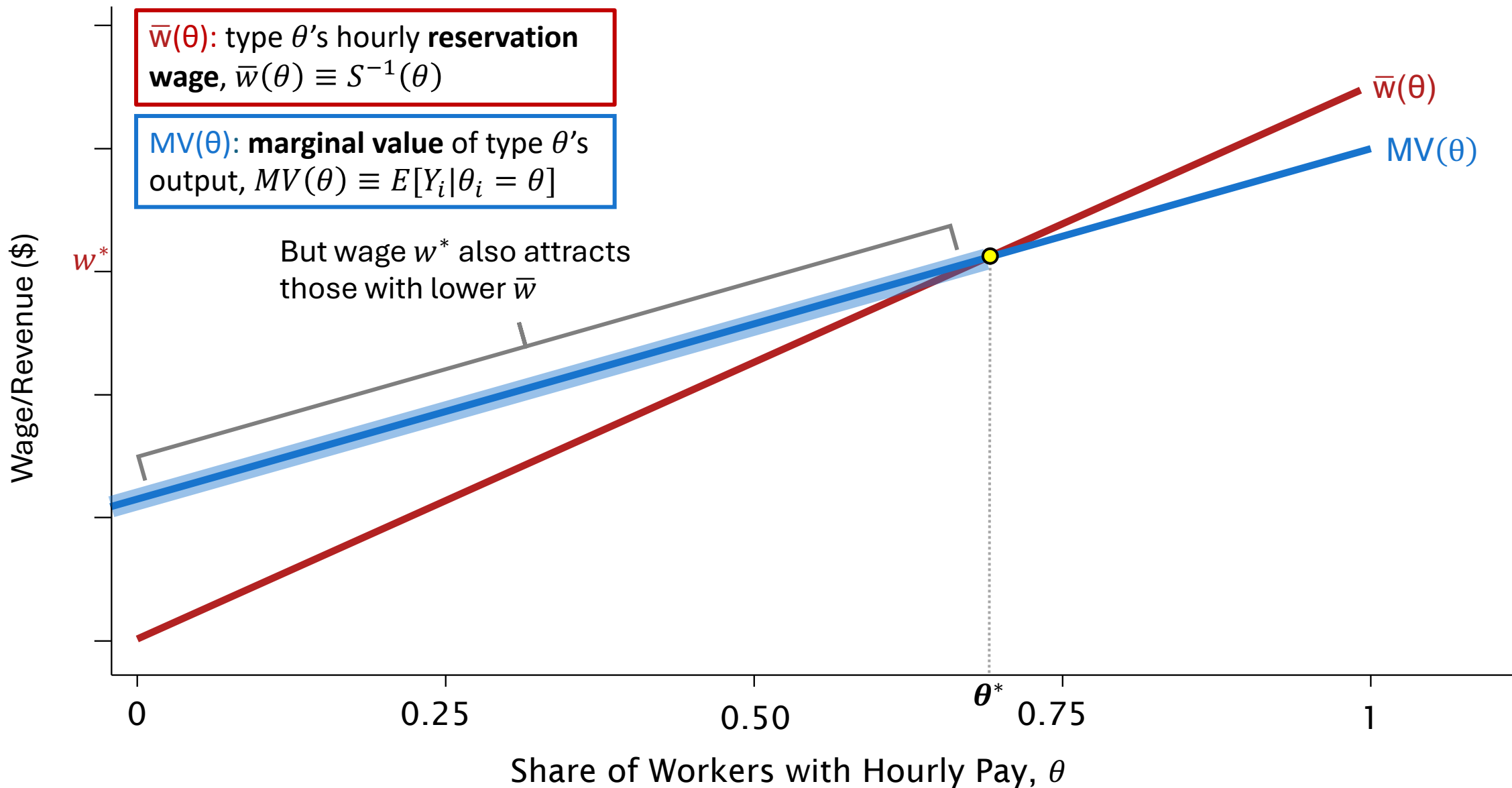


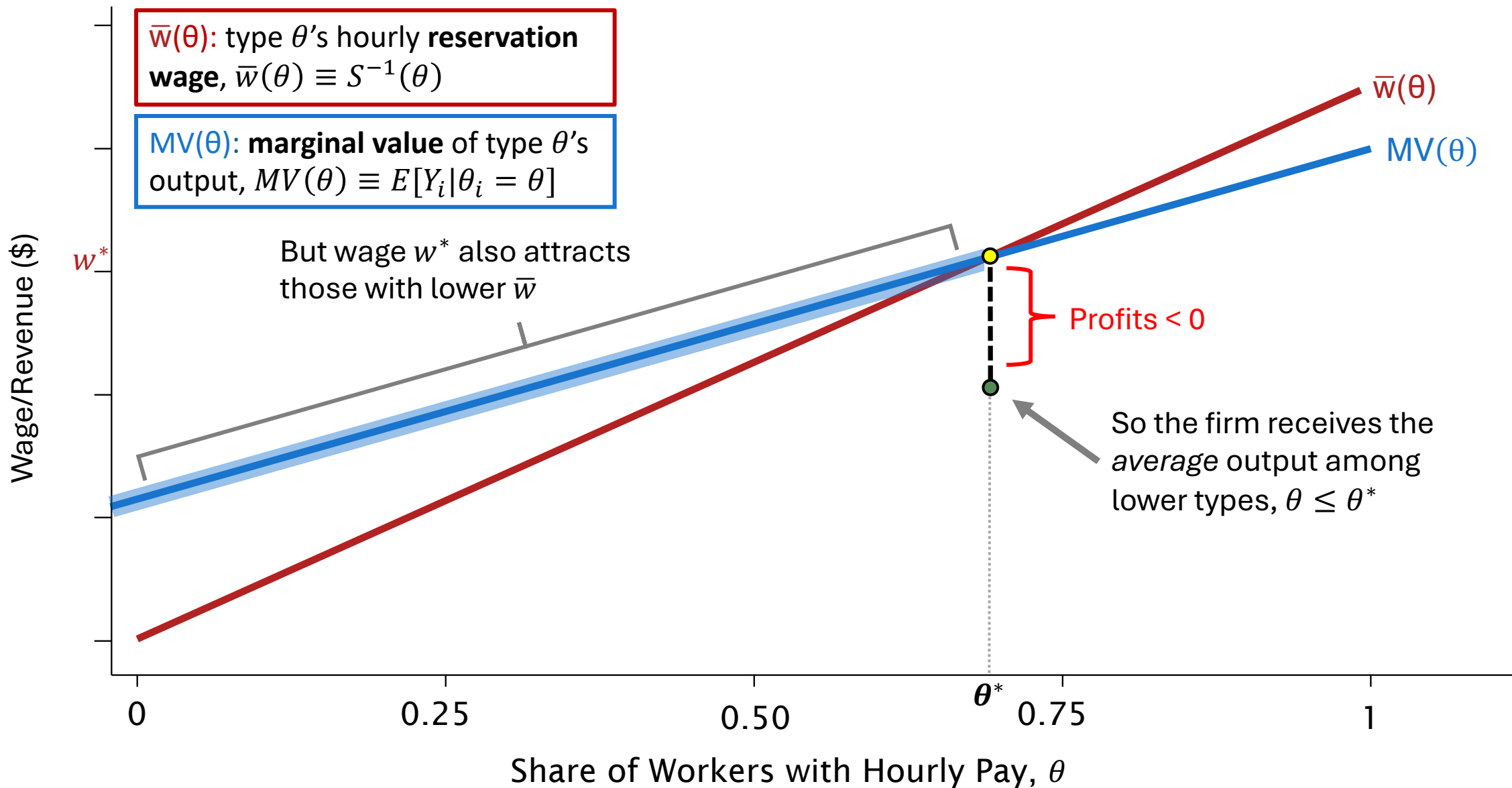


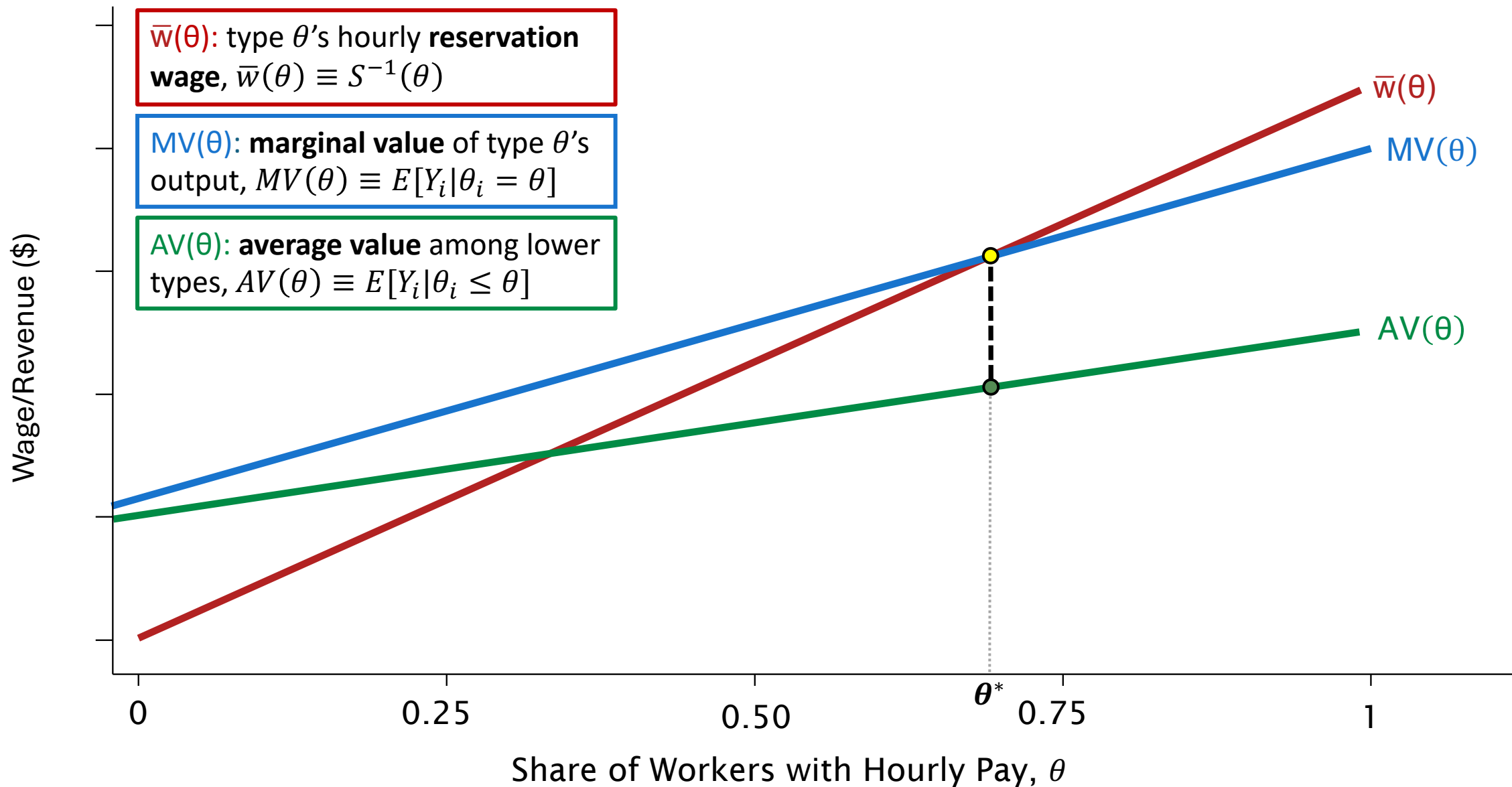


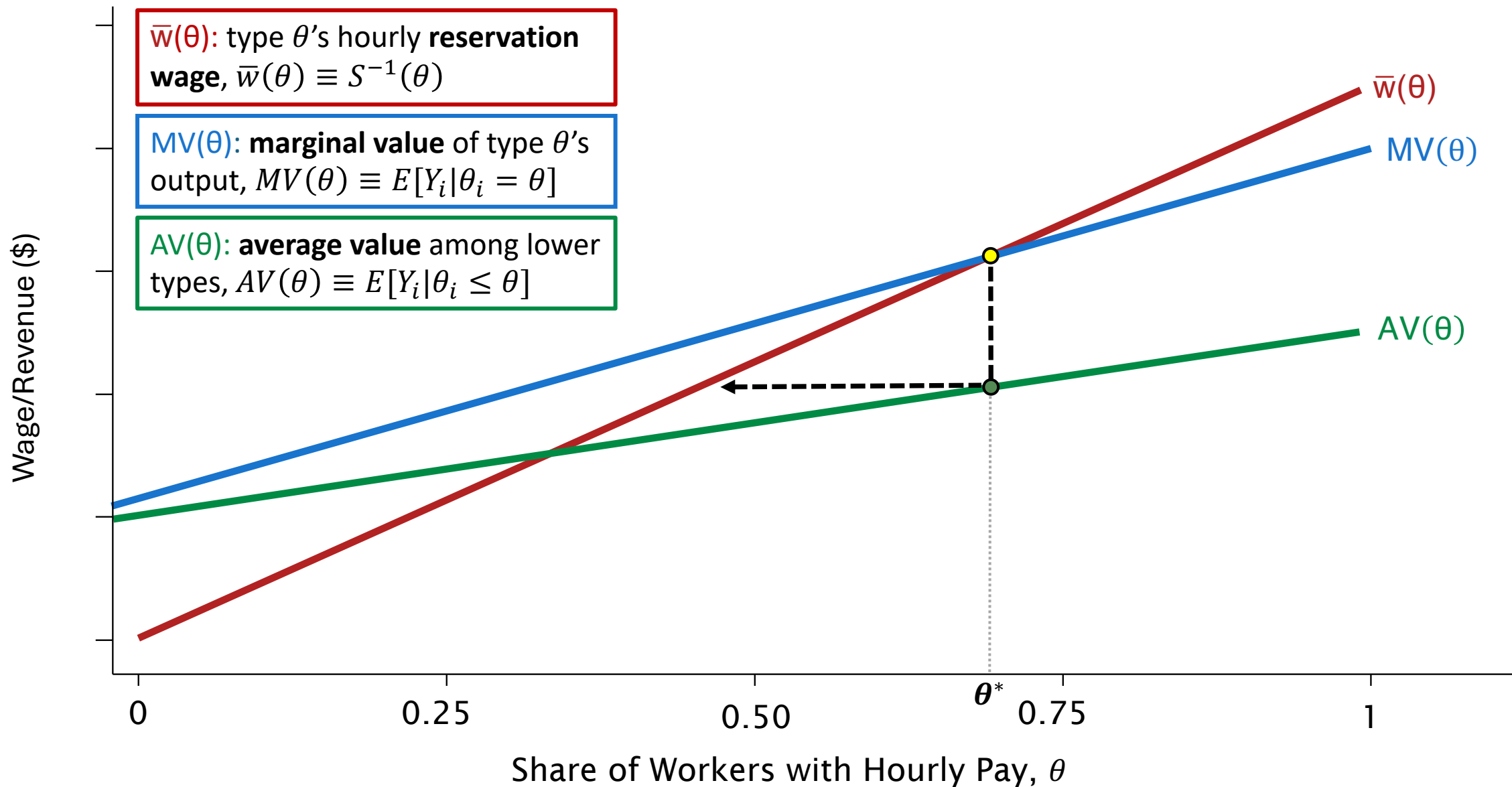


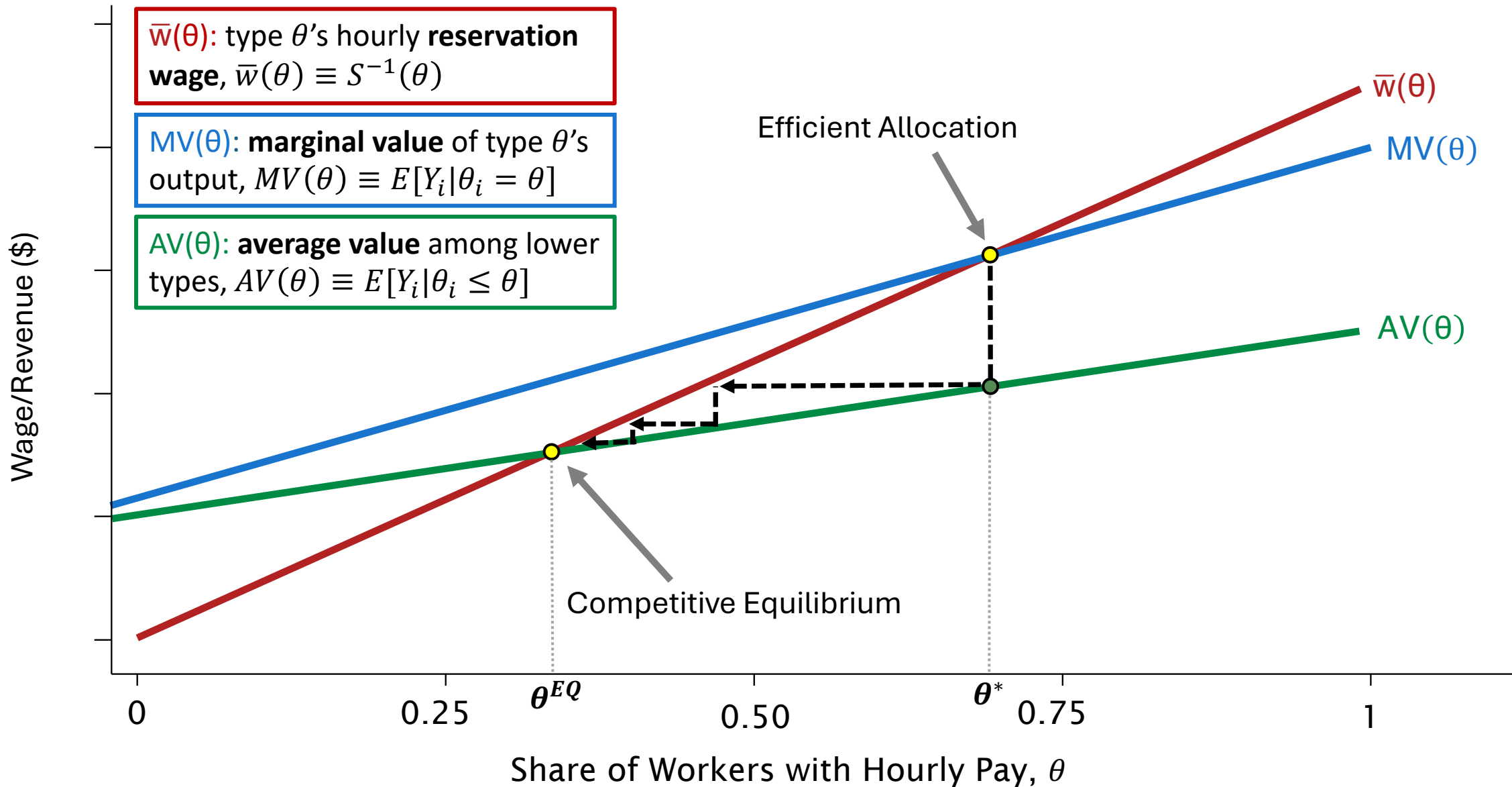


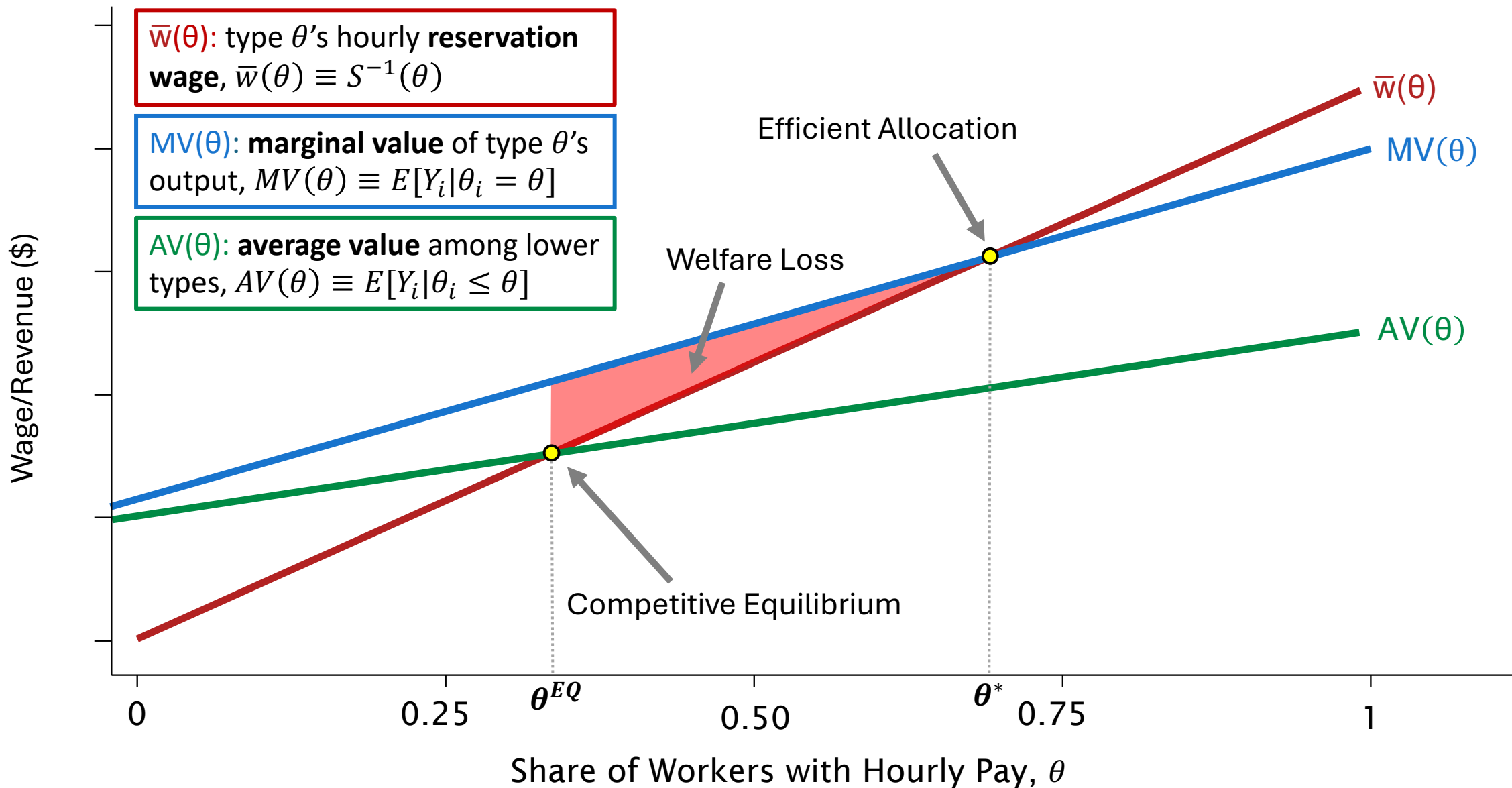












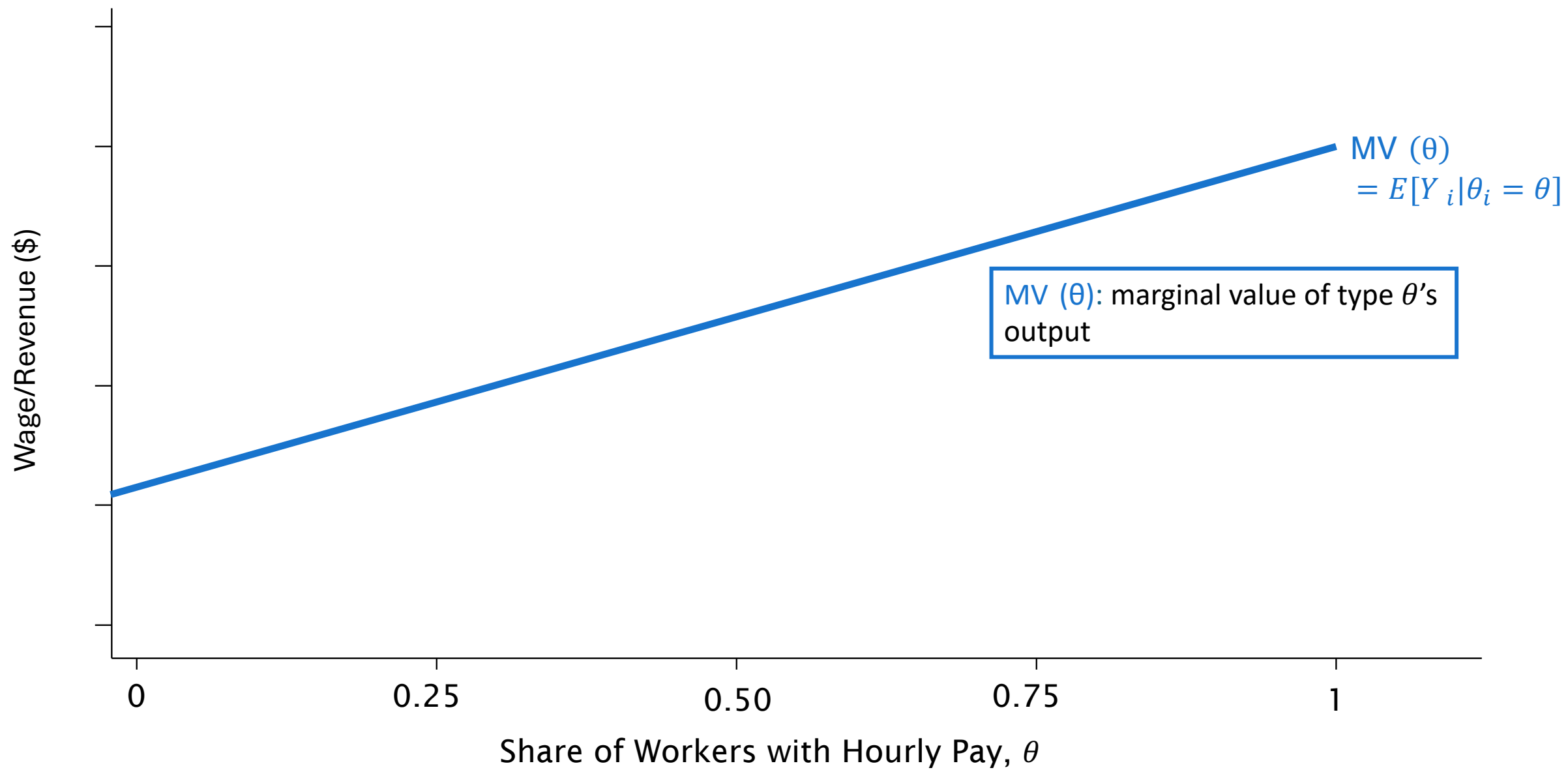
Incorporating Moral Hazard

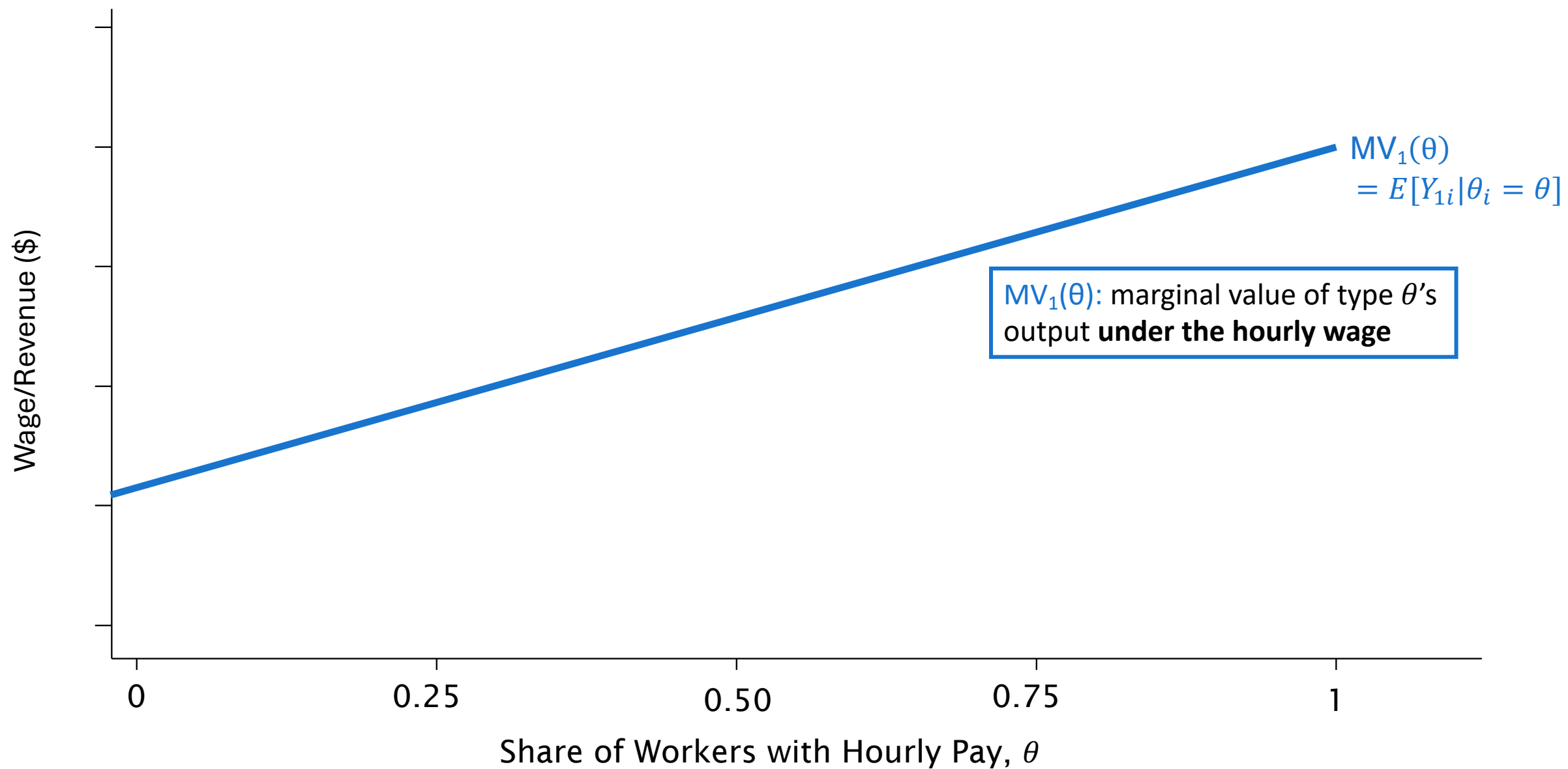
- Where does moral hazard fit in? Consider potential outputs

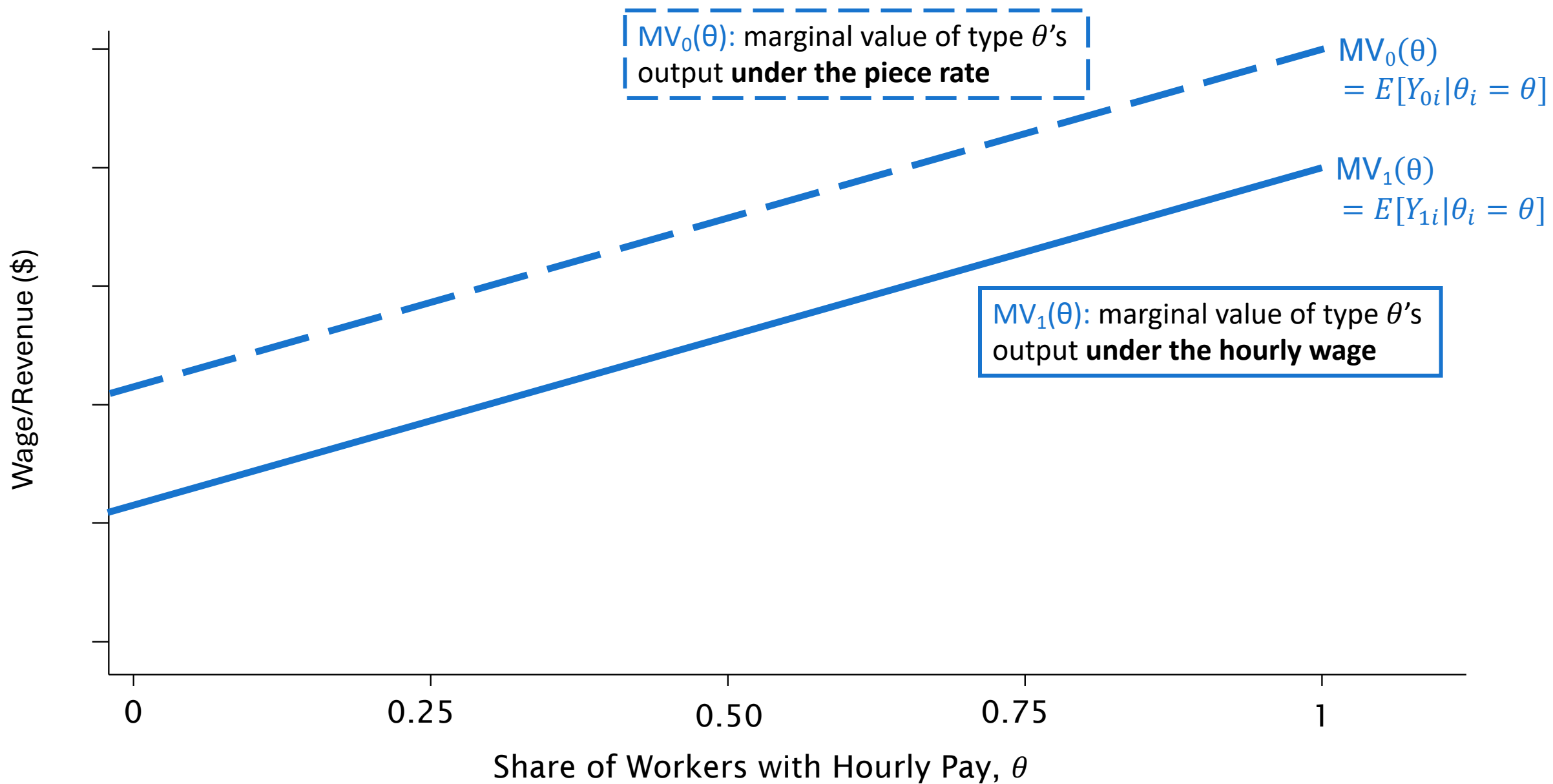
$$\begin{array}{l} Y_{1i}: \text{Potential output value under hourly wage} \\ Y_{0i}: \text{Potential output value under piece rate} \end{array} \quad \} \quad Y_{1i} = Y_{0i} + MH_i$$

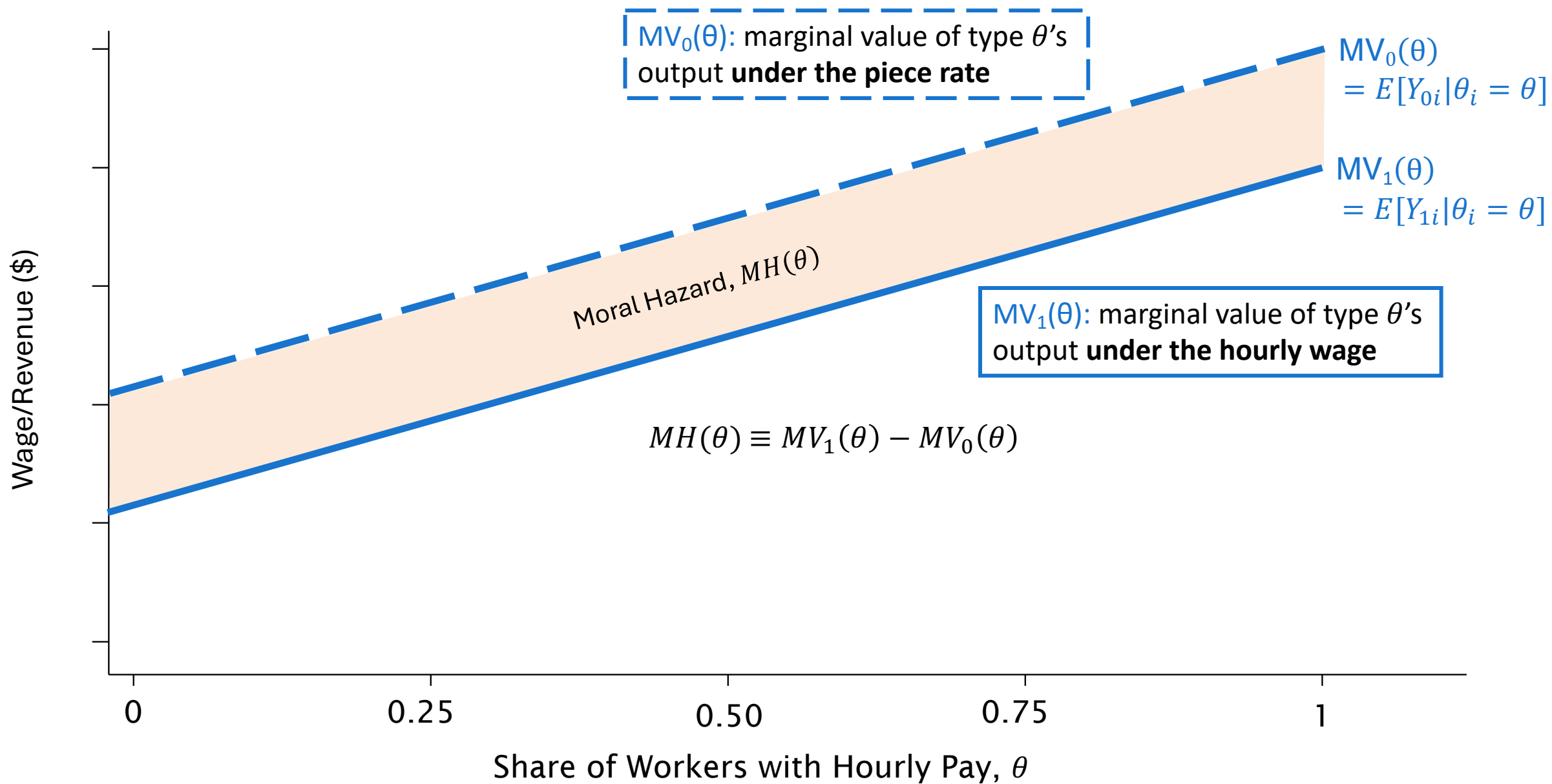
- Firms care about Y_{1i} , but they don't care about Y_{0i}
 - Piece-rate workers sell their output at a constant price per unit, so their productivity has no effect on firm profits
- $MV(\theta)$ & $AV(\theta)$ are defined *conditional* on accepting the hourly wage
$$MV(\theta) \equiv E[Y_i | \theta_i = \theta] = E[Y_{1i} | \theta_i = \theta]$$
$$AV(\theta) \equiv E[Y_i | \theta_i \leq \theta] = E[Y_{1i} | \theta_i \leq \theta]$$

⇒ Equilibrium is inclusive of workers' moral hazard response to the hourly wage
- Still want to separately identify moral hazard
 - Firms might mitigate MH with “partial insurance” (e.g., sales commission, tips)
 - Account for fiscal costs policies promoting fixed wages ($MH \Rightarrow \text{tax revenue} \downarrow$)

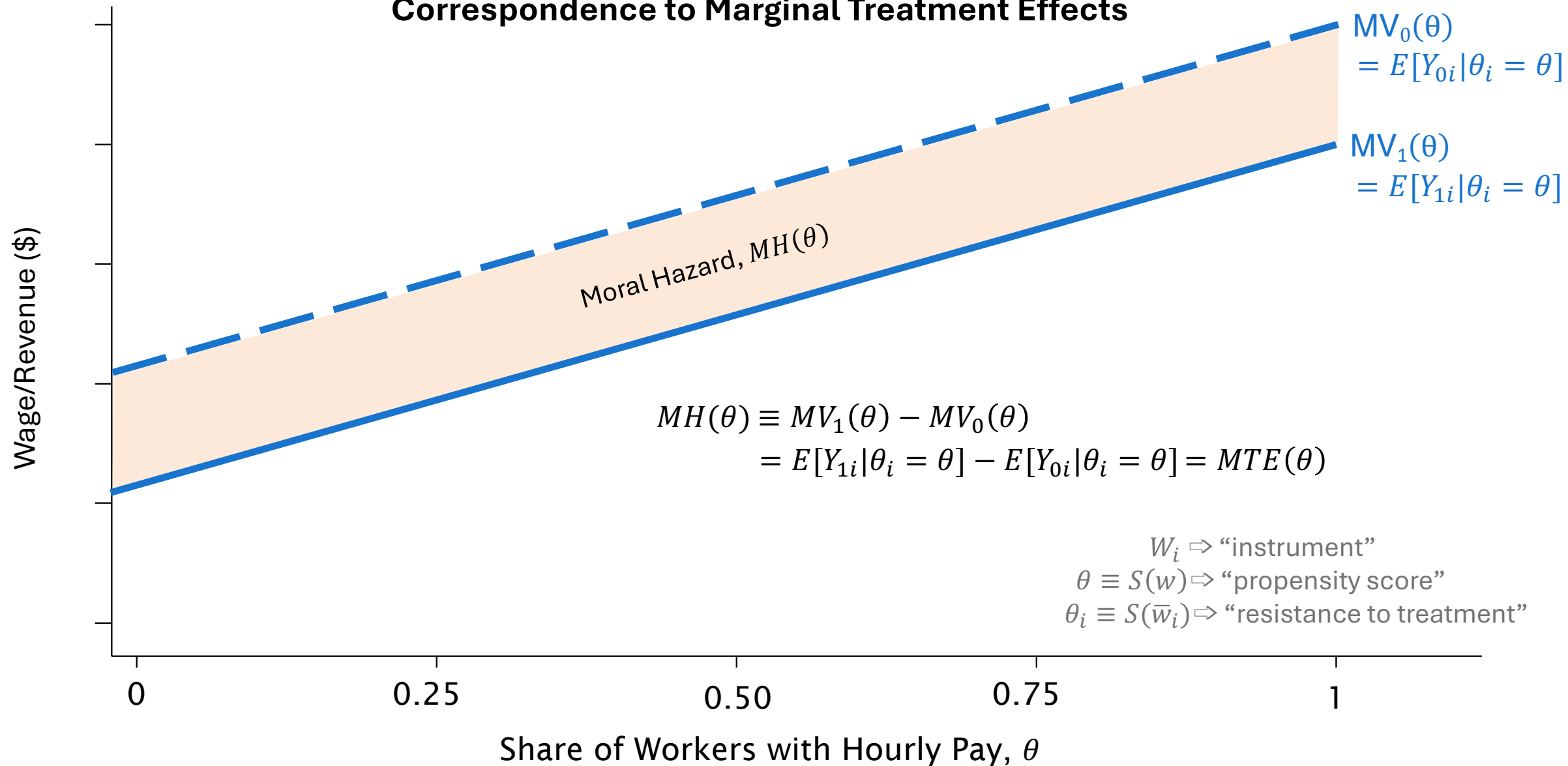


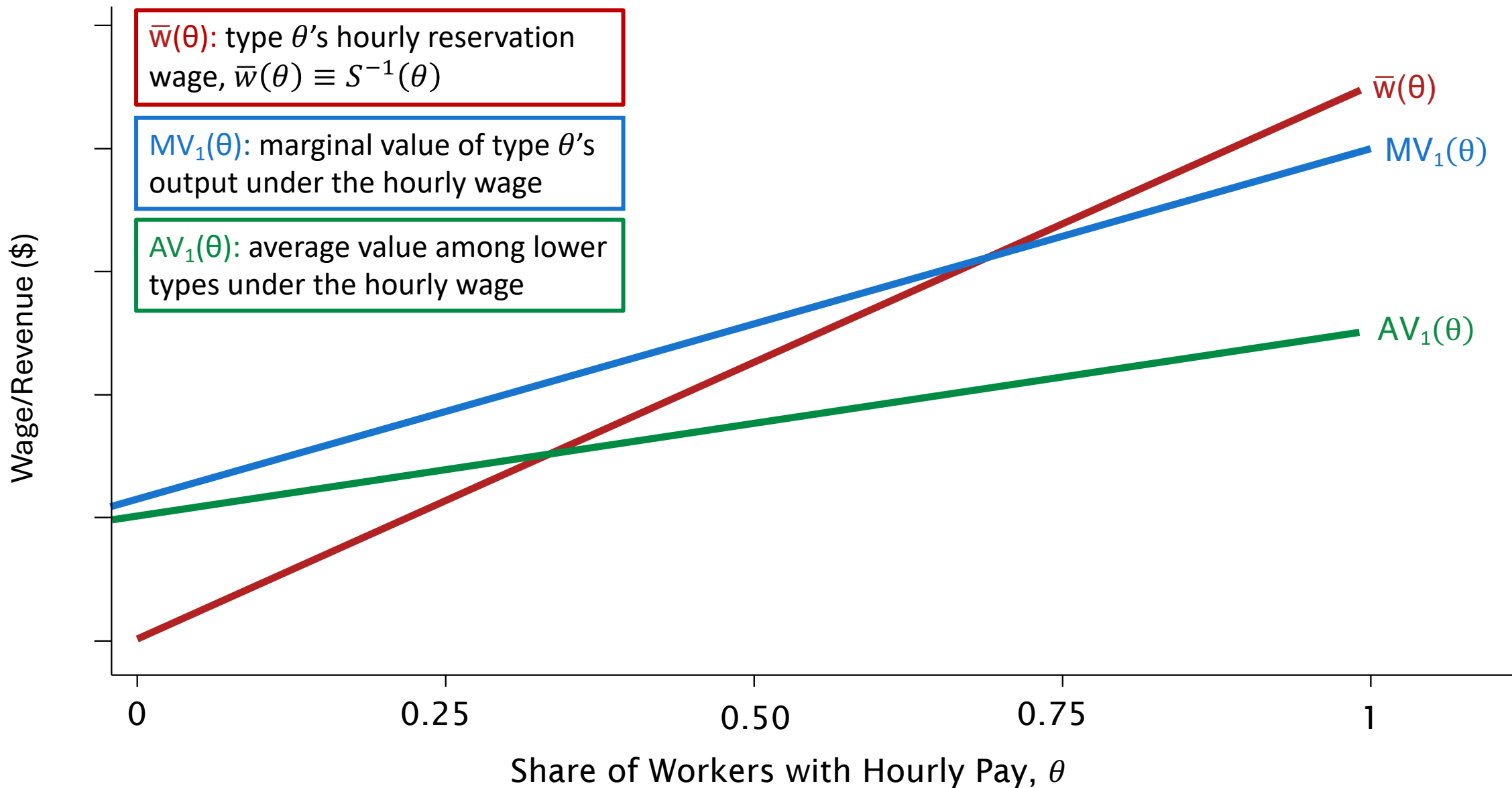


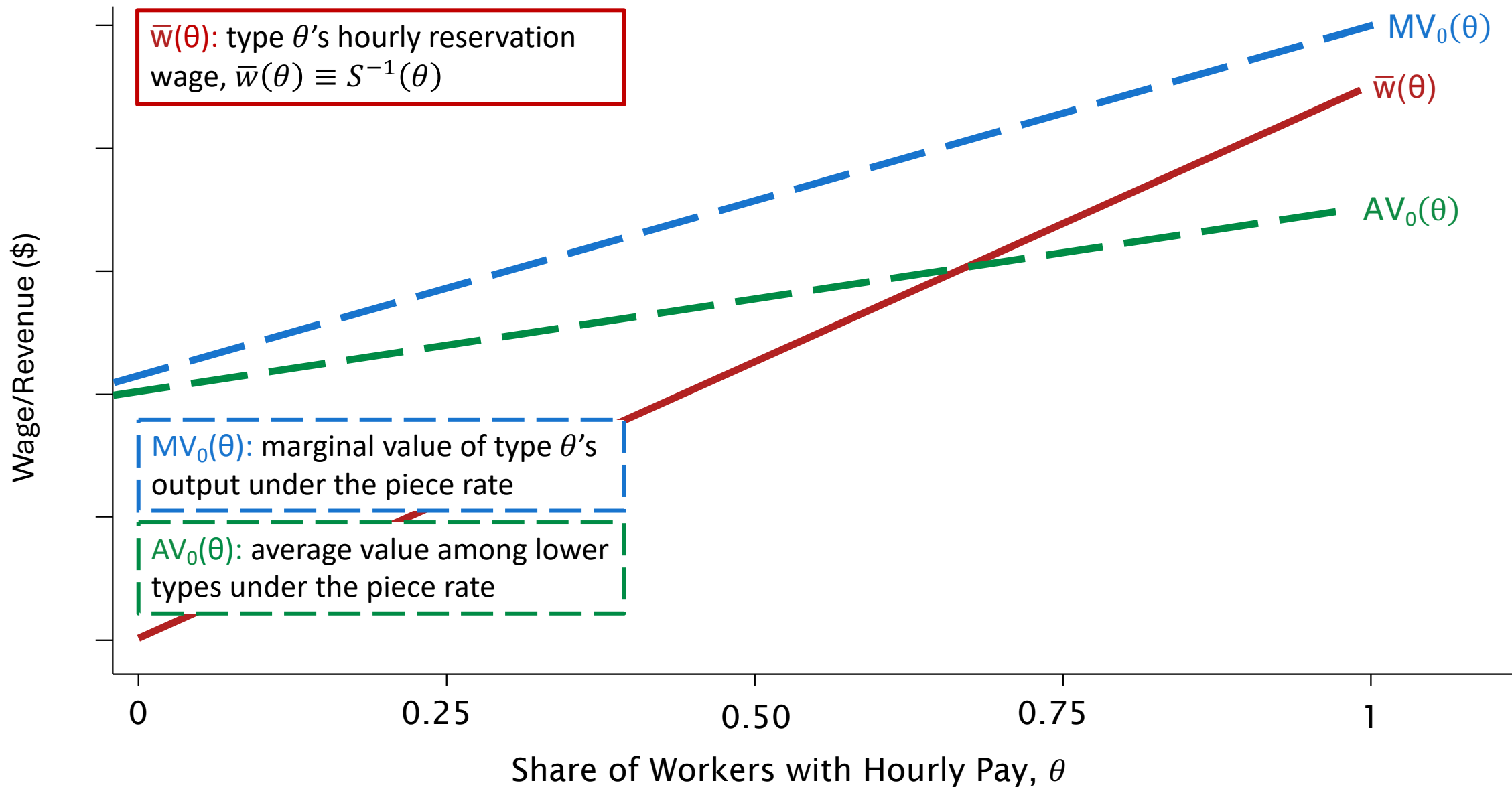




Correspondence to Marginal Treatment Effects

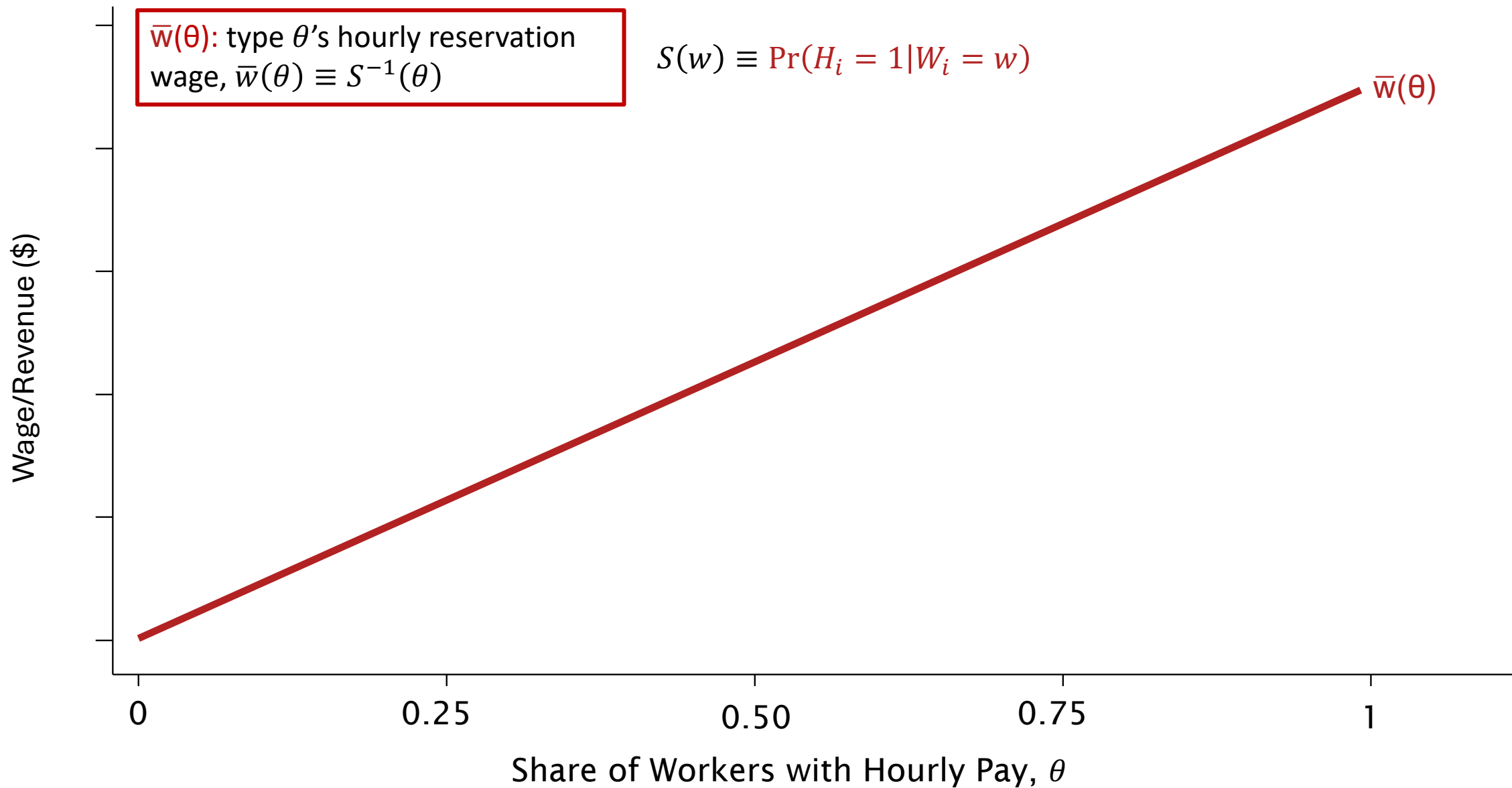


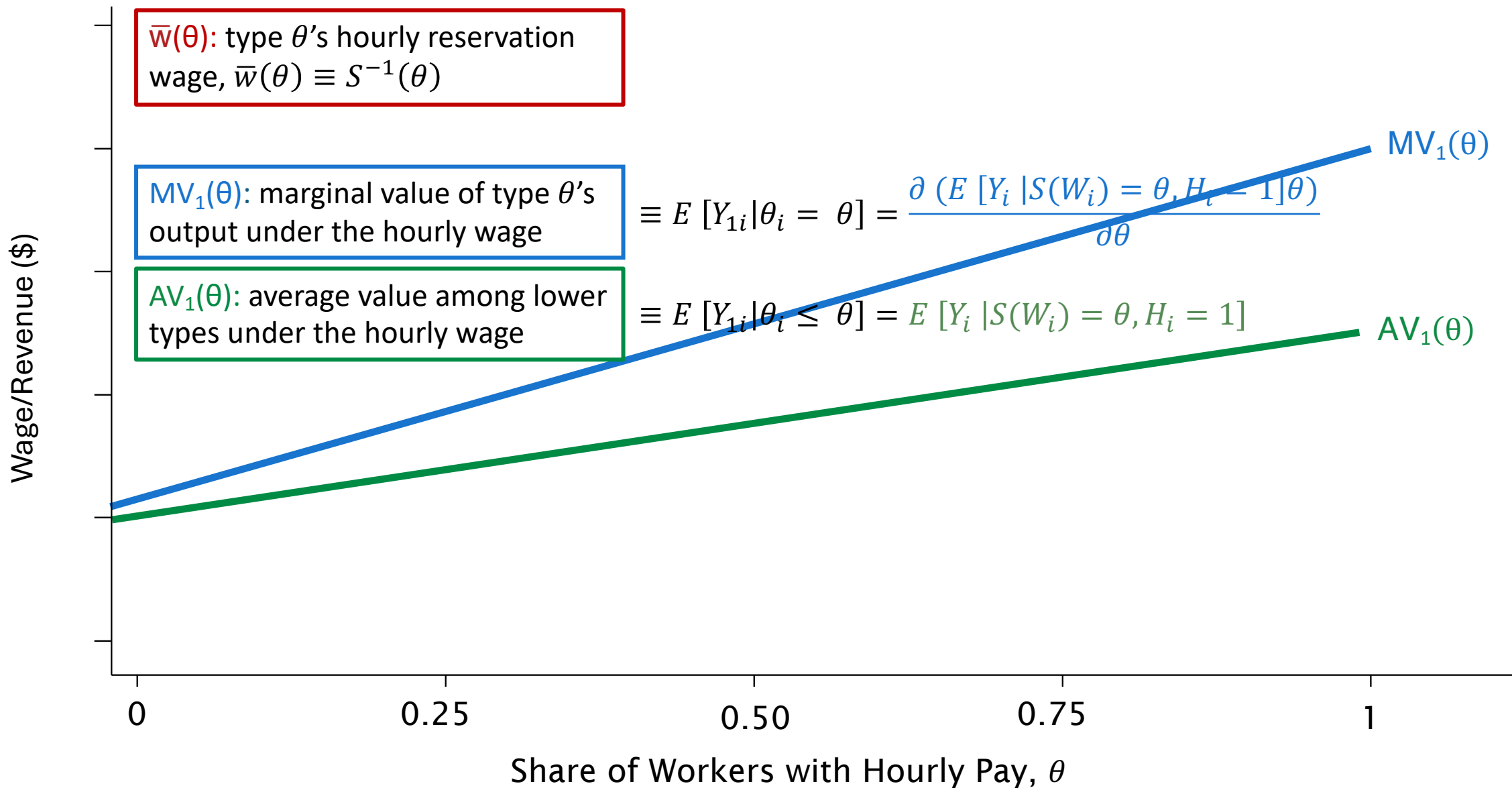


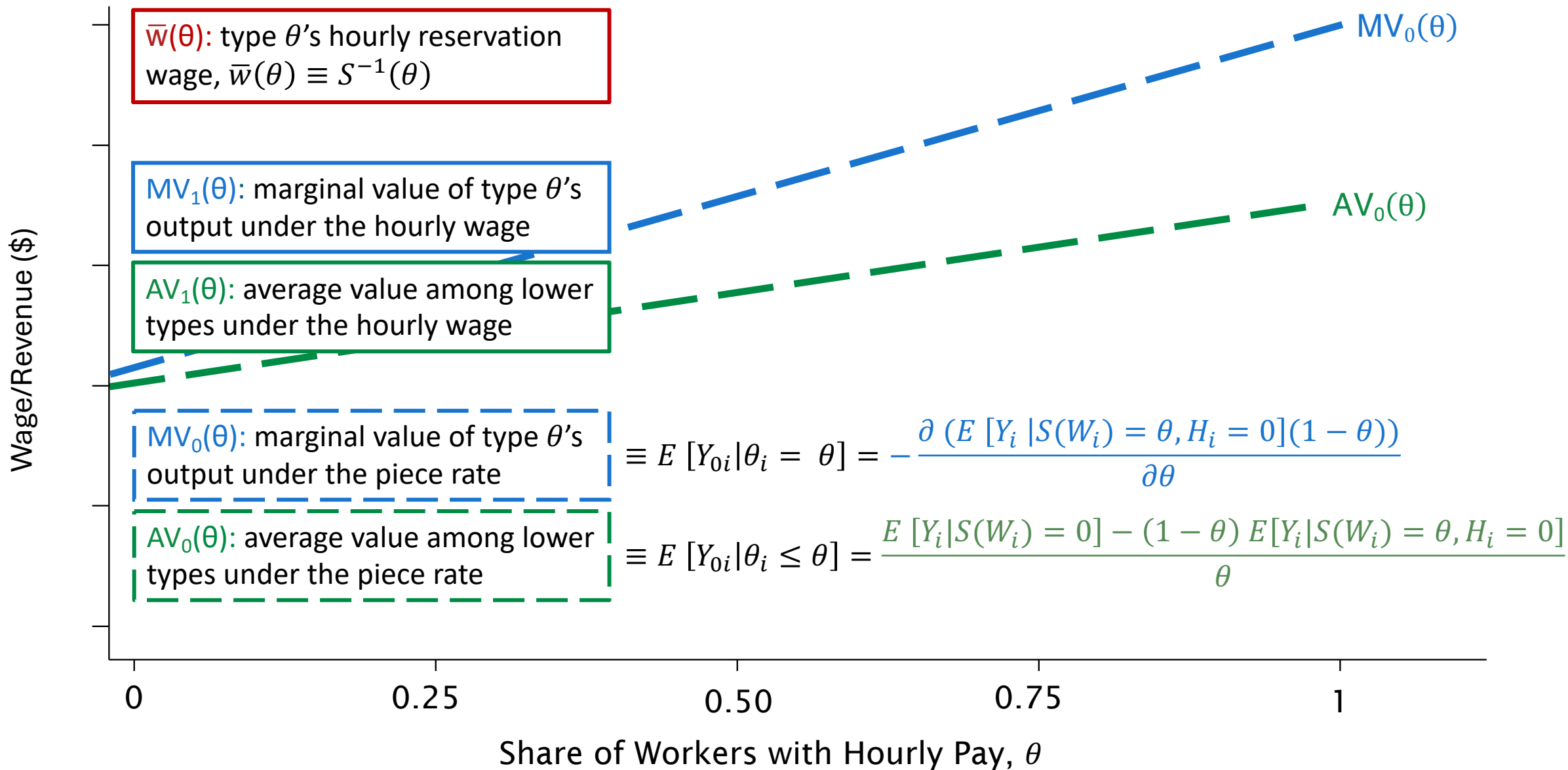


Outline

- ① **Experimental Design**
- ② **Main Results**
- ③ **Model of Asymmetric Information**
- ④ **Estimates of Marginal Value and Welfare Loss**
- ⑤ **Optimal Wage Subsidies**







$\bar{w}(\theta)$: type θ 's hourly reservation wage, $\bar{w}(\theta) \equiv S^{-1}(\theta)$

$$S(w) \equiv \Pr(H_i = 1 | W_i = w)$$

$MV_1(\theta)$: marginal value of type θ 's output under the hourly wage

$$\equiv E[Y_{1i} | \theta_i = \theta] = \frac{\partial (E[Y_i | S(W_i) = \theta, H_i = 1] \theta)}{\partial \theta}$$

$AV_1(\theta)$: average value among lower types under the hourly wage

$$\equiv E[Y_{1i} | \theta_i \leq \theta] = E[Y_i | S(W_i) = \theta, H_i = 1]$$

$MV_0(\theta)$: marginal value of type θ 's output under the piece rate

$$\equiv E[Y_{0i} | \theta_i = \theta] = - \frac{\partial (E[Y_i | S(W_i) = \theta, H_i = 0](1 - \theta))}{\partial \theta}$$

$AV_0(\theta)$: average value among lower types under the piece rate

$$\equiv E[Y_{0i} | \theta_i \leq \theta] = \frac{E[Y_i | S(W_i) = 0] - (1 - \theta) E[Y_i | S(W_i) = \theta, H_i = 0]}{\theta}$$

Estimation

(Carneiro et al. 2011)

1. Estimate $\Pr(H_i = 1|W_i = w)$
(logit regression)

2. Separately estimate

$$E[Y_i | S(W_i) = \theta, H_i = 1]$$

&

$$E[Y_i | S(W_i) = \theta, H_i = 0]$$

(local polynomial regression)

3. Differentiate with respect to θ

$$S(w) \equiv \Pr(H_i = 1|W_i = w)$$

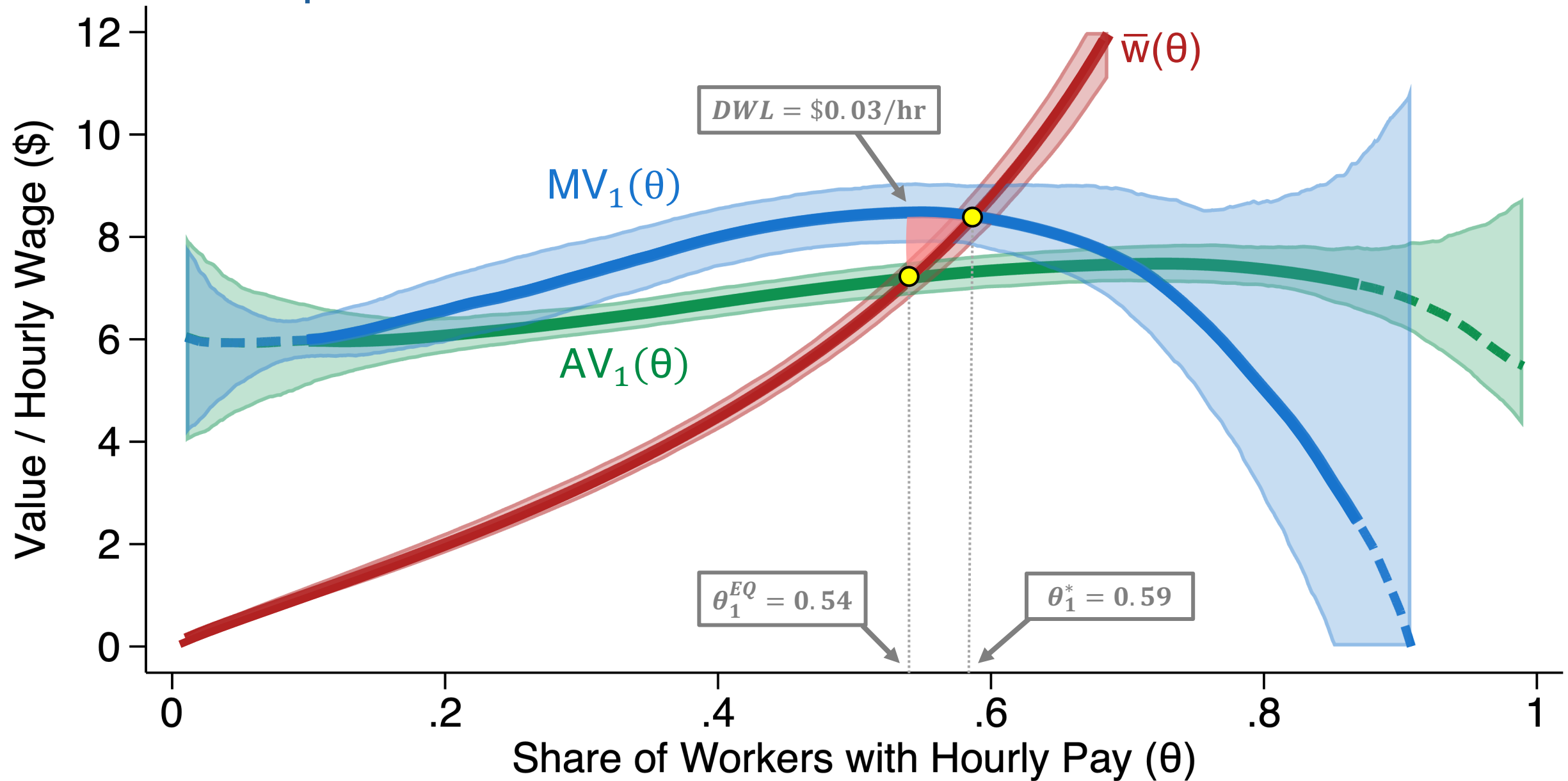
$$MV_1(\theta) \equiv E[Y_{1i} | \theta_i = \theta] = \frac{\partial (E[Y_i | S(W_i) = \theta, H_i = 1])}{\partial \theta}$$

$$AV_1(\theta) \equiv E[Y_{1i} | \theta_i \leq \theta] = E[Y_i | S(W_i) = \theta, H_i = 1]$$

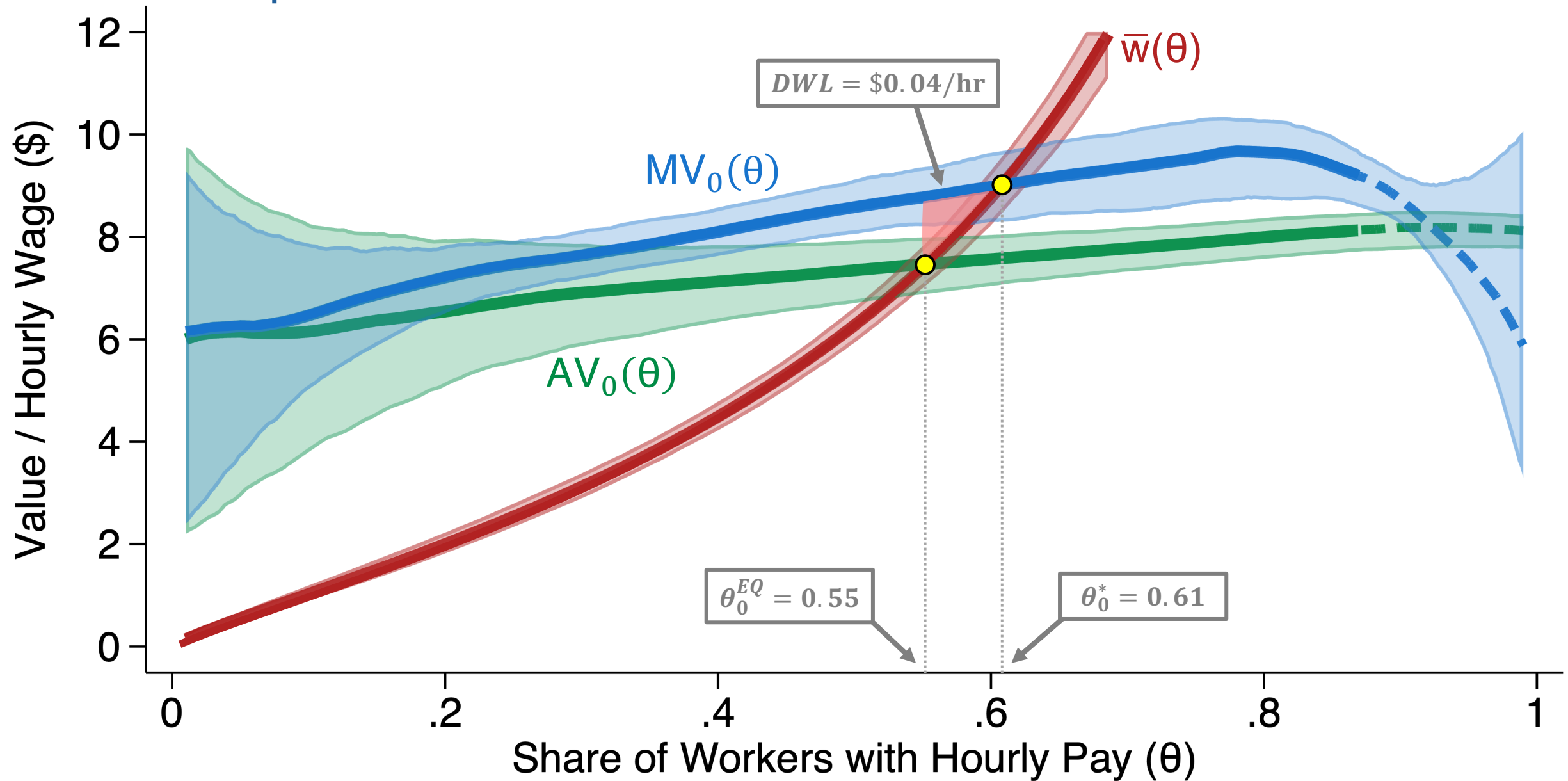
$$MV_0(\theta) \equiv E[Y_{0i} | \theta_i = \theta] = - \frac{\partial (E[Y_i | S(W_i) = \theta, H_i = 0])}{\partial \theta} (1 - \theta)$$

$$AV_0(\theta) \equiv E[Y_{0i} | \theta_i \leq \theta] = \frac{E[Y_i | S(W_i) = 0] - (1 - \theta) E[Y_i | S(W_i) = \theta, H_i = 0]}{\theta}$$

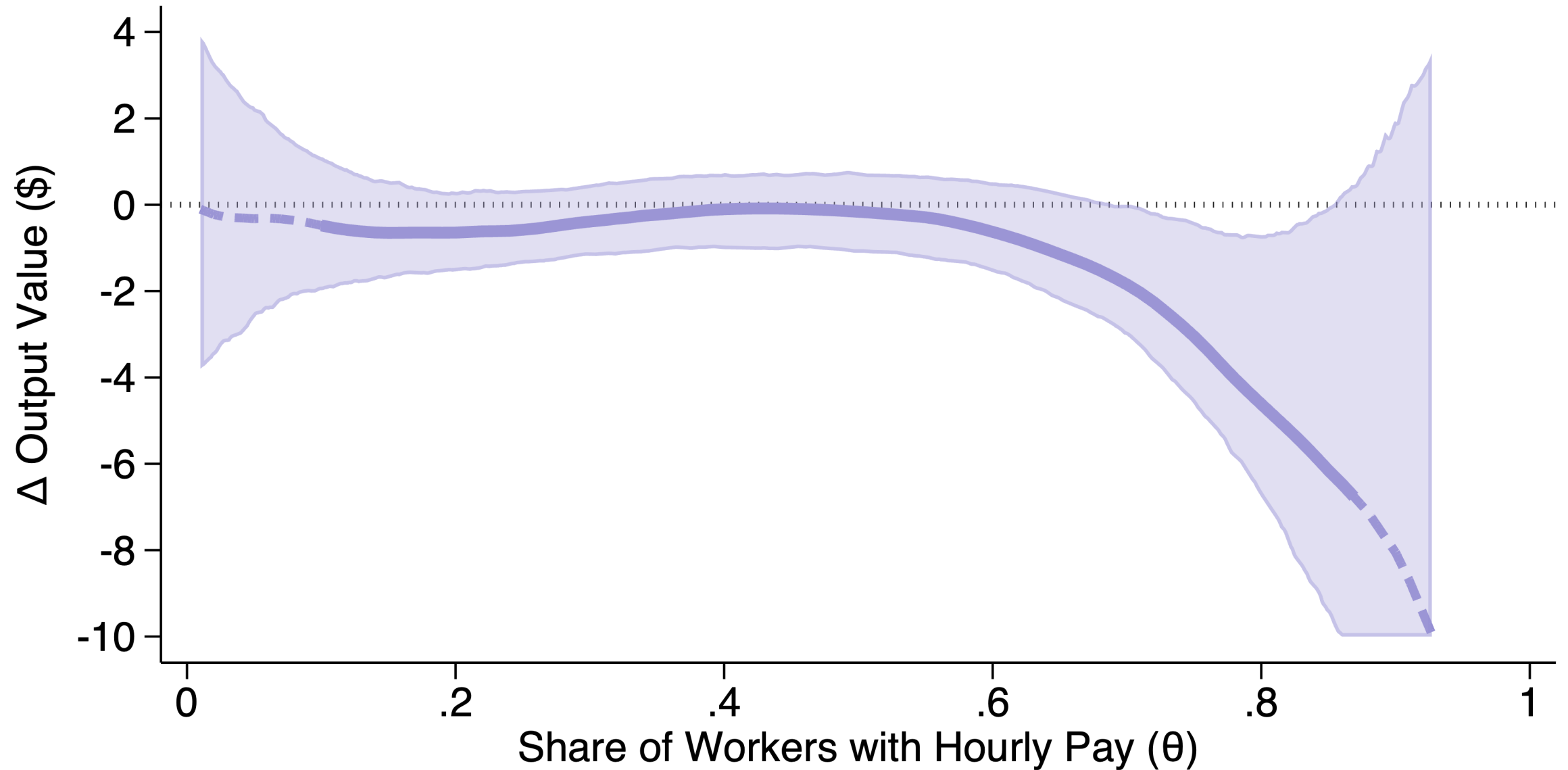
Equilibrium with Adverse Selection and Moral Hazard



Equilibrium with Adverse Selection and no Moral Hazard



Moral Hazard: Estimates of Marginal Treatment Effects



Outline

- 1 Experimental Design
- 2 Main Results
- 3 Model of Asymmetric Information
- 4 Estimates of Marginal Value and Welfare Loss
- 5 Optimal Fixed-Wage Subsidy and Piece-Rate Tax

Policy Solutions

- Welfare loss from adverse selection suggests role for government: promote insurance-like provisions in labor contracts.
- I calculate the **Marginal Value of Public Funds (MVPF)** for two policies
 1. Fixed-wage subsidies
 - e.g., deductions for portion of hourly payments
 2. Piece-rate tax
 - e.g., eliminate tax credits on tipped earnings
- Results may inform broad set of labor policies, even if estimates are specific to data-entry setting

$$MVPF = \frac{Welfare\ Benefit}{Net\ Govt\ Expenditure}$$

Policy #1: Fixed-wage Subsidy

Marginal Value of Public Funds (MVPF) for an hourly wage subsidy:

$$MVPF_{subsidy} = \frac{\textit{Welfare Benefit}}{\textit{Net Govt Expenditure}}$$

Welfare Benefit: The aggregate amount workers are willing to pay for subsidy

Net Govt Expenditure: The aggregate amount spent including fiscal externalities

Policy #1: Fixed-wage Subsidy

Marginal Value of Public Funds (MVPF) for an hourly wage subsidy:

$$MVPF_{subsidy} = \frac{\textit{Transfer} + \textit{Insurance Value}}{\textit{Net Govt Expenditure}}$$

Welfare Benefit: The aggregate amount workers are willing to pay for subsidy

- *Transfer* from government to hourly workers
- *Insurance Value* to workers who move from piece-rate to hourly pay

Net Govt Expenditure: The aggregate amount spent including fiscal externalities

Policy #1: Fixed-wage Subsidy

Marginal Value of Public Funds (MVPF) for an hourly wage subsidy:

$$MVPF_{\text{subsidy}} = \frac{\text{Transfer} + \text{Insurance Value}}{\text{Transfer} + \text{Tax Loss from MH}}$$

Welfare Benefit: The aggregate amount workers are willing to pay for subsidy

- *Transfer* from government to hourly workers
- *Insurance Value* to workers who move from piece-rate to hourly pay

Net Govt Expenditure: The aggregate amount spent including fiscal externalities

- *Transfer* from government to hourly workers
- *Tax Loss from MH* when hourly workers shirk and pay less income tax

Policy #1: Fixed-wage Subsidy

Marginal Value of Public Funds (MVPF) for an hourly wage subsidy:

$$MVPF_{subsidy} = \frac{\delta \theta^\delta + \int_{\theta^{EQ}}^{\theta^\delta} (MV_1(\theta) - \bar{w}(\theta)) d\theta}{\delta \theta^\delta - \int_{\theta^{EQ}}^{\theta^\delta} \tau MH(\theta) d\theta}$$

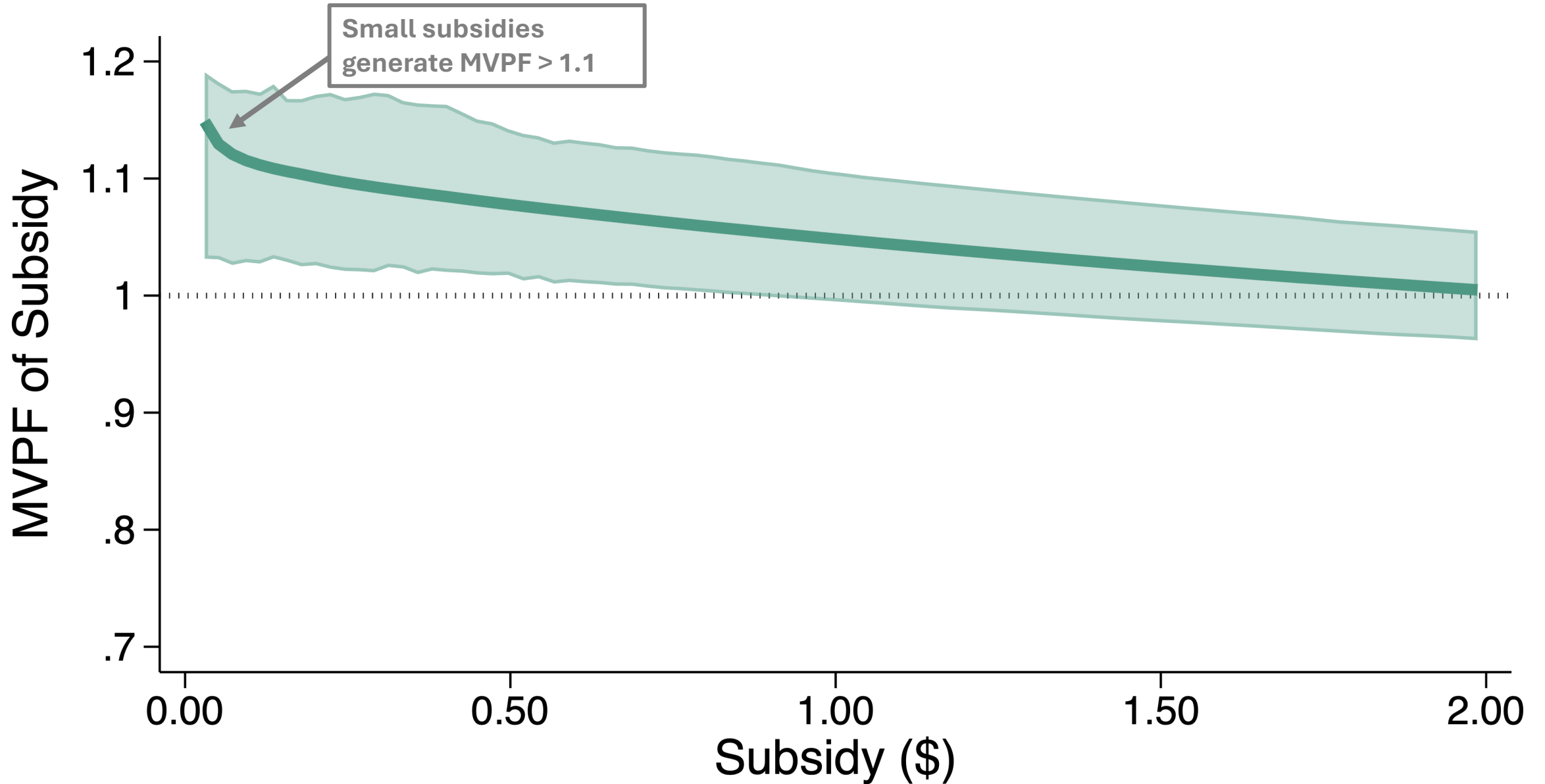
Welfare Benefit: The aggregate amount workers are willing to pay for subsidy

- *Transfer* from government to hourly workers
- *Insurance Value* to workers who move from piece-rate to hourly pay

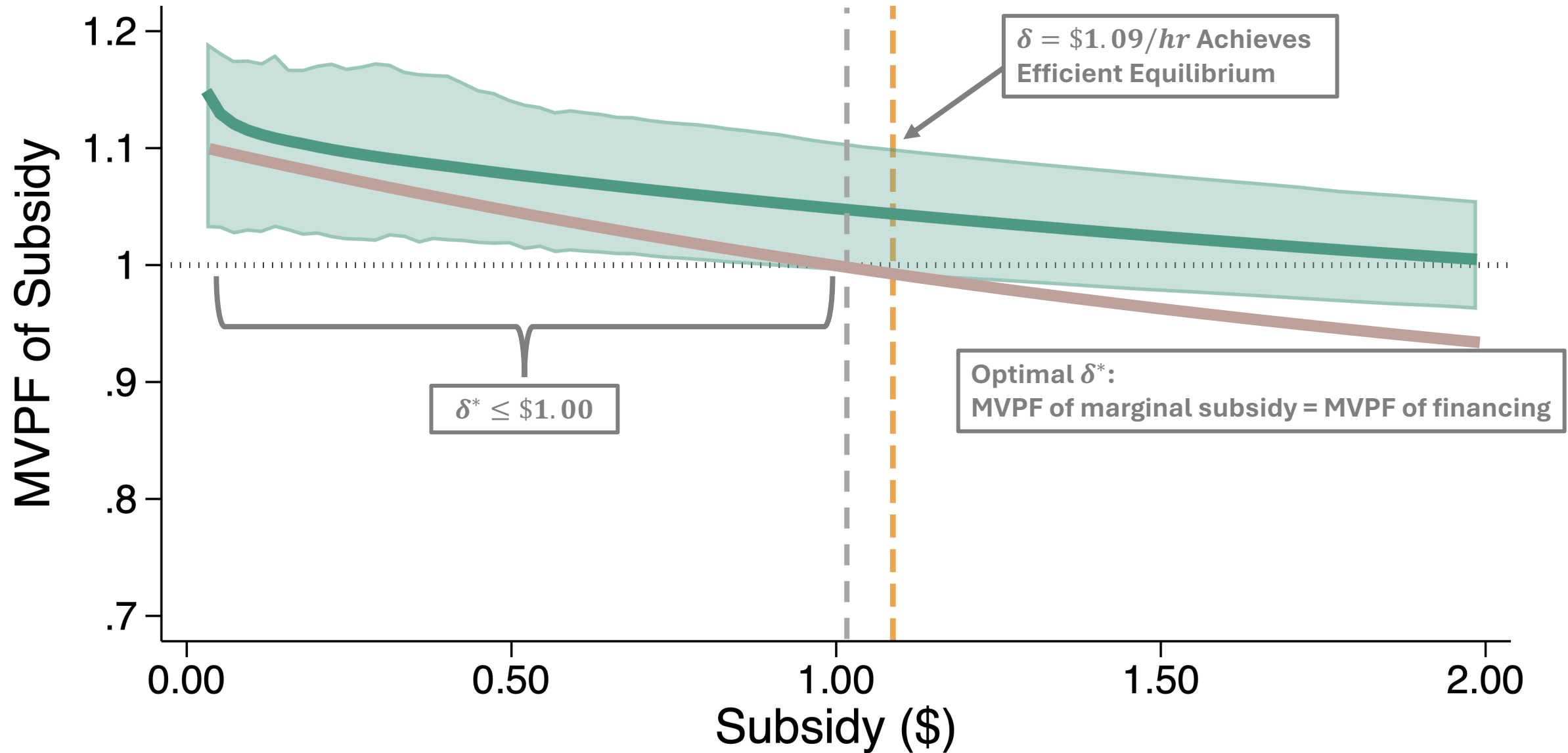
Net Govt Expenditure: The aggregate amount spent including fiscal externalities

- *Transfer* from government to hourly workers
- *Tax Loss from MH* when hourly workers shirk and pay less income tax

MVPF of Hourly Wage Subsidy



MVPF of Hourly Wage Subsidy



Policy #2: Piece-rate Tax

Marginal Value of Public Funds (MVPF) for a piece-rate tax:

$$MVPF_{tax} = \frac{Welfare\ Loss}{Net\ Govt\ Revenue}$$

Welfare Loss: The aggregate amount workers would pay to avoid the tax

Net Govt Revenue: The aggregate revenue raised including fiscal externalities

Socially efficient tax $\Rightarrow$ **lower** MVPF

Policy #2: Piece-rate Tax

Marginal Value of Public Funds (MVPF) for a piece-rate tax:

$$MVPF_{tax} = \frac{\textit{Transfer} - \textit{Insurance Value}}{\textit{Net Govt Revenue}}$$

Welfare Loss: The aggregate amount workers would pay to avoid the tax

- *Transfer* from piece-rate workers to government
- *Insurance Value* to workers who move from piece-rate to hourly pay

Net Govt Revenue: The aggregate revenue raised including fiscal externalities

Socially efficient tax $\Rightarrow$ **lower** MVPF

Policy #2: Piece-rate Tax

Marginal Value of Public Funds (MVPF) for a piece-rate tax:

$$MVPF_{tax} = \frac{\text{Transfer} - \text{Insurance Value}}{\text{Transfer} - \text{Tax Loss from MH}}$$

Welfare Loss: The aggregate amount workers would pay to avoid the tax

- *Transfer* from piece-rate workers to government
- *Insurance Value* to workers who move from piece-rate to hourly pay

Net Govt Revenue: The aggregate revenue raised including fiscal externalities

- *Transfer* from piece-rate workers to government
- *Tax Loss from MH* when hourly workers shirk and pay less income tax

Socially efficient tax $\Rightarrow$ **lower** MVPF

Policy #2: Piece-rate Tax

Marginal Value of Public Funds (MVPF) for a piece-rate tax:

$$MVPF_{tax} = \frac{\int_{\theta^P}^1 \rho MV_0(\theta) d\theta - \int_{\theta^{EQ}}^{\theta^P} (MV_1(\theta) - \bar{w}(\theta)) d\theta}{\int_{\theta^P}^1 \rho MV_0(\theta) d\theta + \int_{\theta^{EQ}}^{\theta^P} \tau MH(\theta) d\theta}$$

Welfare Loss: The aggregate amount workers would pay to avoid the tax

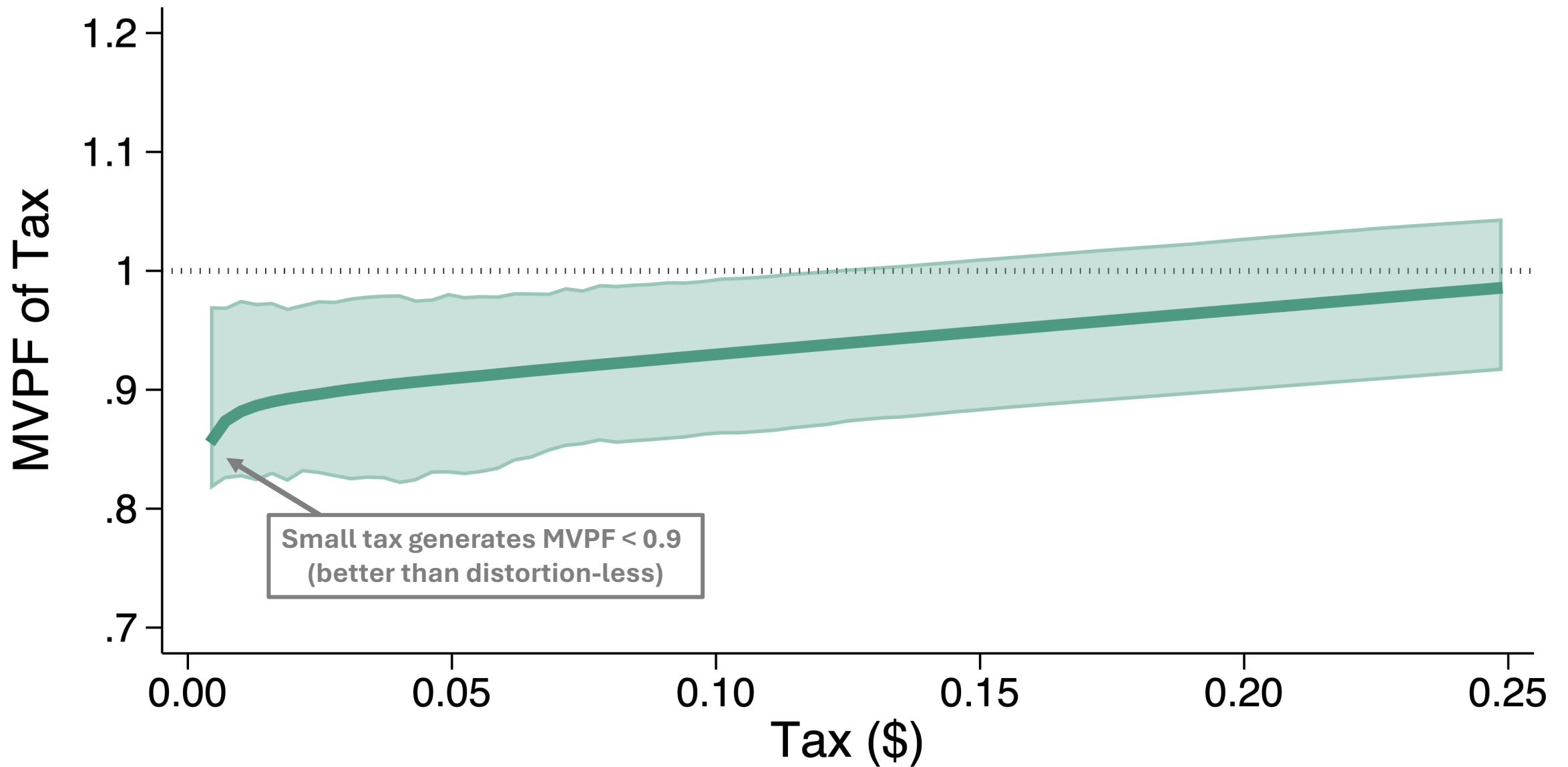
- *Transfer* from piece-rate workers to government
- *Insurance Value* to workers who move from piece-rate to hourly pay

Net Govt Revenue: The aggregate revenue raised including fiscal externalities

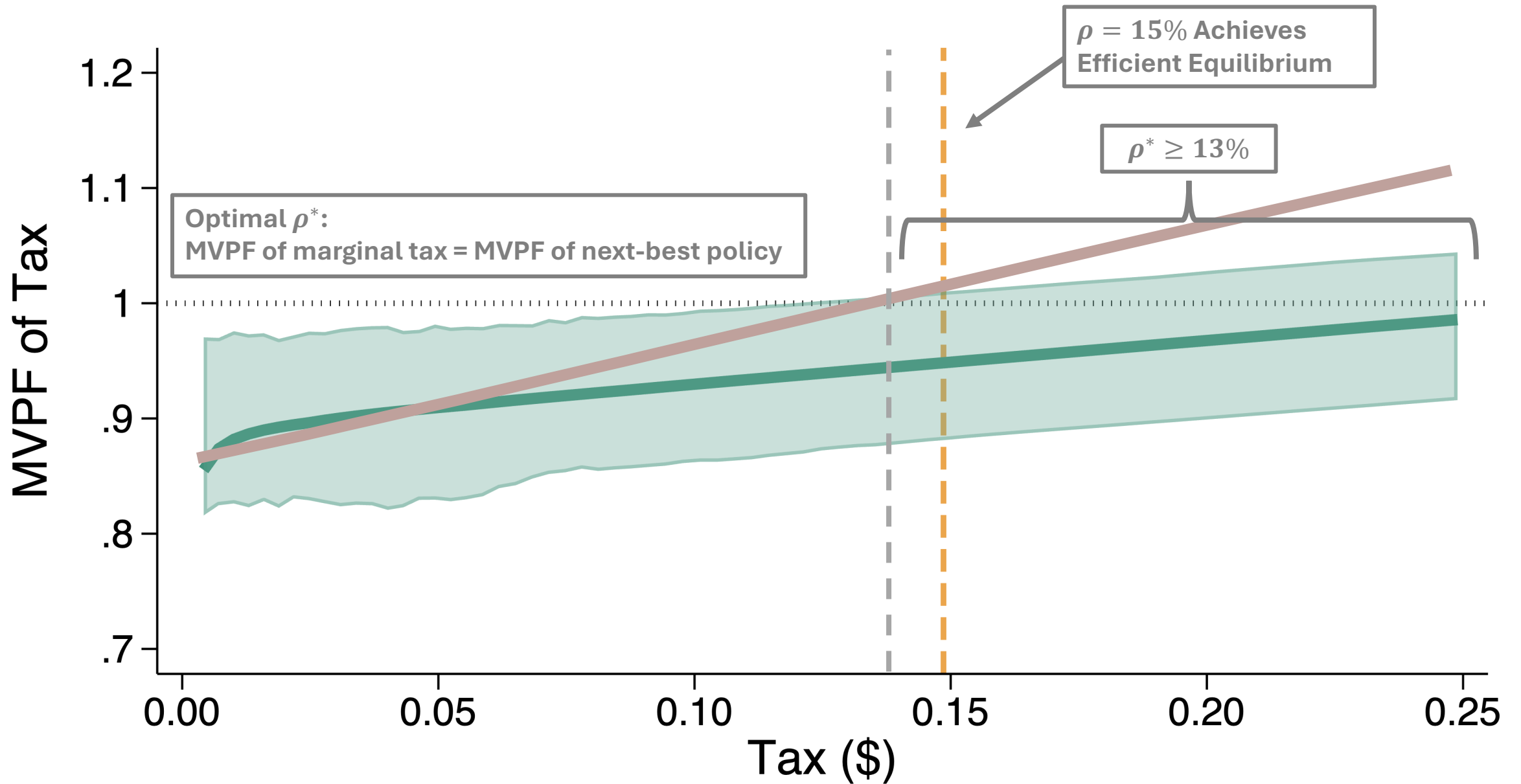
- *Transfer* from piece-rate workers to government
- *Tax Loss from MH* when hourly workers shirk and pay less income tax

Socially efficient tax $\Rightarrow$ **lower** MVPF

MVPF of Piece Rate Tax



MVPF of Piece Rate Tax



Conclusion

- Develop insurance model of wage contracts under asymmetric information
 - Shows how moral hazard and adverse selection affect market equilibrium
 - Parameters map into an MTE framework
- RCT estimates (marginal) selection and treatment effects of hourly pay
 - Separately identifies moral hazard and adverse selection
 - Estimate welfare loss of \$0.05/hour from inefficient contracts
- Policy prescription: increase taxes on piece-rate compensation
 - Tax on commissions, bonuses, or tips can encourage fixed wages and mitigate welfare loss
 - $\tau \approx 13\%$ raises \$1 of government revenue at an estimated social cost of \$0.90 or less
- Future research and extensions
 - Employer learning: screening on past performance
 - Separating contracts: randomize *menu* of offers
 - Alternative approach to measure adverse selection in other settings or contract dimensions
 1. Solicit reservation wage: “What’s the minimum pay raise you would accept to forego your bonus?”
 2. Measure worker productivity by stated willingness-to-accept

Appendix

Experimental Design: Single Wage Offer

Comparing piece-rate workers who faced different offer sets identifies ***selection into treatment***:

$$\frac{E[Y_i|W_i = 0] - E[Y_i|H_i = 0, W_i = 1]}{\pi} = \frac{(\pi E[Y_{0i}|H_i = 1] + (1 - \pi)\pi E[Y_{0i}|H_i = 0]) - E[Y_{0i}|H_i = 0]}{\pi} = E[Y_{0i}|H_i = 1] - E[Y_{0i}|H_i = 0]$$

Identifying Selection Across Multiple Wage Offers

W_i : Randomized treatment-offer group

- $W_i = L$: Offered a choice between low hourly wage and piece rate
- $W_i = H$: Offered a choice between high hourly wage and piece rate
- π^W : Take-up of wage offer $W \in \{L, H\}$

Selection on Y_0 into offer H among those who would reject offer L :

$$\begin{aligned} E[Y_{0i} | H_i^H = 1, H_i^L = 0] - E[Y_{0i} | H_i^H = 0] \\ = \frac{1 - \pi^L}{\pi^H - \pi^L} E[Y_i | H_i = 0, W_i = L] - E[Y_i | H_i = 0, W_i = H] \end{aligned}$$

Selection on Y_1 into offer L among those who would accept offer H :

$$\begin{aligned} E[Y_{1i} | H_i^L = 1] - E[Y_{1i} | H_i^H = 1, H_i^L = 0] \\ = \frac{\pi^H}{\pi^H - \pi^L} E[Y_i | H_i = 1, W_i = L] - E[Y_i | H_i = 1, W_i = H] \end{aligned}$$

Balance Test

	(1) Experimental Wage Offer	(2) Output Value
Number of Previous Tasks/1000	−0.0478 (0.0343)	0.191*** (0.0305)
Age	0.00141 (0.00529)	−0.0683*** (0.00453)
Female	0.0909 (0.124)	0.366*** (0.108)
Minority	−0.0528 (0.125)	−0.896*** (0.109)
Employed	−0.202 (0.138)	0.142 (0.121)
Student	0.0685 (0.169)	−0.474*** (0.149)
F-statistic	1.019	36.492
p-value	0.426	0.000
<i>N</i>	3030	3030

Hourly Wage Take-up

	(1) Accepted Offer	(2) Accepted Offer	(3) Accepted Offer	(4) Accepted Offer
Log Hourly Wage Offer	1.198*** (0.0554)	1.202*** (0.0554)	1.212*** (0.0560)	1.245*** (0.0578)
Task Controls	No	Yes	Yes	Yes
Employment Controls	No	No	Yes	Yes
Demographic Controls	No	No	No	Yes
<i>N</i>	2728	2728	2728	2728

$$\Pr(H_i = 1) = \text{inv logit}(\alpha \ln W_i + \boldsymbol{\nu} \mathbf{X}_i)$$

Suppose supply and average value are linear in the wage offer:

$$Y_i = \xi_0 + \alpha H_i + \beta_0(1 - H_i) \times W_i + \beta_1 H_i \times W_i + \gamma H_i \times W_i^P + \epsilon_i$$

$$H_i = \gamma + \delta w + v_i$$

Let $\bar{w}_i$ be worker i 's hourly reservation wage $\Rightarrow$ CEFs of hourly take-up and output value:

$$E[H_i | \bar{w}_i \leq w] = \gamma + \delta w$$

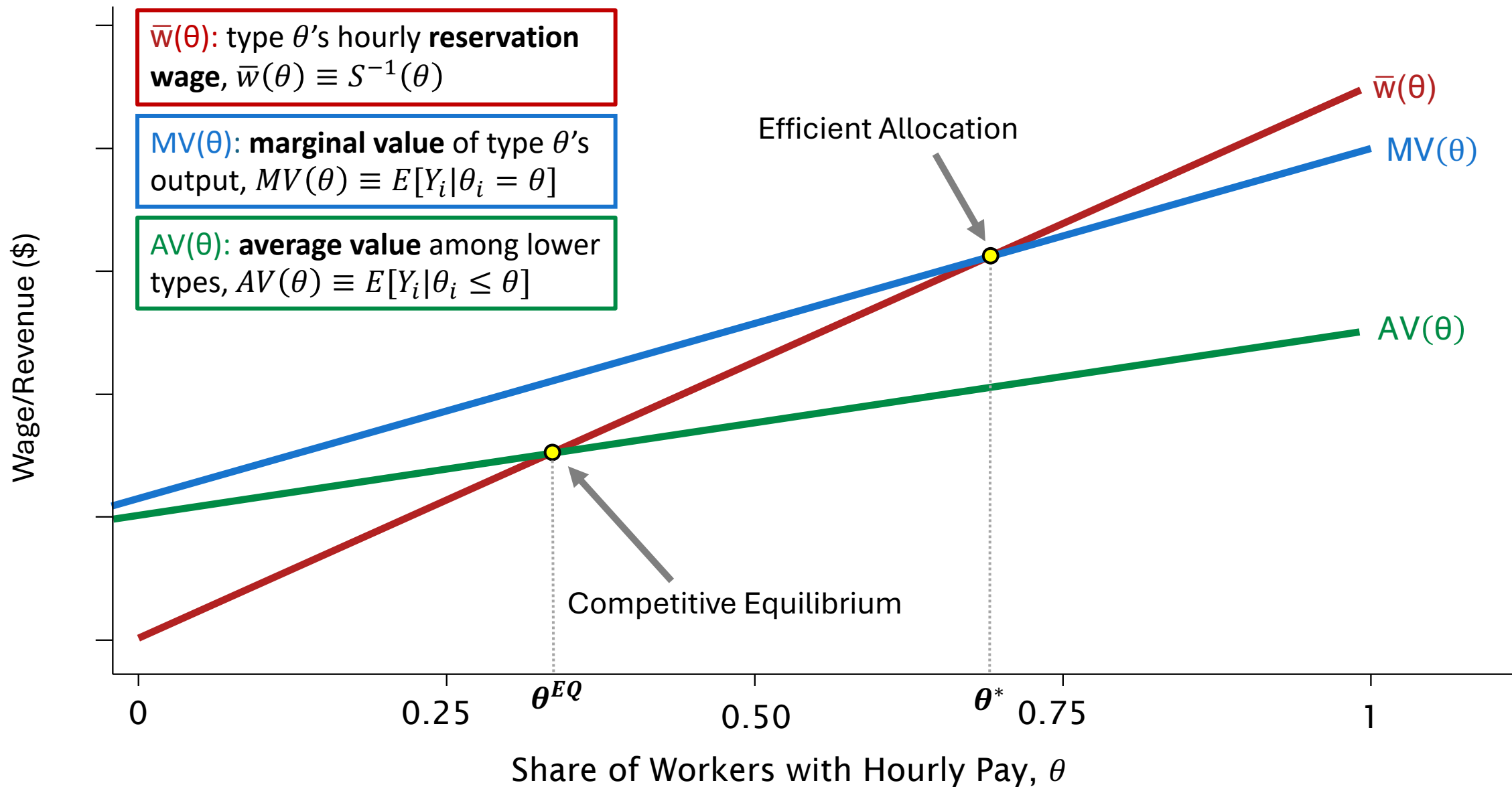
$$E[Y_i | \bar{w}_i \leq w] = \xi_0 + \alpha + \beta_1 H_i \times w$$

$$E[Y_i | \bar{w}_i > w] = \xi_0 + \beta_0(1 - H_i) \times w$$

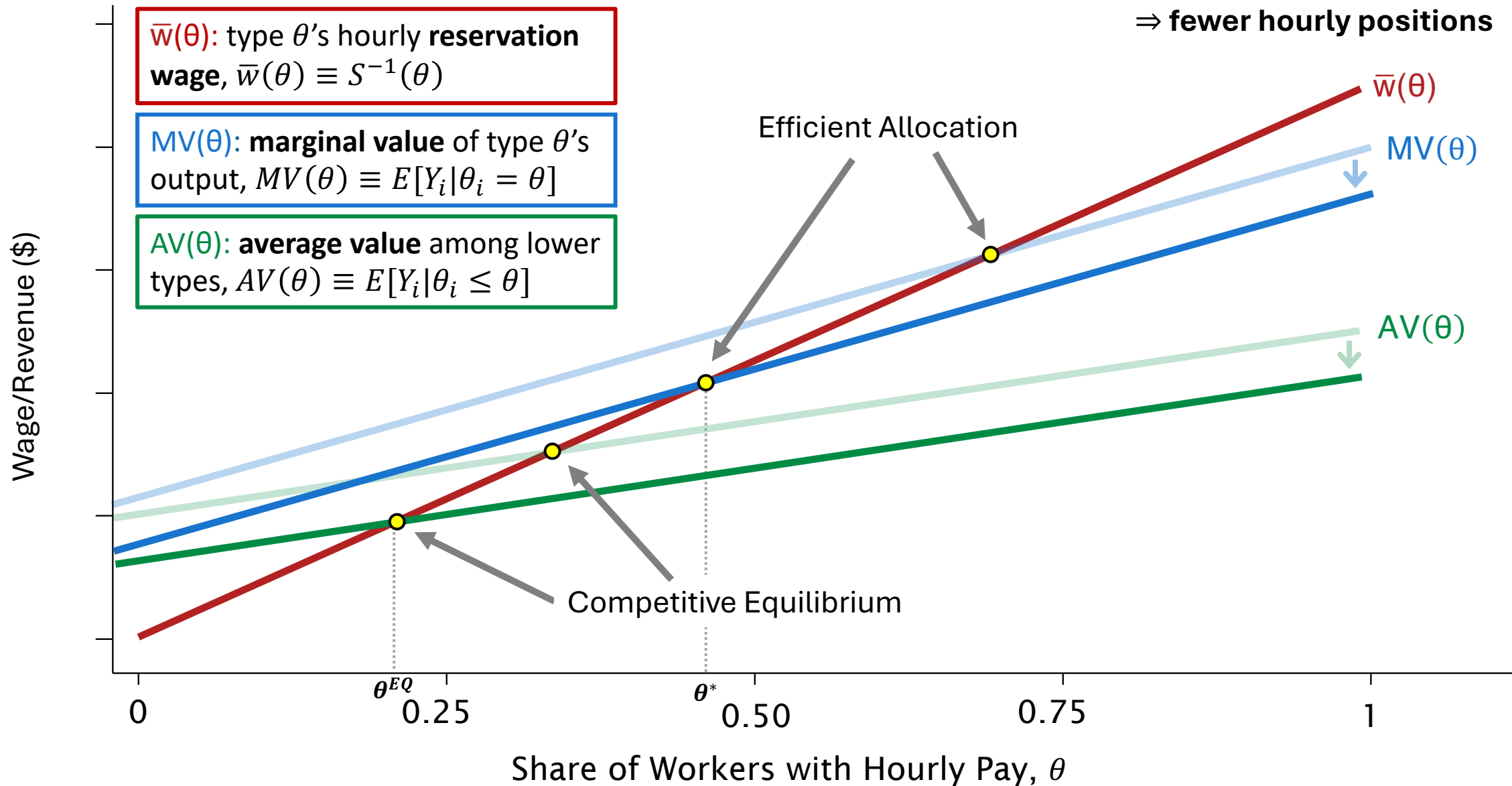
The *marginal* potential output values $E[Y_{1i} | \bar{w}_i = w]$ & $E[Y_{0i} | \bar{w}_i = w]$ are given by

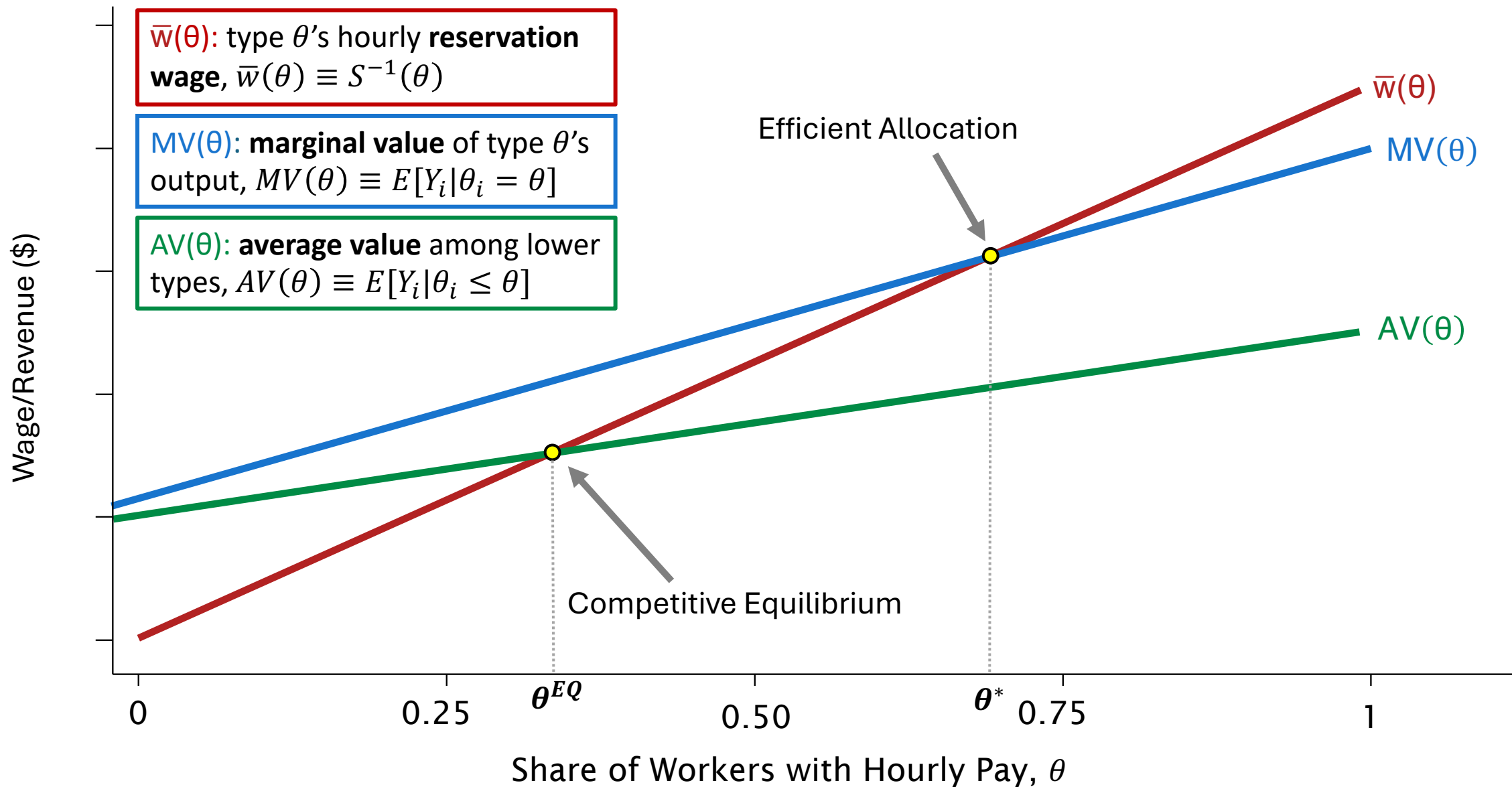
$$E[Y_{1i} | \bar{w}_i = w] = \frac{\partial(E[Y_i | \bar{w}_i \leq w]S(w))}{\partial S(w)} = \xi_0 + \alpha + \frac{\gamma\beta_1}{\delta} + 2\beta_1 w$$

$$E[Y_{0i} | \bar{w}_i = w] = -\frac{\partial(E[Y_i | \bar{w}_i > w](1 - S(w)))}{\partial S(w)} = \xi_0 + \frac{(\gamma - 1)\beta_0}{\delta} + 2\beta_0 w$$

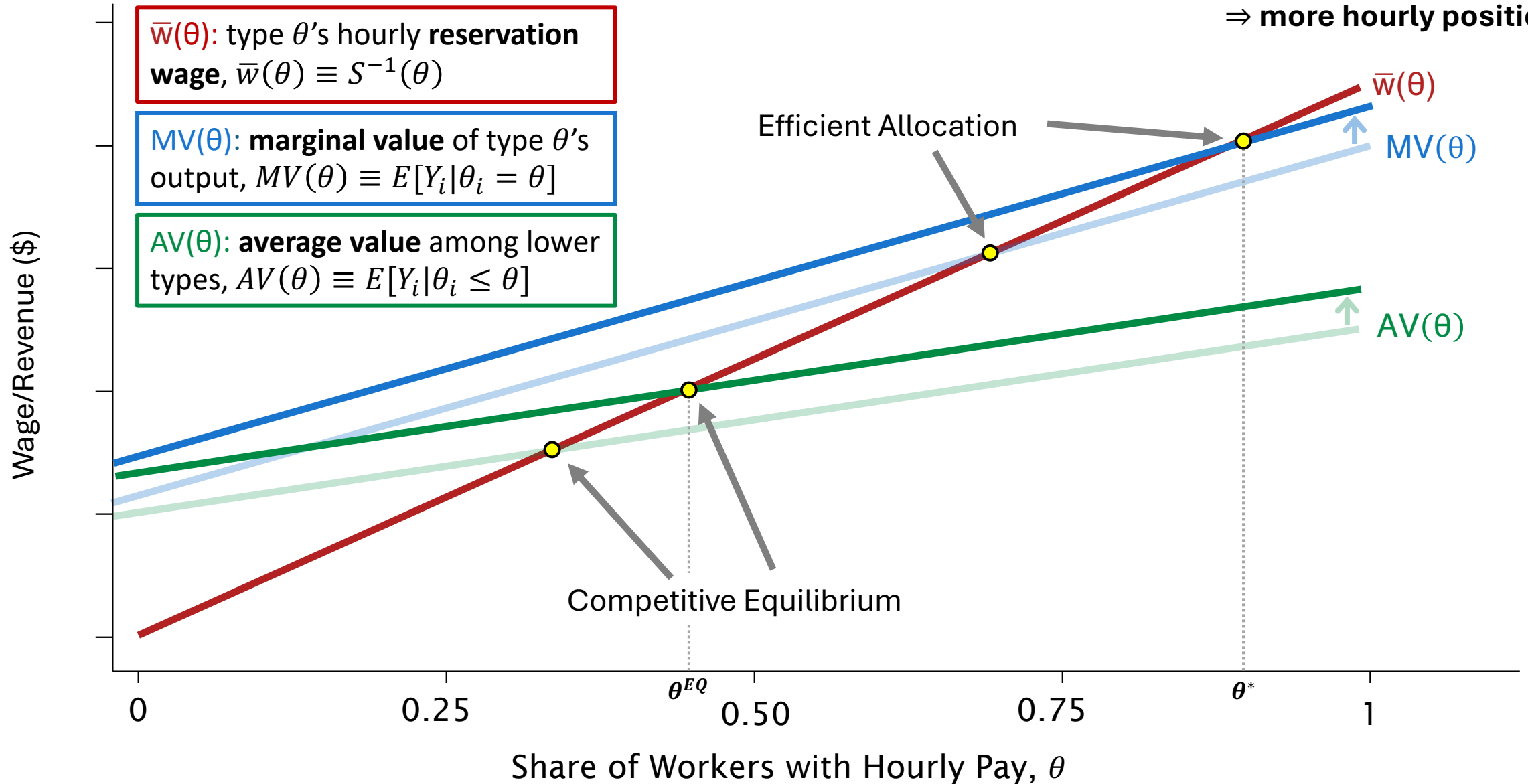


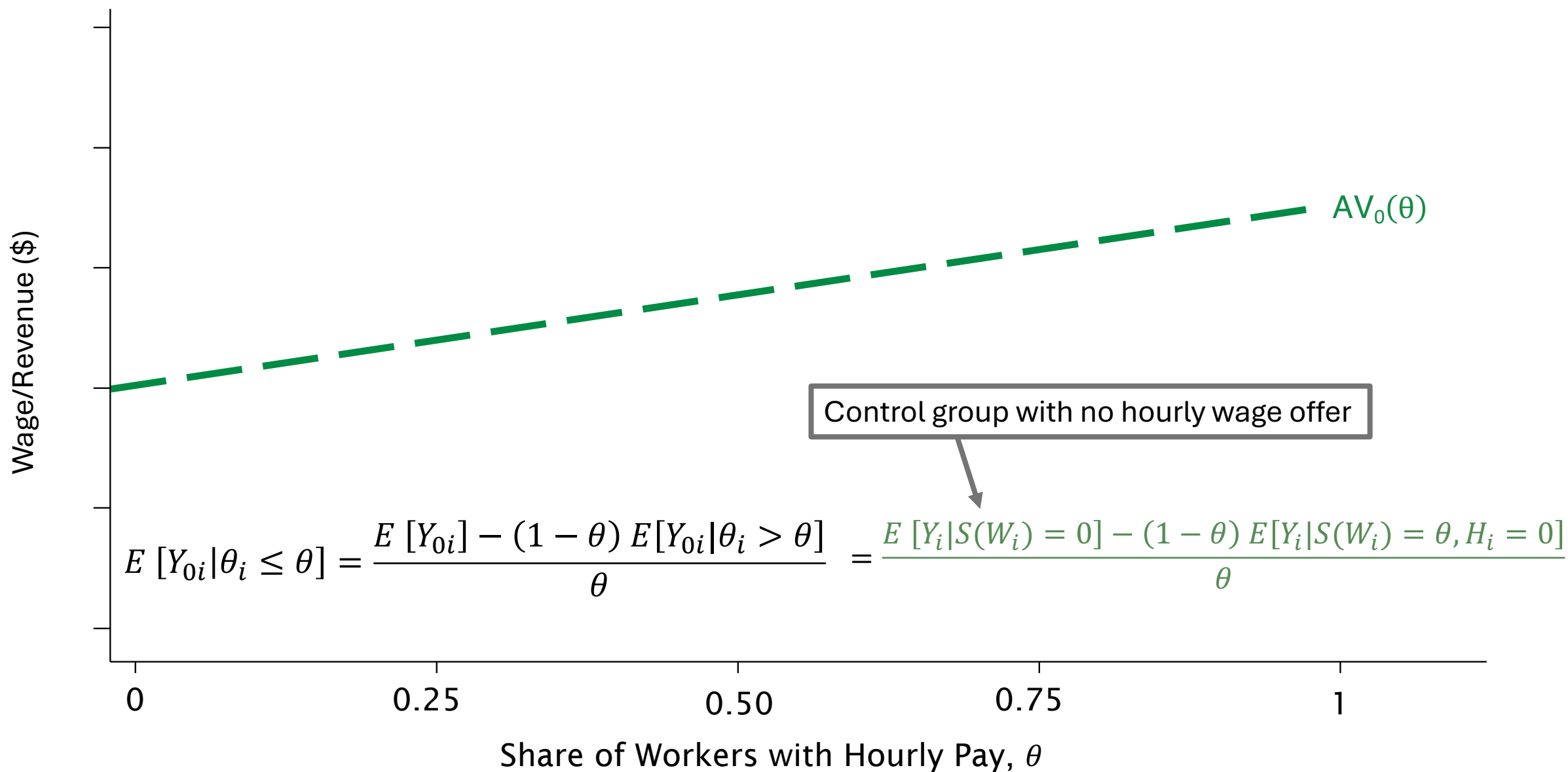
Input monitoring costs reduce firm's value of fixed-wage hire
⇒ **fewer hourly positions**





**Output monitoring costs increase the firm's value of fixed-wage hire
⇒ more hourly positions**





Estimation Details

Estimation Follows Carneiro et al. (2011):

1. Estimate $S(w) \equiv \Pr(H_i = 1 | W_i = w)$ using [logit regression](#)
2. Condition on X using double-residual regression (Robinson 1988)
 - X : Controls for number of previous tasks, task start time, and employment status
 - Assumes additive separability of X : $E[Y_{ji} | X_i = x, \theta_i = \theta] = \xi_j \tilde{X}_i + MV_j(\theta)$
3. Separately estimate $E[Y_i | S(W_i) = \theta, H_i = 1]$ and $E[Y_i | S(W_i) = \theta, H_i = 0]$
 - Local quadratic regressions of $\tilde{Y}$ on $S(W_i)$ for hourly and piece-rate workers ($bw=0.2$)
4. Differentiate with respect to θ

